

Supplementary Material for

Physics-informed neural networks and the Deep Ritz method for diffusion, wave and Laplace problems: a reproducible benchmark and its failure modes

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This document supplements the article named above. Its sections, equations, figures, tables and numbered statements carry the prefix S; numbers without the prefix refer to the main article. Sections S1–S6 follow Sections 2–7 of the article, Section S7 contains the proofs, and Section S8 the settings of every experiment, the validation of the drivers, full tables and the account of the verification. Code, data and the commands that regenerate every figure and table are at <https://github.com/zhouyi-xiaoxiao/pinn-diffusion-wave-laplace>.

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S1 Test problems: exact solutions, uniqueness and evaluation sets

This section completes Section 2. A classical solution is one that is twice continuously differentiable on the closed domain (the closed space–time cylinder for Heat, Wave1 and Wave2). The proofs of the two propositions are in Section S7.

Proposition S1 (Exact solutions).

- (a) For each of the problems Heat, Wave1, Wave2, Lap2, Lap3s, LapD and PoiD of Table 1, the function u^* listed there is infinitely differentiable on the closed domain, satisfies the differential equation at every point, and satisfies every initial and boundary condition pointwise.
- (b) For Lap3, let $k_n = \sqrt{1 + n^2}$ and $b_n = 4n/(\pi(n^2 - 1))$ for even $n \geq 2$. The series (4),

$$u^*(x, y, z) = \sin y \sum_{n \text{ even}} b_n \sin(nz) \frac{\sinh(k_n x)}{\sinh(k_n \pi)},$$

converges, together with all its term-by-term derivatives, uniformly on $[0, \pi - \delta] \times [0, \pi]^2$ for every $\delta \in (0, \pi)$. Its sum is harmonic in $(0, \pi)^3$, belongs to $C^\infty([0, \pi] \times [0, \pi]^2)$, vanishes on the five faces $\{x = 0\}$, $\{y = 0\}$, $\{y = \pi\}$, $\{z = 0\}$, $\{z = \pi\}$, and attains the datum on the sixth face in the mean-square sense,

$$\lim_{x \rightarrow \pi^-} \int_0^\pi \int_0^\pi (u^*(x, y, z) - \sin y \cos z)^2 dy dz = 0.$$

The Dirichlet data of Lap3 are discontinuous along the two edges $\{x = \pi, z = 0\}$ and $\{x = \pi, z = \pi\}$, and u^* has infinite Dirichlet energy, $\int_{(0, \pi)^3} |\nabla u^*|^2 d\mathbf{x} = \infty$.

Proposition S2 (Uniqueness).

- (a) Each of the problems Heat, Wave1, Wave2, Lap2, Lap3s, LapD and PoiD has at most one solution that is twice continuously differentiable on the closed domain (the closed space–time cylinder for Heat, Wave1 and Wave2). Hence the function u^* of Table 1 is the unique such solution.

- (b) *Problem Lap3 has at most one solution u with the following three properties: $u \in C^2([0, \pi) \times [0, \pi]^2)$ and $\Delta u = 0$ in $(0, \pi)^3$; $u = 0$ on the five faces $\{x = 0\}$, $\{y = 0\}$, $\{y = \pi\}$, $\{z = 0\}$, $\{z = \pi\}$ (for $x < \pi$); and $u(x, \cdot, \cdot) \rightarrow \sin y \cos z$ in $L^2((0, \pi)^2)$ as $x \rightarrow \pi^-$. The series (4) is this solution.*

Propositions S1 and S2 together say that each problem has a unique solution in the stated class and that it is the function u^* against which we measure; for Lap3 the class is the one in which the datum is attained in mean square.

Evaluation sets. The evaluation sets are dense grids, or random points drawn from a generator of their own. They are separate from the training points with two small exceptions. Grid nodes on the boundary or the initial line can coincide with boundary or initial training points: the four corner nodes for Heat, Wave1 and Lap2 (of 40 401), and 1241 of the 137 781 nodes for Wave2 (0.9 per cent). And for Lap3 the 512 collocation points of the cell-centred strategy grid-c are among the 64 000 evaluation points (0.8 per cent); at those points the network is trained on the residual, not on values of u^* .

Further checks of the exact solutions. The formulas were also checked by means that do not rely on the proofs. (i) Symbolic differentiation returns a zero residual and a zero mismatch in every initial and boundary condition for the seven closed-form solutions (for LapD and PoiD at $d = 2, 3, 5, 10, 20$); for Lap3 it confirms that every term of the series (4) is harmonic and vanishes on the five faces, and that the b_n are the stated sine coefficients. (ii) Central finite-difference residuals of the coded solutions were evaluated at 2000 random points of the inner 90 per cent of each box (every coordinate at least 5 per cent of the side away from both ends), with step $h = 10^{-4}$ in double precision. They are at most 1.3×10^{-4} for Wave1, 1.0×10^{-6} for Heat and 2.7×10^{-7} for Wave2, Lap2 and Lap3. The first two values are truncation errors of the difference quotients, not errors of the formulas: for Wave1 the mode of temporal frequency 8π leaves $\frac{1}{2}(\omega^4 - c^2 k^4)h^2/12 = 1.25 \times 10^{-4}$ with $\omega = 8\pi$, $k = 4\pi$, $c^2 = 4$; for Heat the quotients leave $h^2(\pi^6/6 + \pi^4/12)|u^*| = 1.68 \times 10^{-6}|u^*|$, and $|u^*| \leq 0.59$ at the sampled points, where $t \geq 0.05$. (iii) The series for Lap3, truncated at $n = 4000$ and at $n = 8000$, gives identical double-precision values at all 64 000 evaluation points. (iv) For LapD and PoiD, automatic differentiation in double precision gives $\max|\Delta u^* + f| \leq 7.11 \times 10^{-15}$ over 2000 random points for $d = 2, 3, 5, 10$, where f is the right-hand side. (v) The closed forms of Proposition 1 agree with Monte-Carlo estimates on 400 000 points to within 5×10^{-4} and with the values on the evaluation sets to within 2×10^{-3} , for $d = 2, 3, 5, 10, 20$.

S2 Methods: further details

S2.1 The fixed point sets of the heat problem

Figure S1 shows the four fixed sets for Heat. The cell-centred grid leaves the strip $0 < t < 1/32$ next to the initial line without a residual point; the node-centred grid has a column on the initial line but its next column is at $t = 1/15$.

S2.2 Implementation of the low-dimensional runs

The multi-seed runs use our own L-BFGS for a stack of networks, with Armijo backtracking (Section S8.1); a stack advances only when its slowest member has accepted a step, and the stacks used between 1.5 and 7.3 evaluations per iteration, an upper bound for each member. Section S8.2 compares the two L-BFGS implementations.

All low-dimensional experiments are run by one PyTorch package (`pinnbench`); the benchmark trains 25 or 50 networks per stack, independent in exact arithmetic, stacked along a batch axis

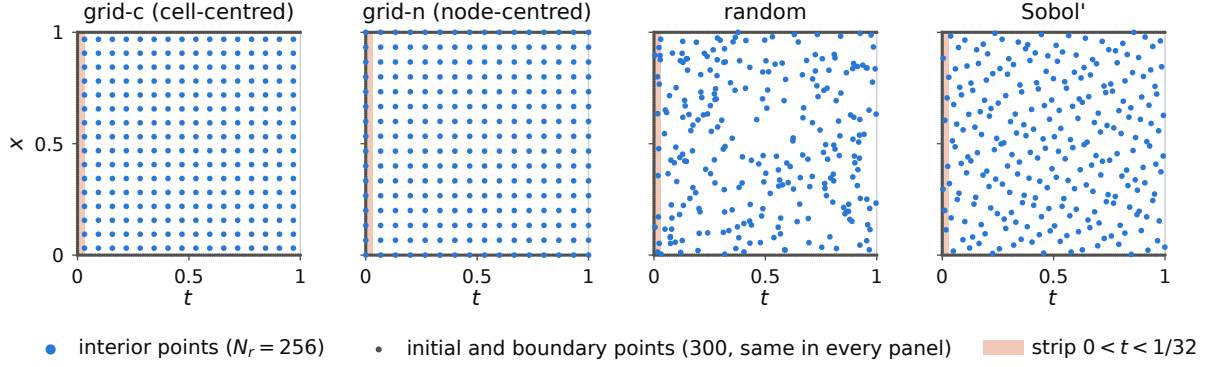


Figure S1: The four fixed interior point sets for problem **Heat** on the space–time square, $N_r = 256$: cell-centred and node-centred 16×16 grids, and the **random** and **Sobol'** sets of seed 0 as generated by the benchmark code. Grey dots: the 300 initial and boundary points, the same in every panel. Shaded: the strip $0 < t < 1/32$ between the initial line and the first column of the cell-centred grid. The strategies **resample** and **RAD** change their sets during training (Table 3) and are not shown.

on the GPU of an Apple-silicon machine (budget sweep of Section S3.3: stacks of 15; control experiment of Section 4.3: stacks of 40 and 20 on the GPU; **Lap3s**: 5 networks on the CPU). The driver is validated against single-network training in Section S8.2; in floating point the result of a network depends on the size of its stack (Section S8.4). The d -dimensional runs use one CPU thread. The machine, with 12 cores, was shared with other jobs throughout, which is why Sections 5 and 6 quote more than one time measurement wherever a cost is stated.

S2.3 The penalty bias for **LapD** at $d = 2$

Numerical observation S3 (Size of the penalty bias for **LapD** at $d = 2$). For **LapD** in dimension $d = 2$, where $u^* = x_1 x_2$ and $f = 0$, the Robin problem of Proposition 2(a) was solved by second-order finite differences (five-point scheme with ghost points; the solutions on grids of 100 and 200 cells per side differ by less than 0.03 per cent in the quantity reported). The relative L^2 distance $\|u_\beta - u^*\|/\|u^*\|$ is 0.288, 5.00×10^{-2} , 5.48×10^{-3} and 5.54×10^{-4} for $\beta = 1, 10, 100$ and 1000, that is, between $0.50/\beta$ and $0.55/\beta$ for $\beta \geq 10$, in line with part (c); the bound of part (c) and the sharper bound of Corollary 7(b) are compared with these values in Observation S6, and the bias in dimension $d = 3$ to 20 is computed in Observation S7 and, exactly, in Proposition S11. Deep Ritz networks trained with the protocol of Section 3.5 (one run for each β , 4000 iterations, with the verification code **V-dim** of Section 3.9) end at $\varepsilon_2 = 0.290$ for $\beta = 1$ and 4.78×10^{-2} for $\beta = 10$, that is, within 5 per cent of the bias of the Robin minimiser and far from zero; at 1.02×10^{-2} for $\beta = 100$, above the bias; and at 5.89×10^{-2} for $\beta = 1000$, about a hundred times the bias, where the error is that of the optimisation. The distance of the networks from u_β itself was not measured.

The bias is the error of the limit of the training, not a lower bound on the error of a trained network: a network at distance $\|v - u_\beta\|$ from the minimiser has an error between $\|u_\beta - u^*\| - \|v - u_\beta\|$ and $\|u_\beta - u^*\| + \|v - u_\beta\|$, and the run with $\beta = 10$ indeed ends 4.5 per cent below the bias. What a small β does is to make training converge to a function that is not the solution; in these runs a large β in turn slowed the optimisation. Section 6 returns to both effects. The proof of Proposition 2(a) also gives the identity $\mathcal{E}_\beta(v) - \mathcal{E}_\beta(u_\beta) = \frac{1}{2}\|\nabla(v - u_\beta)\|^2 + \beta\|v - u_\beta\|_\partial^2$ quoted in Section 3.4; the minimum of the loss is negative for **PoiD** and positive for **LapD**, where $f = 0$.

S2.4 Two ways of computing the Laplacian

The Laplacian in the PINN residual can be computed in two ways. Nested reverse-mode differentiation takes one further backward pass for each of the d coordinates. Alternatively the

triple $(u, \nabla u, \Delta u)$ can be propagated forward through the layers [9]. Both are exact: at $d = 2$ and $d = 10$ the two versions return the same single-precision loss value, digit for digit as printed (against the separately written nested version of V-dim the single-precision losses agree to seven or more significant digits up to $d = 20$), and in double precision the Laplacians differ by less than 10^{-16} ; they differ only in cost (Section 6.4). The main runs of Section 6 use the forward version; five runs of the weight sweep, made earlier, and V-dim use the nested one.

S3 Low-dimensional benchmark: further results

S3.1 Optimiser

In V-bench the Adam $\rightarrow$ L-BFGS arm is the better one in 39 of 40 pairs; the exception is Heat with grid-c and seed 1 (1.303×10^{-2} for Adam against 1.316×10^{-2}). The same network is the closest pair of the benchmark on Heat (1.58×10^{-2} against 1.36×10^{-2} , a ratio of 0.86) and, Wave1 apart, where neither arm succeeds and the closest pair has the ratio 0.995, of the whole benchmark. The logged Adam errors of one run fluctuate by a median factor of 8.0 on Heat (largest factor 41) and of 28 on Lap2 (largest 246) between iterations 1500 and 3000 (Figure S3).

One alternative to the constant learning rate of the Adam arm was tried, in V-bench with random points and seeds 0 to 2: a rate that decays exponentially from 10^{-3} to 10^{-5} over the 3000 iterations. Compared with the constant-rate runs of the same code and seeds, it ended at 3.1×10^{-2} , 2.3×10^{-2} and 2.0×10^{-2} against 3.1×10^{-2} , 2.7×10^{-2} and 9.3×10^{-2} on Heat,

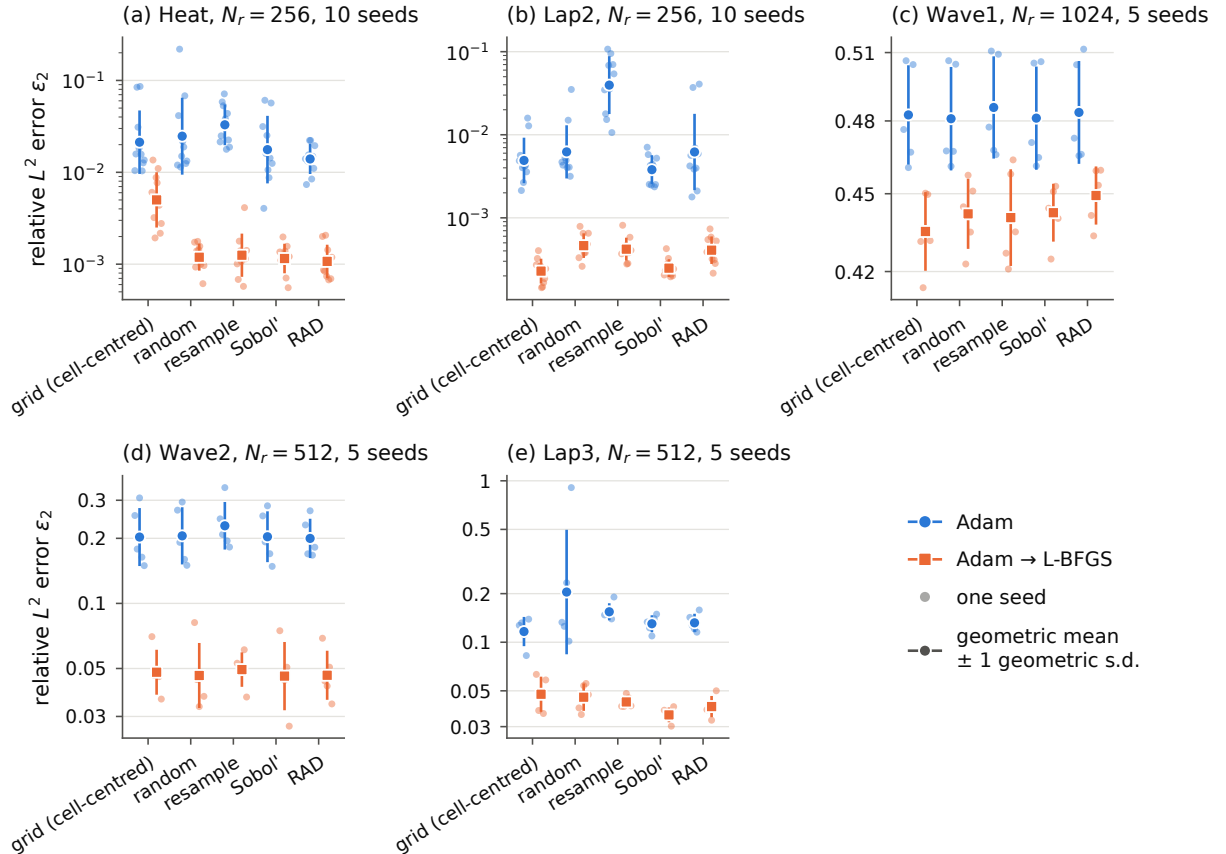


Figure S2: Held-out relative L^2 error ε_2 of every benchmark run, by problem and collocation strategy, for the arms Adam (blue circles, 3000 iterations) and Adam $\rightarrow$ L-BFGS (orange squares, 1500 + 1500 iterations). Small dots: one seed each. Large markers with bars: geometric mean over seeds and one geometric standard deviation either side. “grid (cell-centred)” is grid-c. The evaluation sets are those of Table 1; the vertical scales differ between panels.

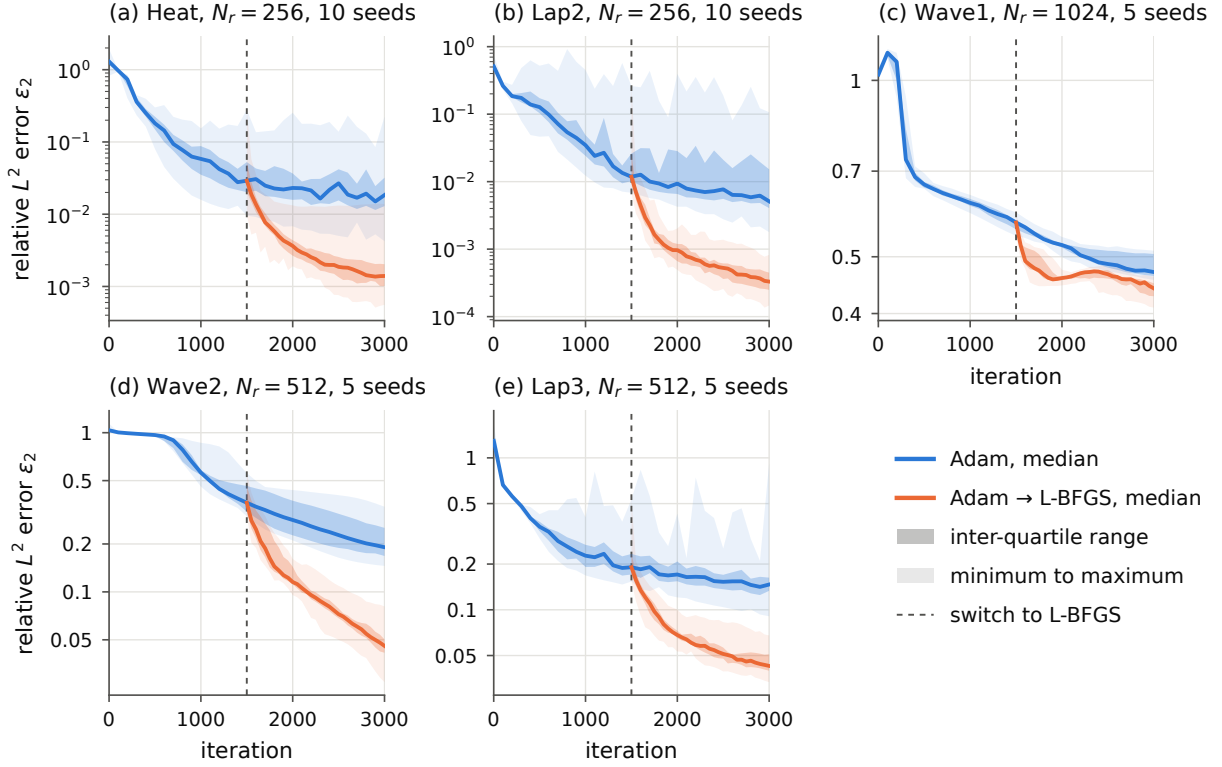


Figure S3: Relative L^2 error during training for the arms Adam (blue) and Adam $\rightarrow$ L-BFGS (orange), measured every 100 Adam iterations and every 50 L-BFGS iterations on a fixed random subset of 4096 points of the held-out evaluation set. Line: median over all strategies and seeds (50 runs per arm for Heat and Lap2, 25 for Wave1, Wave2 and Lap3); dark band: inter-quartile range; light band: minimum to maximum. Dashed line: iteration 1500, where the Adam $\rightarrow$ L-BFGS arm branches from the Adam run.

and at 1.7×10^{-2} , 2.0×10^{-2} and 2.3×10^{-2} against 7.4×10^{-3} , 4.6×10^{-3} and 4.4×10^{-3} on Lap2. On Heat this schedule ended at 1.00, 0.85 and 0.22 times the constant-rate error, which is still more than ten times the error of the Adam $\rightarrow$ L-BFGS arm; on Lap2 it was 2.3 to 5.3 times worse. It did not close the gap; it is not a tuned Adam baseline.

S3.2 Collocation strategy

In the error field of seed 0 (Figure S4) the largest error of **grid-c** on Heat, 6.5×10^{-3} , is at $t = 0.025$. In the control experiment of Section 4.3, in 9 of 10 seeds between 67 and 73 per cent of the squared error of **grid-c** lies in $t \leq 0.05$, against a median of 25 per cent for **random** (Table 6). The per-seed ratios of the node-centred grid **grid-n** to **random** in the control range from 0.52 to 7.2. In V-bench, with 6 seeds, **grid-n** was 1.26 times **random** (per-seed ratios 1.08, 1.13, 5.2, 0.95, 1.27 and 0.52; $p = 0.44$) and 0.32 times **grid-c**, while the same code reproduced the penalty of **grid-c**, 3.94 times **random** (worse in 6 of 6). It is not established that a grid with a column on the initial line loses all of the penalty. On Lap2 the grid result concerns the cell-centred grid; **grid-n** was not run there.

The adaptive strategy. In the Adam $\rightarrow$ L-BFGS arm the set of RAD is the Sobol set for the first 500 iterations and is then redrawn four times only, from a pool of $8N_r$ points: at iterations 500, 1000 and 1500 and once more before L-BFGS, the third set serving a single iteration. It is fixed during the 1500 L-BFGS iterations, in which most of the error is removed. Wu et al. [19], in their Burgers example, first train for 16 000 iterations and then redraw every 2000 iterations, half of which are L-BFGS iterations. On Lap3, the one problem whose solution is not smooth and on which a residual-adaptive strategy would be expected to matter most, RAD has the ratio

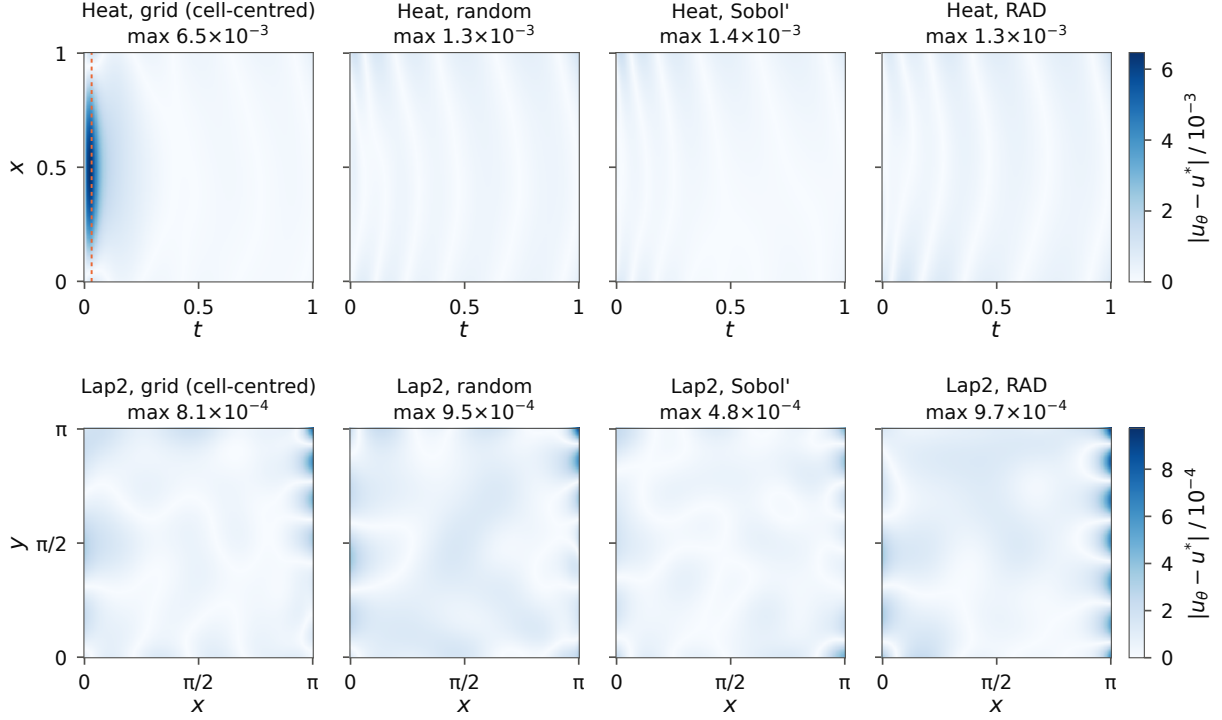


Figure S4: Pointwise absolute error $|u_\theta - u^*|$ on the 201×201 evaluation grid for seed 0 after Adam $\rightarrow$ L-BFGS, with $N_r = 256$. Top row: **Heat** (time horizontal); bottom row: **Lap2**. Columns: **grid-c**, **random**, **Sobol'**, **RAD**. Each row has one linear colour scale, in units of 10^{-3} and 10^{-4} . The dashed line in the first panel is $t = 1/32$, the first column of the cell-centred grid. One seed is shown; the statistics are in Tables 5 and 6.

0.88 to random, better in 3 of 5 seeds.

Sobol' points seed by seed. The ratios of Sobol to random after Adam $\rightarrow$ L-BFGS are 0.97, 0.54, 1.00, 0.99 and 0.78 for Heat, Lap2, Wave1, Wave2 and Lap3; on Heat four of the ten per-seed ratios are above 1 (1.18, 1.12, 1.45 and 1.25), and on Lap3 Sobol is better in 5 of 5 seeds, which five seeds cannot make significant.

Resampling after L-BFGS. The three V-bench seeds on Lap2 give the ratio 1.56 of resample to random after Adam $\rightarrow$ L-BFGS (per seed 2.15, 1.90 and 0.94; better in 1 of 3 seeds).

S3.3 Point budget

With 64 points neither paired difference is distinguishable from none (Table S1). V-bench confirms the agreement at 1024 points on two seeds (grid-c 1.10×10^{-3} and 8.2×10^{-4} ; random 1.29×10^{-3}

Table S1: Budget sweep on problem Heat, arm Adam $\rightarrow$ L-BFGS, seeds 0 to 4: mean $\pm$ sample standard deviation of ε_2 , and the ratio to random with the same seed (geometric mean; in brackets the number of seeds in which the strategy is better and the exact two-sided Wilcoxon p -value, whose smallest attainable value is 0.0625). The row $N_r = 256$ is the benchmark, restricted to the same five seeds.

N_r	ε_2 , mean $\pm$ s.d.			ratio to random	
	grid-c	random	Sobol	grid-c	Sobol
64	$(7.21 \pm 3.09) \times 10^{-2}$	$(4.76 \pm 4.45) \times 10^{-2}$	$(1.64 \pm 1.27) \times 10^{-2}$	2.02 (1/5; 0.312)	0.41 (3/5; 0.438)
256	$(6.71 \pm 4.67) \times 10^{-3}$	$(1.45 \pm 0.34) \times 10^{-3}$	$(1.35 \pm 0.37) \times 10^{-3}$	3.79 (0/5; 0.0625)	0.93 (3/5; 0.438)
1024	$(1.23 \pm 0.45) \times 10^{-3}$	$(1.13 \pm 0.43) \times 10^{-3}$	$(1.15 \pm 0.41) \times 10^{-3}$	1.11 (1/5; 0.438)	1.02 (2/5; 0.812)

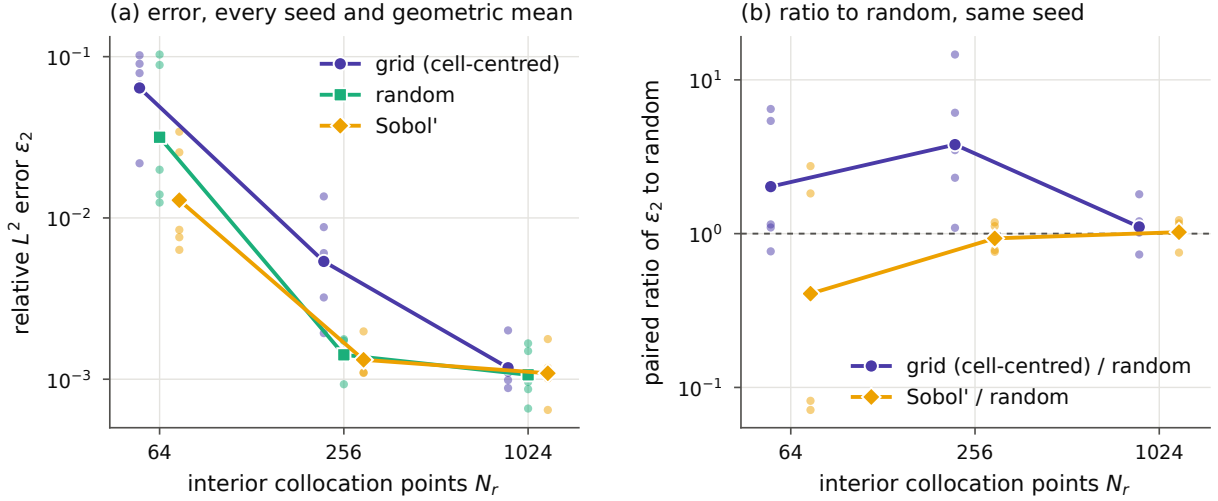


Figure S5: Budget sweep on problem Heat, arm Adam $\rightarrow$ L-BFGS, seeds 0 to 4. (a) Relative L^2 error ε_2 against the number N_r of interior points for grid-c, random and Sobol; dots: single seeds, lines: geometric means. (b) Ratio of ε_2 to that of random with the same seed; lines: geometric mean of the ratios. Markers are displaced horizontally for legibility; the budgets are 64, 256 and 1024.

and 8.1×10^{-4}). We do not extend the conclusion of the sweep to the other problems or to larger budgets.

S3.4 Error on time slices

The error itself is not concentrated at early times: in these runs the time levels $t \leq 0.25$ carry 42 to 64 per cent of the squared error (medians of the four combinations of strategy and arm). Six of the 20 L-BFGS phases stopped on the tolerance of the PyTorch optimiser after 1365 to 1493 of their 1500 iterations.

Table S2: Problem Heat, $N_r = 256$, 10 seeds: relative L^2 error in per cent on the whole evaluation grid (ε_2) and on single time slices ($\varepsilon_2^{\text{slice}}(t)$); geometric means over seeds, and the range over seeds at $t = 1$. Figure 1 shows every run.

strategy	arm	$100 \varepsilon_2$	$100 \varepsilon_2^{\text{slice}}(t)$ at $t =$						range at $t = 1$
			0	0.1	0.25	0.5	0.75	1	
random	Adam	1.58	0.572	1.09	3.82	24.9	420	5849	2360–52 582
random	Adam $\rightarrow$ L-BFGS	0.107	0.0436	0.0777	0.235	2.26	20.9	479	156–1241
Sobol	Adam	2.06	0.625	1.50	5.36	41.2	536	7242	1965–107 521
Sobol	Adam $\rightarrow$ L-BFGS	0.116	0.0421	0.0886	0.349	3.49	32.6	567	198–1298

S4 Harder problems and classical baselines: further results

S4.1 Longer runs on Wave2

At iteration 3000 the longer runs of Section 5.1 agree with the benchmark runs of the same seeds (7.4×10^{-2} , 4.5×10^{-2} , 4.5×10^{-2} against 7.3×10^{-2} , 4.6×10^{-2} , 4.4×10^{-2} , both on the 4096-point subset of the evaluation grid on which the training curves are logged). The L-BFGS phase stops on the termination test of the optimiser (gradient below 10^{-9} , or step or loss change below 10^{-12}). Between iterations 2000 and 3000 the error falls by a factor of 2.1 to 3.3; over the last 1000 logged iterations before iteration 9500 it still falls by 8 to 15 per cent (Figure S6).

S4.2 The two modes of Wave1

We projected the ten stored predictions of seed 0 of the benchmark (five strategies, two arms) onto the two spatial modes, $a_k(t) = \langle u_\theta(t, \cdot), \sin k\pi x \rangle / \langle \sin k\pi x, \sin k\pi x \rangle$ for $k = 1, 4$. The amplitude of the slow mode is reproduced to 8 to 12 per cent after Adam and 4 to 9 per cent after Adam $\rightarrow$ L-BFGS (relative L^2 error of a_1 over t); the amplitude of the fast mode has a root-mean-square value of 6 per cent of the exact one after Adam and 23 to 25 per cent after Adam $\rightarrow$ L-BFGS, and the fast mode carries 84 to 93 per cent of the squared error. At iteration 3000 the longer runs of Section 5.3 agree with the benchmark runs of the same seeds (0.442, 0.414 and 0.452 against 0.442, 0.419 and 0.448, both on the 4096-point subset on which the curves are logged). In the six longer runs the root-mean-square amplitude of the fast mode is 0.35 to 0.43 of the exact one.

S4.3 Finite differences

On Lap3 the observed order of the seven-point scheme is 1.97, 2.70 and 1.89 (Table S18): the value 2.70 occurs where the evaluation changes from cubic interpolation ($n = 20, 40$) to nodal values ($n = 80, 160$) of the cell-centred evaluation grid, and the last value, between two nodal evaluations, is somewhat below two. The datum on the face $x = \pi$ is the cause of both irregularities: with the compatible single-mode datum of Lap3s the same solver and the same

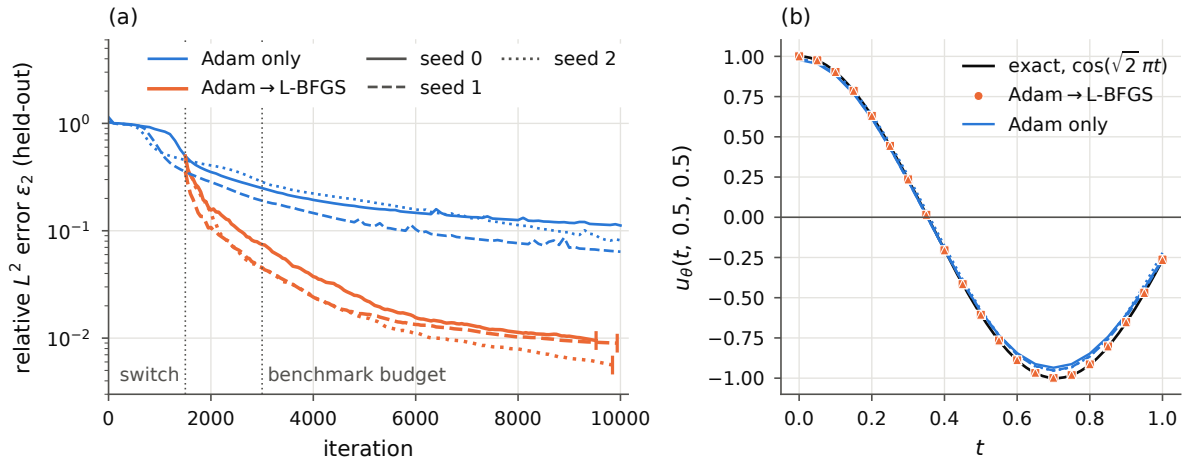


Figure S6: Problem Wave2, Sobol points, $N_r = 512$, seeds 0, 1, 2 (longer runs of Section 5.1). (a) Relative L^2 error on a fixed random subset of 4096 points of the $21 \times 81 \times 81$ held-out grid against the iteration number, for 10 000 Adam iterations (blue) and for 1500 Adam iterations followed by L-BFGS (orange); the dotted vertical lines mark the switch and the 3000-iteration budget of the benchmark, and the short vertical ticks the iteration at which L-BFGS stopped. (b) Value of the trained networks at the point $(x, y) = (0.5, 0.5)$ at $t = 0, 0.05, \dots, 1$: markers, Adam $\rightarrow$ L-BFGS (three seeds, one marker shape each); blue lines, Adam only; black line, exact solution.

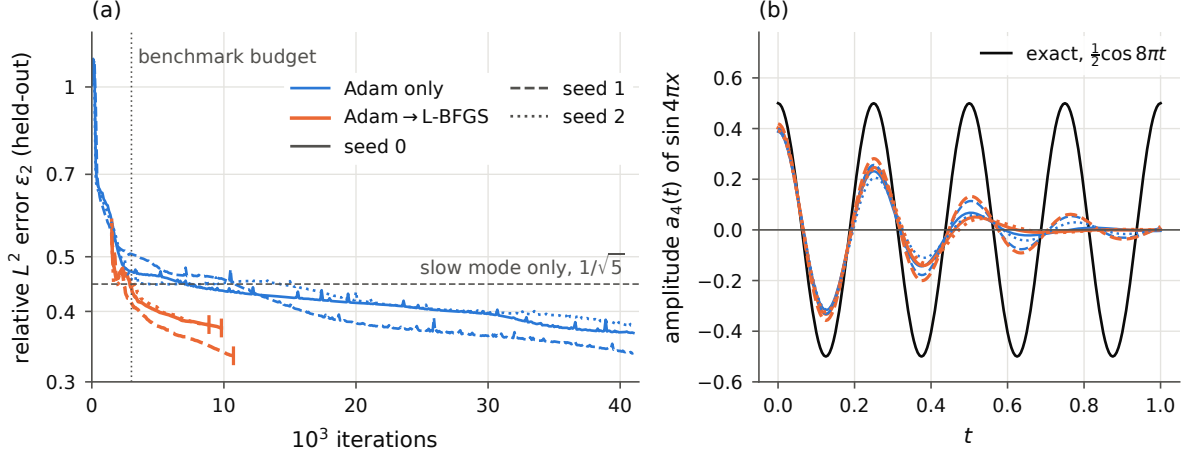


Figure S7: Problem Wave1, Sobol points, $N_r = 1024$, seeds 0, 1, 2 (longer runs of Section 5.3). (a) Relative L^2 error on a fixed random subset of 4096 points of the 201×201 held-out grid against the iteration number, for 41 000 Adam iterations (blue) and for 1500 Adam iterations followed by L-BFGS (orange); the short vertical ticks mark the iteration at which L-BFGS stopped, the dotted vertical line the budget of the benchmark, and the dashed horizontal line the error $1/\sqrt{5}$ of a predictor that contains the slow mode only. (b) Amplitude $a_4(t)$ of the fast mode $\sin 4\pi x$ in the trained networks (same colours and line styles), against the exact amplitude $\frac{1}{2}\cos 8\pi t$ (black).

evaluation give $\varepsilon_2 = 2.11 \times 10^{-3}$, 5.29×10^{-4} , 1.32×10^{-4} and 3.31×10^{-5} for $n = 20, 40, 80$ and 160 , that is, the orders 1.99, 2.00 and 2.00. Since Lap3s also makes the solution a single mode, the comparison does not separate the compatibility of the data from the modal content; that the discontinuity is what lowers the order is what the theory of the scheme leads one to expect, since the scheme is of second order for smooth solutions. The finite-difference errors do not rest on one code: the verification re-derived them from closed-form solutions of the discrete schemes and from a separate sparse solve. For the grids of Table 8 it obtained 4.918×10^{-4} for Heat (on the nodes of the finite-difference grid, not on the interpolated evaluation grid), 7.508×10^{-5} for Lap2, 5.121×10^{-3} for Wave2 and 9.968×10^{-3} for Lap3 (3.900×10^{-2} for $n = 20$); Wave1 was not re-derived.

S4.4 Cost

(A) The single-network driver was timed for 400 Adam and 200 L-BFGS iterations while the load average was 95 to 167 and extrapolated to 3000 iterations. (B) The runs of V-bench comprise 21 runs for Heat, 12 for Lap2 and one for each other problem. Twelve of these 36 runs, all on Heat and Lap2, stopped on the tolerance of L-BFGS after 1178 or more of its 1500 iterations, which makes the lower ends of the ranges, if anything, too low. (C) The training time of the Adam $\rightarrow$ L-BFGS arm of a stack divided by the number of networks in it is 12.4 s for Heat (50 networks), 11.5 and 13.9 s for Lap2 (two stacks of 25), 6.9 s for Wave1, 7.4 s for Wave2 and 11.7 s for Lap3; the load during these runs was not recorded. Table S19 keeps the three measurements apart.

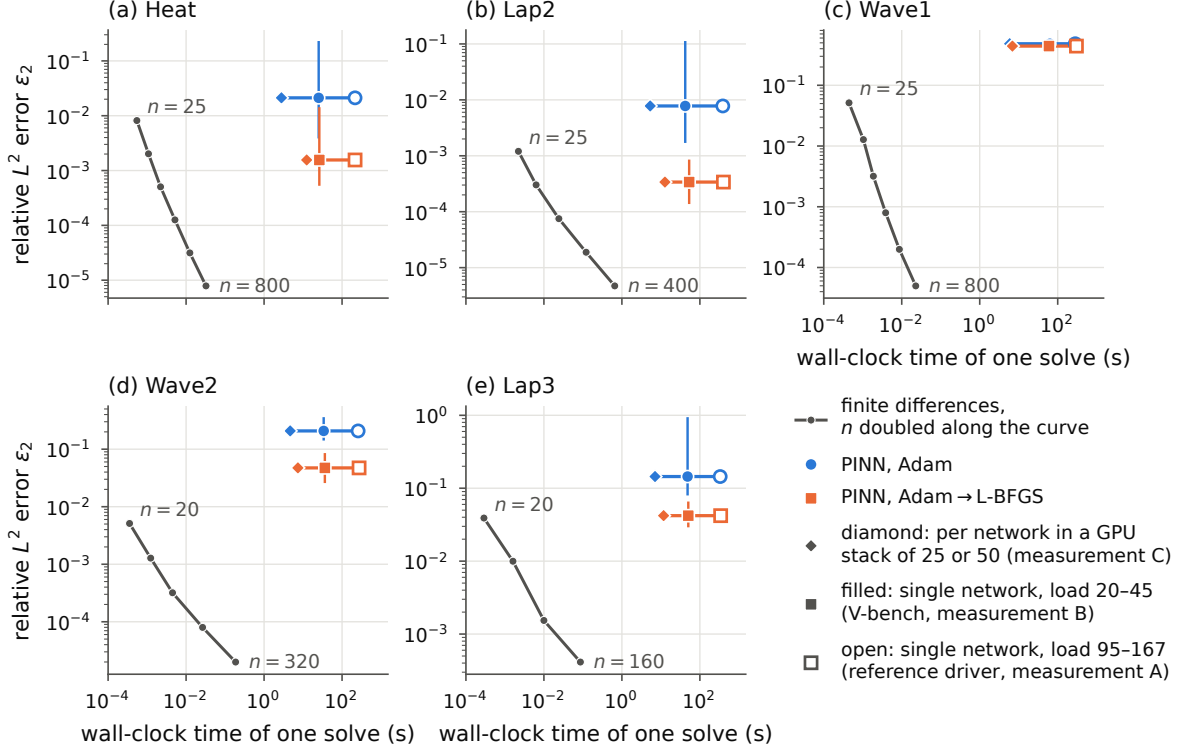


Figure S8: Accuracy against wall-clock time of one solve, one panel per problem. Grey: second-order finite differences, the grid parameter n doubling along each curve (time: best of three solves). Blue circles and orange squares: Adam and Adam → L-BFGS arms of the benchmark (3000 iterations); the ordinate is the geometric mean of ε_2 over all strategies and seeds (50 runs for Heat and Lap2, 25 otherwise) and the vertical bar its range over those runs. Each PINN arm is drawn at three times, the measurements C, B and A of Section 5.4; diamond, time per network in the GPU stacks of the benchmark; filled marker, median of the single-network runs of V-bench; open marker, single-network driver under heavy load. The horizontal bar spans the three. All errors are on the held-out sets of Table 1.

S5 Dimension study: further results

S5.1 Errors at equal iterations, verification runs and training curves

The verification code V-dim (separately written, nested Laplacian, its own test points, seeds 7 to 9, and 7 to 11 for Deep Ritz on LapD at $d = 20$; Table S5) also shows that three seeds understate the spread. On LapD at $d = 2$ the gap between its worst PINN seed and its best Deep Ritz seed is narrow (9.19×10^{-3} against 1.02×10^{-2}). On LapD at $d = 20$ one seed gave 3.36×10^{-2} (PINN) and 3.59×10^{-2} (Deep Ritz), below both ranges of Table S3, and its five Deep Ritz seeds have a standard deviation of 4.1×10^{-3} against 1.8×10^{-3} for our three. The standard deviations in the tables are therefore rough. The slope of Deep Ritz on LapD in Section 6.2 is flattened by errors that are not monotone in d (Table S3); we did not isolate the cause.

S5.2 The networks at $d = 20$ and the effect of the budget

On PoiD at $d = 10$ the mean validation error of Deep Ritz is 0.143, 0.141, 0.138, 0.137 at iterations 250, 500, 1000, 1500 (Figure S9). On LapD at $d = 20$ the descent has begun, but the learning-rate reduction at iteration 2000 arrests it (0.111 at iteration 250, 0.052 to 0.057 at 2000, 0.046 to 0.047 at 4000). The 1024 boundary points of an iteration are shared among 40 faces at $d = 20$. With 16 000 iterations the errors at $d = 10$ on LapD are 5.88×10^{-3} (PINN) and 7.28×10^{-3} (Deep Ritz), and at $d = 20$ on LapD 7.57×10^{-3} and 8.70×10^{-3} ($\varepsilon_2^C = 2.8 \times 10^{-2}$ and

Table S3: Problem LapD after 4000 iterations, three seeds: errors on the 50 000 held-out points in units of 10^{-2} (so 1.260 means 1.260×10^{-2}). For ε_2 : mean $\pm$ sample standard deviation, with [minimum, maximum] underneath; for ε_{H^1} and the centred error ε_2^c : means. Ratio: mean ε_2 of Deep Ritz divided by that of the PINN. Floors: ε_2 of the best affine and of the best constant approximation of u^* (Proposition 1).

d	ε_2		ratio	ε_{H^1}		ε_2^c		floor of ε_2	
	PINN	Deep Ritz		PINN	Deep Ritz	PINN	Deep Ritz	affine	constant
2	0.608 ± 0.187 [0.392, 0.721]	1.882 ± 0.347 [1.527, 2.220]	3.10	2.14	6.05	0.92	2.85	25.0	66.1
3	0.672 ± 0.255 [0.457, 0.954]	1.421 ± 0.337 [1.160, 1.802]	2.11	2.40	5.43	1.01	2.14	25.0	66.1
5	0.760 ± 0.055 [0.698, 0.804]	1.113 ± 0.077 [1.039, 1.192]	1.46	3.29	5.05	1.44	2.11	20.0	52.9
10	1.260 ± 0.084 [1.171, 1.337]	1.896 ± 0.159 [1.730, 2.047]	1.50	6.45	9.63	3.42	5.15	13.9	36.7
20	4.663 ± 0.248 [4.381, 4.848]	4.613 ± 0.178 [4.416, 4.764]	0.99	24.51	26.71	17.30	17.11	10.2	26.9

Table S4: Problem PoiD after 4000 iterations, three seeds; quantities and units as in Table S3. Here $\varepsilon_2^c = \varepsilon_2$ because u^* has zero mean, and the best constant has $\varepsilon_2 = 1$.

d	ε_2		ratio	ε_{H^1}		floor of ε_2
	PINN	Deep Ritz		PINN	Deep Ritz	
2	0.134 ± 0.049 [0.102, 0.190]	0.574 ± 0.121 [0.441, 0.678]	4.29	0.73	2.70	12.0
3	0.380 ± 0.038 [0.347, 0.421]	1.426 ± 0.183 [1.215, 1.536]	3.76	1.82	5.83	12.0
5	0.929 ± 0.053 [0.875, 0.982]	1.509 ± 0.041 [1.462, 1.539]	1.62	4.10	6.72	12.0
10	2.168 ± 0.482 [1.710, 2.671]	7.488 ± 2.553 [5.057, 10.148]	3.45	8.58	23.41	12.0
20	14.424 ± 0.176 [14.231, 14.576]	12.633 ± 0.022 [12.614, 12.658]	0.88	47.86	45.82	12.0

3.2×10^{-2}). On PoiD at $d = 20$ the PINN seeds end at 4.90×10^{-2} , 6.58×10^{-2} and 6.02×10^{-2} and the Deep Ritz seeds at 9.17×10^{-2} , 9.87×10^{-2} and 8.26×10^{-2} . The PINN's remainder after its affine fit now correlates 0.84 to 0.91 with that of u^* and its residual is 0.12 to 0.15; the Deep Ritz remainder has reached 0.63 to 0.76 of the right norm, with residual 0.60 to 0.70. The lower residual of the PINN cannot by itself lower its error by more than the bound of Remark 4.

Table S5: Runs of the verification code V-dim (Section 3.9) after 4000 iterations: ε_2 of each seed, in units of 10^{-2} , on its own 50 000 test points; weights as in the main study. For LapD at $d = 20$ two further PINN seeds (8 and 9), run with the forward Laplacian, gave 4.50 and 4.46.

	d	PINN	seeds	Deep Ritz	seeds
LapD	2	0.201, 0.919, 0.146	7–9	1.02, 2.29, 1.64	7–9
LapD	10	1.09, 1.15, 1.39	7–9	1.52, 1.69, 2.52	7–9
LapD	20	3.36	7	3.59, 4.53, 4.53, 4.50, 4.40	7–11
PoiD	2	0.124, 0.123, 0.126	7–9	0.469, 0.506, 0.662	7–9
PoiD	10	2.46, 2.09, 2.56	7–9	9.54, 9.59, 7.11	7–9
PoiD	20	14.2	7	12.7, 12.7	7–8

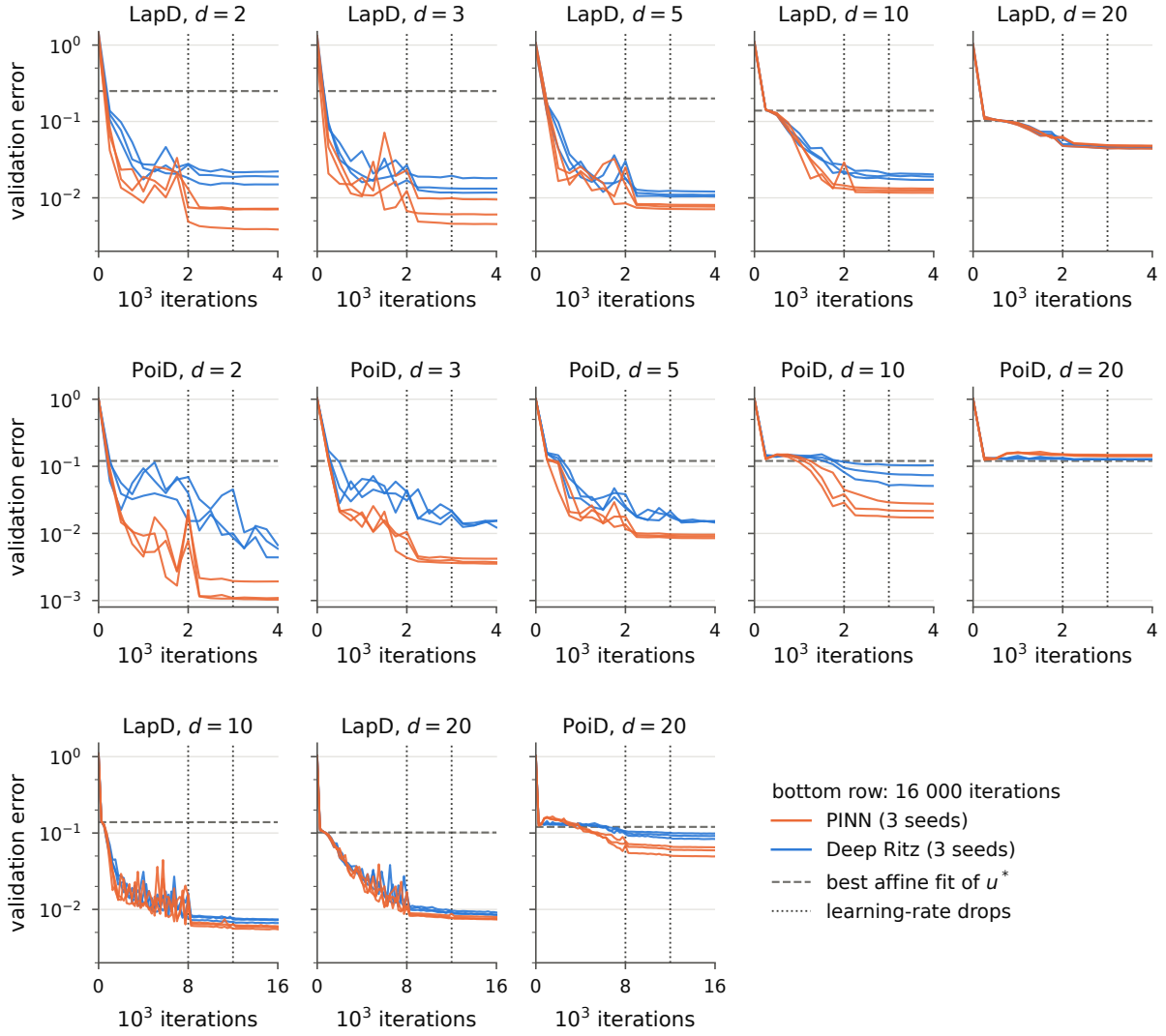


Figure S9: Relative L^2 error on a validation set of 5000 points (distinct from the test points), recorded every 250 iterations, for each of the three seeds; PINN in orange, Deep Ritz in blue. Top two rows: the 4000-iteration runs of Tables S3 and S4. Bottom row: the 16 000-iteration runs of Table 9. Dashed: error of the best affine approximation of u^* . Dotted verticals: the learning-rate reductions at 50 and 75 per cent of the run.

S5.3 Cost per iteration

At $d = 100$ the medians of the ratio of a PINN iteration to a Deep Ritz iteration are 10.7 and 13.3 with the forward Laplacian and 29.5 and 32.2 with the nested one; for $d \geq 10$ the nested Laplacian costs 1.9 to 2.9 and 1.8 to 2.3 times the forward one in the two measurements. Process CPU time on this machine depends on the load: seven training runs repeated bit for bit took 1.1 to 1.8 times the CPU seconds stored for the originals. The Deep Ritz times of the primary measurement at $d = 50$ and 100 (8.0 and 8.4 ms, with rounds ranging from 5.8 to 9.7 ms at $d = 100$) appear inflated in this way; the quieter measurement gives 4.8 and 5.1 ms.

S5.4 Equal compute

With the smaller count of 9000 Deep Ritz iterations on LapD at $d = 10$, V-dim still gives 9.24×10^{-3} , 9.41×10^{-3} , 9.65×10^{-3} , below every PINN run (Figure S10(b,c)). The equal-compute comparison was made in the study only at $d = 10$, with three seeds and three seeds of V-dim, and we do not extrapolate it: the cost ratio grows with d (Table S6), but so does the

Table S6: CPU time of one training iteration in milliseconds (one thread, network and batch of Section 6; median of 7 rounds of 10 iterations, the three variants interleaved in every round) for Deep Ritz and for the PINN with the Laplacian computed by forward propagation or by nested reverse-mode differentiation. “Ratio”: time relative to Deep Ritz, computed round by round; median, then [minimum, maximum]. Last column: nested divided by forward (ratio of the medians). (a) Timing of the primary study, load average about 90 to 110 on 12 cores. (b) Second timing, made with separate code for the verification, load average about 20; $d = 3$ was not measured. Absolute times depend on the load.

d	Deep Ritz	PINN, forward Laplacian			PINN, nested			nested/forward
	ms	ms	ratio	[min, max]	ms	ratio	[min, max]	
<i>(a) primary measurement</i>								
2	4.22	6.50	1.54	[1.45, 1.64]	7.48	1.72	[1.56, 1.92]	1.15
3	4.33	7.06	1.62	[1.50, 1.90]	9.06	2.12	[1.96, 2.26]	1.28
5	4.55	8.08	1.77	[1.74, 1.90]	12.47	2.70	[2.56, 2.94]	1.54
10	4.68	10.66	2.23	[2.10, 2.34]	20.61	4.44	[3.91, 4.53]	1.93
20	4.98	15.23	3.09	[2.90, 3.16]	37.80	7.72	[7.11, 8.74]	2.48
50	8.02	48.82	6.11	[5.66, 7.82]	139.26	18.21	[14.78, 19.94]	2.85
100	8.36	90.31	10.73	[9.76, 15.19]	243.45	29.46	[23.55, 42.29]	2.70
<i>(b) second measurement (verification)</i>								
2	3.29	5.44	1.59	[1.44, 1.82]	6.09	1.81	[1.67, 1.98]	1.12
5	3.69	6.47	1.87	[1.65, 2.25]	9.97	2.83	[2.52, 3.17]	1.54
10	4.03	9.86	2.48	[1.87, 2.99]	17.93	4.39	[3.55, 5.66]	1.82
20	4.47	14.26	3.15	[2.54, 4.09]	33.26	6.80	[6.07, 9.09]	2.33
50	4.77	36.99	8.44	[7.01, 8.70]	76.74	16.71	[14.00, 18.45]	2.07
100	5.12	69.03	13.34	[12.00, 16.16]	159.63	32.20	[28.15, 35.47]	2.31

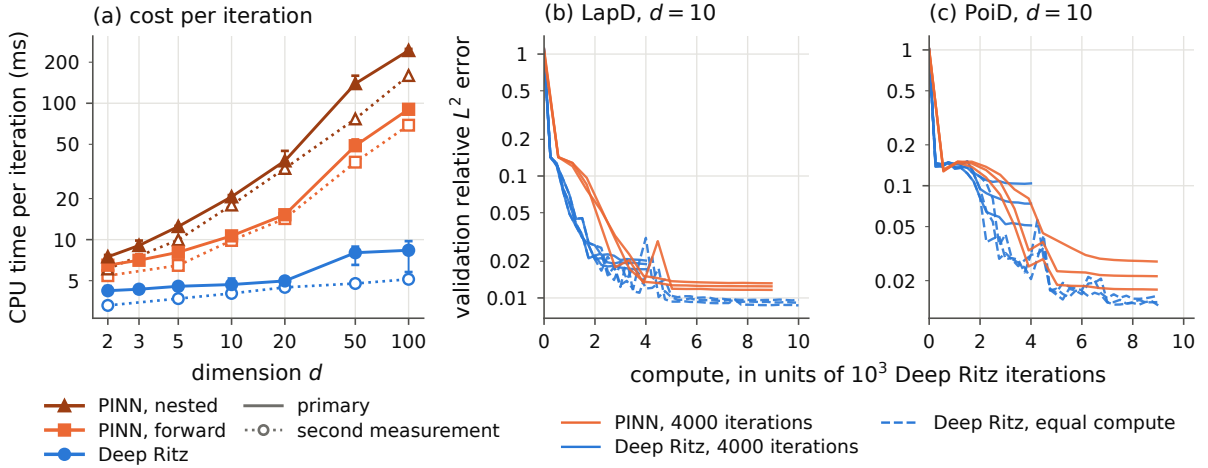


Figure S10: (a) CPU time per training iteration against d (Table S6): filled symbols and full lines are the primary measurement (median; bars from the fastest to the slowest of 7 rounds), open symbols and dotted lines the second one. (b, c) Validation error (5000 points) against compute at $d = 10$ for the three seeds of each method. The abscissa counts compute in Deep Ritz iterations; one PINN iteration is counted as 2.23 of them, the ratio of the primary measurement (with the ratio 2.48 of the second measurement the PINN curves would end at 9.9 instead of 8.9). Dashed: Deep Ritz with 10 000 iterations (LapD) and 9000 (PoiD).

number of iterations that either method needs (Table 9), and no equal-compute run was made at the larger budget.

S5.5 Penalty weights and penalty bias

The PINN on LapD changes little with λ (Table S7). At $d = 10$ on PoiD (V-dim, seeds 7–9) Deep Ritz with $\beta = 10$ or 100 stays on the affine plateau for the whole run (ε_2 between 0.119 and 0.126, against 0.071 to 0.096 with $\beta = 1$), and so does the PINN with $\lambda = 10^4$ (0.111 to 0.120, against 0.021 to 0.026 with $\lambda = 10^3$); on LapD at $d = 10$, $\lambda = 10^4$ is 14 per cent worse in the mean. A large weight lengthens the plateau. Smaller values of λ , and values of β other than 100 on LapD, were not tried at $d = 10$.

At $d = 2$ on LapD the minimiser of the penalised energy with $\beta = 100$ is itself 5.5×10^{-3} away from u^* (Observation S3). The bias is not a lower bound for a run that has not converged, which can end on either side of it: the run with $\beta = 10$ of Observation S3 ends 4.5 per cent below the bias for that β . After 16 000 iterations the bias is 20 and 8 per cent of the Deep Ritz error at $d = 10$ and 20 (Table S8). By the triangle inequality the Deep Ritz networks at $d = 10$ are between 5.8×10^{-3} and 8.7×10^{-3} from their own target u_β , against 5.9×10^{-3} for the distance of the PINN from u^* .

S5.6 The constant and the error bound: protocol, outcomes and re-checks

Protocol. The 88 saved networks are the main runs at $d = 2, \dots, 20$, the 16 000-iteration and equal-compute runs at $d = 10$ and the weight sweep at $d = 5$. Besides H1 to H3, two descriptive outputs were fixed before the confirmatory evaluation: η and the share against the penalty weight in the sweep at $d = 5$ (H4), and a comparison with the bound from the maximum principle. Thirteen networks of an earlier pilot had been evaluated before the freeze; seven of the 88 saved networks are bit-identical to some of them and are treated as non-confirmatory.

Outcomes. The margins of H1 are at least 222 standard errors on the new points and 75 on the test points. All failures of H2 are at $d = 5$ except the three equal-compute Deep Ritz networks of PoiD at $d = 10$ ($\eta = 3.36$ to 3.44); at $d = 2$ the efficiency of Deep Ritz on PoiD is 14.7 to 24.8. For the 4000-iteration networks at $d = 10$ and 20 and the 16 000-iteration networks at $d = 10$ the bound is within a factor 1.93 to 2.76 of the error (Table S10). No explanation of the failures was pre-registered or tested. The post-hoc conjecture that the failures are exactly the networks with the smallest boundary shares does not hold: the share ranges of failures and passes overlap. H3 holds also without the seven non-confirmatory networks. At $d = 10$ the interior term B_{int} is up to 0.81 of the error, so there Remark 4 does not show the error to be dominated by the boundary misfit. In the weight sweep at $d = 5$ (H4) a larger boundary weight lowers the boundary share and raises η : for the PINN from 1.85 and 1.89 at weight 1 (share 0.97 and 0.99) to 4.15 and 3.02 at weight 1000 (share 0.48 and 0.57) on PoiD and LapD, and for Deep Ritz on LapD from 1.74

Table S7: Choice of the weights at $d = 5$: relative L^2 error on the validation set of 5000 points after 4000 iterations, in units of 10^{-2} , for one run (seed 100) per weight. The weight is λ for the PINN and β for Deep Ritz; they are normalised differently (Remark 3).

	weight			
	1	10	100	1000
LapD, PINN (λ)	1.03	0.917	0.792	0.716
LapD, Deep Ritz (β)	16.0	2.33	1.03	1.10
PoiD, PINN (λ)	4.08	1.31	1.12	0.976
PoiD, Deep Ritz (β)	1.60	1.61	1.68	1.68

Table S8: Penalty bias of Deep Ritz on LapD against the trained networks, $\beta = 100$, in units of 10^{-2} . Bias: $\|u_\beta - u^*\|/\|u^*\|$ (Observation S7). PINN, Deep Ritz: mean relative L^2 error of the three seeds (Tables S3 and 9); share: bias over the Deep Ritz error, in per cent.

d	bias	4000 iterations			16 000 iterations		
		PINN	Deep Ritz	share	PINN	Deep Ritz	share
2	0.548	0.608	1.882	29	–	–	–
3	0.470	0.672	1.421	33	–	–	–
5	0.299	0.760	1.113	27	–	–	–
10	0.146	1.260	1.896	8	0.588	0.728	20
20	0.074	4.663	4.613	2	0.757	0.870	8

Table S9: The constant of Theorem 5. $1/(2d)$ and L_d : proved lower bounds of κ_d^2 ; “comp.”: the largest Rayleigh quotient found on finite-dimensional subspaces of sine modes, a lower bound of κ_d^2 up to rounding (Observation S10); U_d : proved upper bound. $\sqrt{(1/2)/U_d}$: the factor by which $\sqrt{U_d}$ improves Payne’s constant $\sqrt{1/2}$. $\sqrt{2dU_d}$: factor of the root-mean-square boundary misfit in Corollary 7(c). $2C_d$: the constant of Proposition 2(c) for the same map. m_d : the bound of κ_d^2 through the torsion function of the cube (Remark S9), computed by quadrature, and the factor $\sqrt{m_d/U_d}$ by which $\sqrt{U_d}$ improves it.

d	$1/(2d)$	L_d	comp.	U_d	comp./ U_d	$\sqrt{(1/2)/U_d}$	$\sqrt{U_d}$	$\sqrt{2dU_d}$	$2C_d$	m_d	$\sqrt{m_d/U_d}$
1	0.5000	0.5000	0.5000	0.5000	1.000	1.000	0.707	1.000	2.099	0.5000	1.000
2	0.2500	0.1857	0.2875	0.2919	0.985	1.309	0.540	1.081	2.050	0.3377	1.075
3	0.1667	0.1214	0.2120	0.2198	0.964	1.508	0.469	1.149	2.033	0.2819	1.132
5	0.1000	0.0811	0.1477	0.1586	0.932	1.776	0.398	1.259	2.020	0.2350	1.217
10	0.0500	0.0531	0.0927	0.1061	0.874	2.171	0.326	1.457	2.010	0.1953	1.357
20	0.0250	0.0365	0.0581	0.0730	0.796	2.617	0.270	1.709	2.005	0.1703	1.527
50	0.0100	0.0227	0.0314	0.0455	0.691	3.316	0.213	2.132	2.002	0.1485	1.807
100	0.0050	0.0160	0.0204	0.0320	0.637	3.953	0.179	2.529	2.001	0.1366	2.066

(share 0.91) to 3.12 (share 0.41), while for Deep Ritz on PoiD η stays between 4.99 and 5.46 (share 0.22 to 0.25). The sampled maximum-principle quantity $\max_{\partial\Omega}|v - g| + \frac{1}{8}\max_{\Omega}|\Delta v + f|$, an upper bound of the error in the maximum norm and hence in L^2 if the maxima were exact, was 5.7 to 42 times the bound of Corollary 7(c) on these networks. At $d = 10$, descriptively, the seed-mean bound orders the two methods as the seed-mean error does, on both problems, at equal iterations (PINN lower) and at equal compute (Deep Ritz lower).

Re-checks. The re-check evaluated all 88 networks with its own network evaluator, the Laplacian from the trace of the Hessian and new Monte-Carlo points. Its two rounds retrained 45 networks with separately written trainers and other seeds (24 and 21; Section S8.5), on LapD, PoiD and two problems outside the study, for $d = 2$ to 20 and including Deep Ritz at the CPU time of the PINN at $d = 20$. At $d = 20$ and equal CPU time the second round gives Deep Ritz $\varepsilon_2 = 0.0092$ to 0.0104 after 12 900 to 14 600 iterations and the PINN 0.0459 to 0.0484 after 4000; at equal iterations Deep Ritz gives 0.0456 to 0.0514.

S5.7 The unit square

For $d = 2$ Theorem 5 gives $\delta_1 \geq 1/U_2 > 3.4253$. A Rayleigh quotient bounds κ_2^2 from below, and hence δ_1 from above, up to rounding and, where quadrature is used, quadrature error. The largest one found on sine modes, 0.287459, gives $\delta_1 \leq 3.4788$, consistent with the published bound $\delta_1 < 1.96256$ for the square of area π [6], which is $\delta_1 < 3.4786$ after rescaling. A Rayleigh–Ritz computation on the harmonic side in the re-check (combinations of harmonic extensions of sine modes on single sides, quadrature with 200, 400 and 800 nodes agreeing to 10^{-13}) gives the larger quotient 0.2877121, that is $\delta_1 \leq 3.47570$. The value $\delta_1 \approx 1.96179$ computed for the square of

Table S10: Corollary 7(c) on the main networks (4000 iterations), new Monte-Carlo set: mean over seeds 0–2 of the relative error, of the two absolute terms of the bound and, with [minimum, maximum], of the efficiency η and of the boundary share $B_{\text{bd}}/(B_{\text{int}} + B_{\text{bd}})$. The relative standard error of η is at most 0.33 per cent on this set.

problem	method	d	ε_2	B_{int}	B_{bd}	η [min, max]	share [min, max]
LapD	PINN	2	6.079×10^{-3}	8.85×10^{-3}	2.11×10^{-3}	5.39 [5.24, 5.49]	0.191 [0.186, 0.199]
LapD	PINN	3	6.704×10^{-3}	5.42×10^{-3}	3.37×10^{-3}	3.97 [3.81, 4.07]	0.391 [0.367, 0.437]
LapD	PINN	5	7.603×10^{-3}	5.71×10^{-3}	7.64×10^{-3}	2.98 [2.89, 3.09]	0.571 [0.546, 0.595]
LapD	PINN	10	1.254×10^{-2}	6.00×10^{-3}	3.10×10^{-2}	2.20 [2.15, 2.23]	0.838 [0.824, 0.857]
LapD	PINN	20	4.634×10^{-2}	4.75×10^{-3}	2.28×10^{-1}	1.93 [1.93, 1.94]	0.979 [0.977, 0.982]
LapD	Deep Ritz	2	1.881×10^{-2}	2.21×10^{-2}	5.39×10^{-3}	4.44 [3.86, 4.79]	0.197 [0.189, 0.209]
LapD	Deep Ritz	3	1.421×10^{-2}	1.42×10^{-2}	5.58×10^{-3}	4.27 [3.62, 4.63]	0.282 [0.273, 0.288]
LapD	Deep Ritz	5	1.114×10^{-2}	1.15×10^{-2}	9.50×10^{-3}	3.20 [3.14, 3.24]	0.453 [0.435, 0.474]
LapD	Deep Ritz	10	1.888×10^{-2}	1.58×10^{-2}	3.83×10^{-2}	2.13 [2.13, 2.13]	0.708 [0.700, 0.721]
LapD	Deep Ritz	20	4.594×10^{-2}	3.26×10^{-2}	2.14×10^{-1}	2.07 [2.06, 2.08]	0.868 [0.859, 0.881]
PoiD	PINN	2	1.341×10^{-3}	6.37×10^{-3}	2.64×10^{-3}	9.91 [8.02, 11.02]	0.286 [0.250, 0.357]
PoiD	PINN	3	3.753×10^{-3}	7.95×10^{-3}	6.47×10^{-3}	5.43 [5.21, 5.86]	0.449 [0.417, 0.466]
PoiD	PINN	5	9.257×10^{-3}	1.43×10^{-2}	1.25×10^{-2}	4.10 [4.03, 4.13]	0.466 [0.449, 0.486]
PoiD	PINN	10	2.181×10^{-2}	1.06×10^{-2}	2.99×10^{-2}	2.65 [2.48, 2.76]	0.736 [0.720, 0.762]
PoiD	PINN	20	1.449×10^{-1}	1.25×10^{-2}	1.94×10^{-1}	2.01 [2.01, 2.02]	0.939 [0.939, 0.940]
PoiD	Deep Ritz	2	5.733×10^{-3}	7.07×10^{-2}	4.16×10^{-3}	19.19 [14.70, 24.82]	0.056 [0.050, 0.066]
PoiD	Deep Ritz	3	1.424×10^{-2}	6.52×10^{-2}	1.10×10^{-2}	7.60 [7.07, 8.31]	0.144 [0.140, 0.149]
PoiD	Deep Ritz	5	1.507×10^{-2}	4.33×10^{-2}	1.28×10^{-2}	5.27 [5.11, 5.39]	0.228 [0.219, 0.238]
PoiD	Deep Ritz	10	7.509×10^{-2}	3.80×10^{-2}	8.08×10^{-2}	2.25 [2.21, 2.27]	0.674 [0.645, 0.701]
PoiD	Deep Ritz	20	1.270×10^{-1}	3.18×10^{-2}	1.65×10^{-1}	2.19 [2.18, 2.19]	0.838 [0.838, 0.838]

area π by Antunes and Gazzola [1], ≈ 3.4772 after rescaling by $\sqrt{\pi}$, lies 4.3×10^{-4} relative above this upper bound, so it overestimates δ_1 by at least 4.3×10^{-4} relative (up to the quadrature error of the bound).

S5.8 The penalty bias, exactly

The exact representation of Proposition S11 was evaluated on the grid $d \in \{2, 3, 5, 10, 20, 50, 100\}$, $\beta \in \{1, 3, 10, 100, 10^3, 10^4, 10^5\}$ (Table S11). For $\beta \leq 1000$ two truncations agree to 2.7×10^{-9} relative and a wider quadrature range changes the entries by at most 1.0×10^{-8} ; at $\beta = 10^4$ and 10^5 , computed with one truncation, the wider range changes them by at most 4.5×10^{-5} . The values agree with the series values of Observation S7 to within 4.3×10^{-4} relative in all 20 shared cells, and with the finite-difference solution at $d = 2$ (ratios 0.99998, 0.99992, 0.99989 for $\beta = 10, 100, 1000$). The separately written implementation of the re-check reproduces all 49 cells to within 3.9×10^{-5} relative. A first-order expansion $u_\beta - u^* = v_1/\beta + O(\beta^{-3/2})$ with an explicit remainder (Theorem S12) was checked through its consequence $\|u_\beta - u^*\| - \|v_1\|/\beta \leq K_d N \beta^{-3/2}$, which holds in all 49 cells (the norm of the remainder itself was not computed); its leading term overestimates the bias at moderate β , the more so the larger d (exact over leading term at $\beta = 100$: 0.988 at $d = 2$, 0.807 at $d = 100$).

Conjecture S4. For LapD let $c(d) = \lim_{\beta \rightarrow \infty} \beta \|u_\beta - u^*\| / \|u^*\|$. Then $dc(d) \rightarrow \sqrt{8/3} = 1.632993\dots$ as $d \rightarrow \infty$.

Evidence (numerical, not a proof): from the exact representation at $\beta = 10^9$, extrapolated in the number of one-dimensional Robin roots ($2 \times 10^5, 4 \times 10^5, 8 \times 10^5$), $dc(d)$ is 1.1098, 1.5276, 1.5200, 1.5773, 1.6112, 1.6222, 1.6294, 1.6319, 1.6326 and 1.6329 for $d = 2, 5, 10, 20, 50, 100, 300, 10^3, 3 \times 10^3$ and 10^4 . The sequence is not monotone ($d = 5$ above $d = 10$) and increases from $d = 10$ on. The two extrapolations agree to 2.5×10^{-5} relative, and without the extrapolation the values at large d are too low. A second computation, made in the re-check, expands $v_1 = \lim_{\beta \rightarrow \infty} \beta(u_\beta - u^*)$ in the Dirichlet sine basis, with the one-dimensional sums in closed form so that no modes are truncated. It agrees with the values above to all printed digits

Table S11: Relative penalty bias $\|u_\beta - u^*\|/\|u^*\|$ of LapD from Proposition S11, and the ratio of two upper bounds to it at $\beta = 1$ and $\beta = 100$: (b) the bound $\sqrt{U_d}\|\partial_n u^*\|_\partial/(2\beta)$ of Corollary 7(b); (c) the bound $C_d\|\partial_n u^*\|_\partial/\beta$ of Proposition 2(c).

d	relative bias, $\beta =$					(b)/exact		(c)/exact	
	1	3	10	100	1000	$\beta = 1$	100	$\beta = 1$	100
2	2.883×10^{-1}	1.386×10^{-1}	5.002×10^{-2}	5.483×10^{-3}	5.542×10^{-4}	3.25	1.71	12.3	6.5
3	2.534×10^{-1}	1.181×10^{-1}	4.244×10^{-2}	4.698×10^{-3}	4.760×10^{-4}	3.21	1.73	13.9	7.5
5	1.640×10^{-1}	7.359×10^{-2}	2.645×10^{-2}	2.991×10^{-3}	3.048×10^{-4}	3.36	1.84	17.1	9.4
10	7.837×10^{-2}	3.383×10^{-2}	1.232×10^{-2}	1.463×10^{-3}	1.513×10^{-4}	3.99	2.14	24.6	13.2
20	3.665×10^{-2}	1.557×10^{-2}	5.799×10^{-3}	7.369×10^{-4}	7.817×10^{-5}	5.19	2.58	38.5	19.1
50	1.233×10^{-2}	5.240×10^{-3}	2.007×10^{-3}	2.812×10^{-4}	3.155×10^{-5}	7.87	3.45	73.9	32.4
100	5.253×10^{-3}	2.243×10^{-3}	8.730×10^{-4}	1.308×10^{-4}	1.560×10^{-5}	11.03	4.43	123.4	49.6

and continues them: 1.632887 at $d = 10^4$, 1.632983 at 10^5 and 1.632992 at 10^6 . A leading-order Laplace evaluation of the same integral gives $\|v_1\|^2 \sim m/(12d)$ and $\|u^*\|^2 \sim m^2/16$ for the $m = \lfloor d/2 \rfloor$ products of LapD, hence the limit $\sqrt{8/3}$; this asymptotic argument has not been written out with error terms, so the statement remains a conjecture.

S5.9 The two remedies: design and further results

Design. An exploratory lift1 arm on the first four problems was added after the frozen analysis, and a fifth problem, with lift1 as a pre-registered arm, in a separately frozen addendum; every later choice was recorded with its time. The network $\mathcal{N}$ of Section 6 takes $\mathbf{x} \in [0, 1]^d$ as it is; per-axis (separable) networks [4] address the cost of high-dimensional collocation rather than the plateau. The network, optimiser, schedule, point sampling and test sets are those of Section 3.5, imported from the study’s code, and the plain runs with seeds 0 to 2 on LapD and PoiD reproduce the runs of Tables S3 and S4 at $d = 20$ bit for bit. Both remedies had been tried on LapD and PoiD at $d = 20$ in exploratory pilots with seeds 0 to 2 before the freeze, as had the centred input (on LapD at $d = 20$, on PoiD up to $d = 16$) and, for Deep Ritz on LapD, rep3, so their direction on these two problems was known; only RidgeD and CosPairD were held out. They have $f = 16u^*$ (RidgeD) and $f = 2\pi^2 u^*$ (CosPairD); a related ridge had been used in an earlier pilot. On LapD, RidgeD and CosPairD an affine function of the lift features approximates u^* no better than an affine function of $\mathbf{x}$, or by about one per cent: the relative floors at $d = 20$ are 0.1015 for both on LapD, 0.7136 and 0.7072 on RidgeD, and 1 and 1 on CosPairD (Proposition S16). For the PINN the affine part of presolve is the $L^2(\partial\Omega)$ projection of g (Proposition S14). The Deep Ritz weight is that of the study on LapD and PoiD ($\beta = 100$ and 1) and was chosen by the same protocol on the held-out problems ($\beta = 100$ on RidgeD, 1 on CosPairD); the PINN uses $\lambda = 1000$ throughout. Two controls were run for Deep Ritz only (Table S20): *lift3u*, which spans the same functions as lift3c but uses the uncentred linear feature x_i in place of $P_1(t_i)$, and *rep3*, the featureless input (t, t, t) . Under Adam, rep3 is exactly the centred-input network $\mathcal{N}(t)$ whose first-layer weight is trained with a three times larger step and whose first bias starts with a third of the variance (Proposition S18); it is a reparametrisation control, not a pure control of the number of inputs. At equal CPU time lift3c runs fewer iterations (3200 to 3750 instead of 4000, schedule rescaled). Presolve costs as much as plain per iteration, and its linear solve takes between 0.0005 and 0.036 s of CPU time; its 4000-iteration runs count as equal-CPU runs where their CPU time is within 5 per cent of plain, which held except for Deep Ritz on RidgeD, run again with 3700 iterations.

Presolve. On RidgeD one Deep Ritz run with presolve ends at an error of 0.295 against 0.042 without the shift (Figure S11). In a secondary pre-registered run with a constant learning rate and 12 000 iterations presolve also suppressed the late exit of Deep Ritz from the plateau on

Table S12: The remedies at $d = 20$, seeds 10–14: ε_2 on the held-out test points in units of 10^{-2} , mean (sample standard deviation), and the seed-paired ratio $\rho = \varepsilon_2(\text{plain})/\varepsilon_2(\text{arm})$: median [minimum, maximum] and the number of seeds with $\rho > 1$. “eq. CPU”: the number of iterations (in brackets) that matches the CPU time of plain; presolve on RidgeD with Deep Ritz was rerun with 3700 iterations because its 4000-iteration CPU time exceeded plain by more than 5 per cent; elsewhere presolve’s 4000-iteration runs are within 5 per cent of plain. f.-m.: feature-matched, excluded from the decision on the lift. The controls lift3u and rep3 are in Table S20.

problem	method	arm	iterations	ε_2 , mean (s.d.)	ρ : median [min, max], wins
LapD	Deep Ritz	plain	4000	4.75 (0.37)	–
LapD	Deep Ritz	presolve	4000	1.93 (0.55)	2.67 [1.67, 3.40] 5/5
LapD	Deep Ritz	lift3c	4000	2.04 (0.17)	2.33 [2.08, 2.66] 5/5
LapD	Deep Ritz	lift3c	eq. CPU (3250)	2.55 (0.22)	1.85 [1.67, 2.13] 5/5
LapD	PINN	plain	4000	4.80 (0.38)	–
LapD	PINN	presolve	4000	1.82 (0.54)	2.85 [1.72, 3.84] 5/5
LapD	PINN	lift3c	4000	1.65 (0.10)	2.86 [2.63, 3.36] 5/5
LapD	PINN	lift3c	eq. CPU (3650)	1.87 (0.13)	2.54 [2.34, 3.01] 5/5
PoiD (f.-m.)	Deep Ritz	plain	4000	12.6 (0.037)	–
PoiD (f.-m.)	Deep Ritz	presolve	4000	12.7 (0.0079)	1.00 [1.00, 1.00] 2/5
PoiD (f.-m.)	Deep Ritz	lift3c	4000	1.92 (0.025)	6.59 [6.49, 6.69] 5/5
PoiD (f.-m.)	Deep Ritz	lift3c	eq. CPU (3350)	2.03 (0.016)	6.22 [6.15, 6.28] 5/5
PoiD (f.-m.)	PINN	plain	4000	14.5 (0.28)	–
PoiD (f.-m.)	PINN	presolve	4000	15.4 (0.50)	0.94 [0.91, 0.98] 0/5
PoiD (f.-m.)	PINN	lift3c	4000	1.07 (0.19)	14.60 [10.35, 15.05] 5/5
PoiD (f.-m.)	PINN	lift3c	eq. CPU (3550)	1.18 (0.21)	13.33 [9.53, 13.99] 5/5
RidgeD	Deep Ritz	plain	4000	3.74 (0.33)	–
RidgeD	Deep Ritz	presolve	4000	10.2 (11)	0.66 [0.14, 0.87] 0/5
RidgeD	Deep Ritz	presolve	eq. CPU (3700)	10.7 (11)	0.59 [0.14, 0.80] 0/5
RidgeD	Deep Ritz	lift3c	4000	1.57 (0.082)	2.27 [2.12, 2.75] 5/5
RidgeD	Deep Ritz	lift3c	eq. CPU (3200)	1.89 (0.096)	1.91 [1.73, 2.26] 5/5
RidgeD	PINN	plain	4000	2.47 (0.16)	–
RidgeD	PINN	presolve	4000	3.18 (0.63)	0.79 [0.63, 0.98] 0/5
RidgeD	PINN	lift3c	4000	1.47 (0.096)	1.77 [1.45, 1.83] 5/5
RidgeD	PINN	lift3c	eq. CPU (3750)	1.54 (0.10)	1.66 [1.39, 1.74] 5/5
CosPairD	Deep Ritz	plain	4000	20.6 (0.79)	–
CosPairD	Deep Ritz	presolve	4000	20.6 (0.79)	1.00 [1.00, 1.00] 0/5
CosPairD	Deep Ritz	lift3c	4000	7.36 (0.30)	2.83 [2.51, 2.94] 5/5
CosPairD	Deep Ritz	lift3c	eq. CPU (3350)	8.13 (0.36)	2.56 [2.25, 2.67] 5/5
CosPairD	PINN	plain	4000	24.6 (0.78)	–
CosPairD	PINN	presolve	4000	24.6 (0.78)	1.00 [1.00, 1.00] 0/5
CosPairD	PINN	lift3c	4000	6.11 (0.30)	4.02 [3.83, 4.23] 5/5
CosPairD	PINN	lift3c	eq. CPU (3750)	6.37 (0.31)	3.85 [3.67, 4.06] 5/5

PoiD: the plain networks capture 84 to 93 per cent of the non-affine part of u^* (test errors 0.048 to 0.065), the shifted ones at most 5.4 per cent (test errors 0.124 to 0.127; three seeds).

Lift. The median ratios of lift3c over plain are 2.33 and 2.86 on LapD, 2.27 and 1.77 on RidgeD, 2.83 and 4.02 on CosPairD (Deep Ritz and the PINN) at 4000 iterations, 1.85 and 2.54, 1.91 and 1.66, 2.56 and 3.85 at equal CPU time, and 6.59 and 14.60 on the feature-matched PoiD. On LapD the uncentred variant lift3u is less accurate than plain in 5 of 5 seeds (median 0.52), so the pre-registered centring clause was triggered and no statement about the Legendre features is made from it. Centring alone does not give the gain on LapD either: lift1 gives 0.92 (2 of 5 seeds better) for Deep Ritz and rep3 0.88 (0 of 5), whereas lift1 gives 1.78 (5 of 5) for the PINN. Against the exploratory lift1 arm (Table S13(b)), lift3c is more accurate on LapD (median ratio

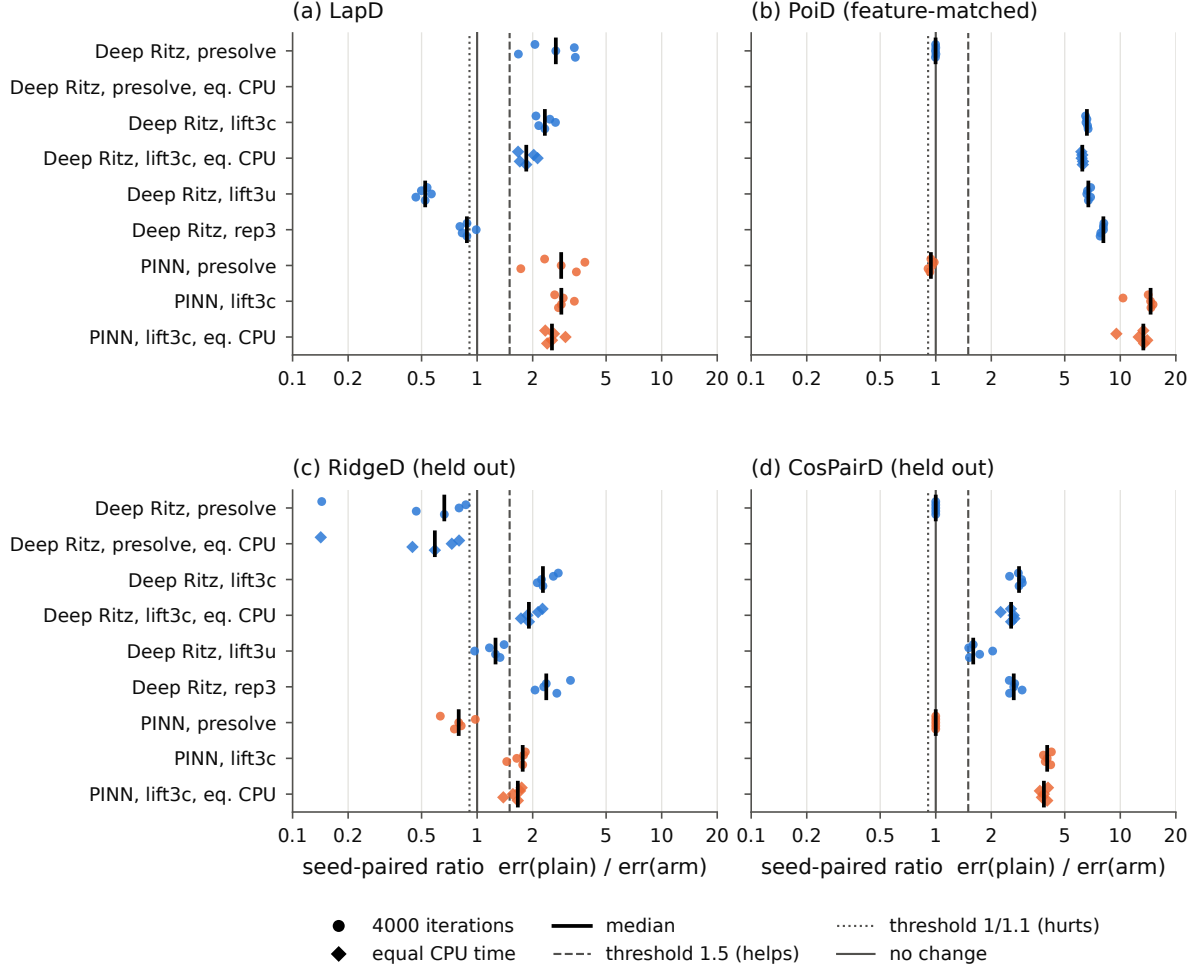


Figure S11: Seed-paired ratios $\rho = \varepsilon_2(\text{plain}) / \varepsilon_2(\text{arm})$ at $d = 20$ on the four pre-registered problems, one row per method and arm, including the controls lift3u and rep3 (right of 1: the arm is more accurate than plain). Circles: 4000 iterations; diamonds: equal CPU time; bars: medians. Dashed: the threshold 1.5 of the decision rule; dotted: 1/1.1.

2.43 for Deep Ritz, 1.61 for the PINN) and on CosPairD (2.23 and 2.39), 5 of 5 seeds each, but not on RidgeD (0.98 with 1 of 5 seeds better, and 0.77 with 0 of 5): there the whole gain of lift3c over plain is matched by centring the input (lift1 against plain: 2.38 and 2.22, 5 of 5). On PoiD, RidgeD and CosPairD rep3 gains about as much as lift3c for Deep Ritz (medians 8.10, 2.37, 2.65 against 6.59, 2.27, 2.83).

A fifth problem. AltRidgeD is $u^* = \cos(\pi s' + 1)$ with $s' = d^{-1/2} \sum_i e_i(2x_i - 1)$, $e_i = +1$ for odd and -1 for even i , and $f = 4\pi^2 u^*$; its affine floor at $d = 20$ is 0.8965 and the floor of the lift features 0.8951 (Proposition S16(e)). It was run with seeds 10 to 14, both remedies and the arms lift3u, rep3 and lift1, with $\beta = 1000$ chosen by the same protocol. For the PINN at equal CPU time lift3c runs 3700 iterations (median ratio 0.72, 0 of 5 seeds better; 0.88 with 1 of 5 at 4000 iterations); for Deep Ritz 3200 (0.32). For Deep Ritz with three seeds the sign is the same at $d = 10$ (0.65) and $d = 5$ (0.39), 0 of 3 seeds better in both. Presolve gives 0.73 for Deep Ritz and 0.83 for the PINN, 1 of 5 each (CPU time 1.042 and 1.012 times plain, so its 4000-iteration runs count as equal-CPU runs). Against lift1, lift3c is less accurate on AltRidgeD for both methods (0.69 with 1 of 5 seeds, 0.68 with 0 of 5) (Table S13, Figure S12). The frozen verdict that lift3c is a general remedy holds on LapD, RidgeD and CosPairD only. The code paths of the first four problems were unchanged by the addition of the fifth: one stored plain run and one stored lift1

run were reproduced bit for bit afterwards.

Two further problems. The second round of the re-check, under a plan frozen before its runs, used fresh seeds 40 to 42 and two problems on which neither remedy had been tried before: **SinLinD**, $u^* = \sum_{k \leq d/2} \sin(\pi x_{2k-1}) x_{2k}$ with $f = \pi^2 u^*$, and **ExpCosD**, $u^* = \sum_{k \leq d/2} e^{x_{2k-1}} \cos(x_{2k})$ with $f = 0$. Their relative affine floors are 0.172 and 0.022 and the floors of the lift features 0.086 and 0.015 (Monte Carlo), so on **SinLinD** the best affine function of the lift features has half the error of the best affine function of $\mathbf{x}$. The Deep Ritz weight was chosen by the study’s protocol on that code’s own validation set ($\beta = 100$ and 1000). For Deep Ritz at $d = 20$ (three seeds each, seed-paired ratios against plain): on **SinLinD** lift3c gives 2.94 (3 of 3 seeds better), 2.53 at equal CPU time (3350 iterations) and 2.14 at $d = 10$, but the centred-input network lift1 alone gives 3.02, and lift3c against lift1 gives 1.01, so the whole gain is matched by centring; presolve gives 1.16 (3 of 3, below the threshold 1.5 of the plan). On **ExpCosD**, which is on the plateau ($\varepsilon_2 = 0.025$ for plain, affine floor 0.022), presolve gives 4.51 ($\varepsilon_2 = 0.0053$ to 0.0063), lift3c 1.85 and lift1 1.33 (3 of 3 each), and lift3c against lift1 1.34. For the PINN on **LapD** at $d = 20$, that code gives ratios of the same size as the study’s (presolve 2.81 and 3.96, lift3c 2.86 and 2.68 for seeds 40 and 41; study medians 2.85 and 2.86). These descriptive results are separate from the pre-registered study. The PINN comparison below uses all three seeds and calibrates its computational budget on one machine.

Table S13: (a) **AltRidgeD** at $d = 20$, seeds 10–14, quantities and units as in Table S12; pre-registered in the addendum, not blind. (b) The lift against the centred-input network lift1 = $\mathcal{N}(2\mathbf{x} - 1)$ at $d = 20$: median [minimum, maximum] of the seed-paired ratio and the number of seeds with ratio above 1. The lift1 arm is exploratory on **LapD** to **CosPairD** and pre-registered on **AltRidgeD**.

(a) *AltRidgeD*, $d = 20$

method	arm	iterations	ε_2 , mean (s.d.)	ρ : median [min, max], wins
Deep Ritz	plain	4000	2.27 (0.11)	–
Deep Ritz	presolve	4000	4.84 (3.2)	0.73 [0.22, 1.10] 1/5
Deep Ritz	lift3c	4000	3.81 (0.36)	0.59 [0.57, 0.64] 0/5
Deep Ritz	lift3c	eq. CPU (3200)	7.11 (0.57)	0.32 [0.30, 0.35] 0/5
Deep Ritz	lift3u	4000	2.77 (0.36)	0.82 [0.66, 1.01] 1/5
Deep Ritz	rep3	4000	2.59 (0.55)	0.93 [0.62, 1.19] 1/5
Deep Ritz	lift1	4000	2.86 (0.64)	0.84 [0.57, 1.01] 1/5
PINN	plain	4000	1.30 (0.067)	–
PINN	presolve	4000	1.61 (0.32)	0.83 [0.69, 1.01] 1/5
PINN	lift3c	4000	1.45 (0.083)	0.88 [0.80, 1.01] 1/5
PINN	lift3c	eq. CPU (3700)	1.76 (0.16)	0.72 [0.68, 0.87] 0/5
PINN	lift1	4000	1.01 (0.048)	1.29 [1.21, 1.36] 5/5

(b) *lift3c against lift1*, $d = 20$, 4000 iterations

problem	method	$\varepsilon_2(\text{lift1})/\varepsilon_2(\text{lift3c})$	$\varepsilon_2(\text{plain})/\varepsilon_2(\text{lift1})$
LapD	Deep Ritz	2.43 [1.92, 2.99] 5/5	0.92 [0.83, 1.17] 2/5
LapD	PINN	1.61 [1.24, 2.01] 5/5	1.78 [1.46, 2.27] 5/5
PoiD	Deep Ritz	4.22 [3.62, 4.79] 5/5	1.57 [1.39, 1.80] 5/5
PoiD	PINN	7.51 [4.74, 8.15] 5/5	1.95 [1.85, 2.19] 5/5
RidgeD	Deep Ritz	0.98 [0.88, 1.09] 1/5	2.38 [2.15, 3.12] 5/5
RidgeD	PINN	0.77 [0.74, 0.90] 0/5	2.22 [1.89, 2.39] 5/5
CosPairD	Deep Ritz	2.23 [1.72, 2.36] 5/5	1.30 [1.25, 1.46] 5/5
CosPairD	PINN	2.39 [1.99, 2.52] 5/5	1.70 [1.63, 1.92] 5/5
AltRidgeD	Deep Ritz	0.69 [0.56, 1.09] 1/5	0.84 [0.57, 1.01] 1/5
AltRidgeD	PINN	0.68 [0.65, 0.74] 0/5	1.29 [1.21, 1.36] 5/5

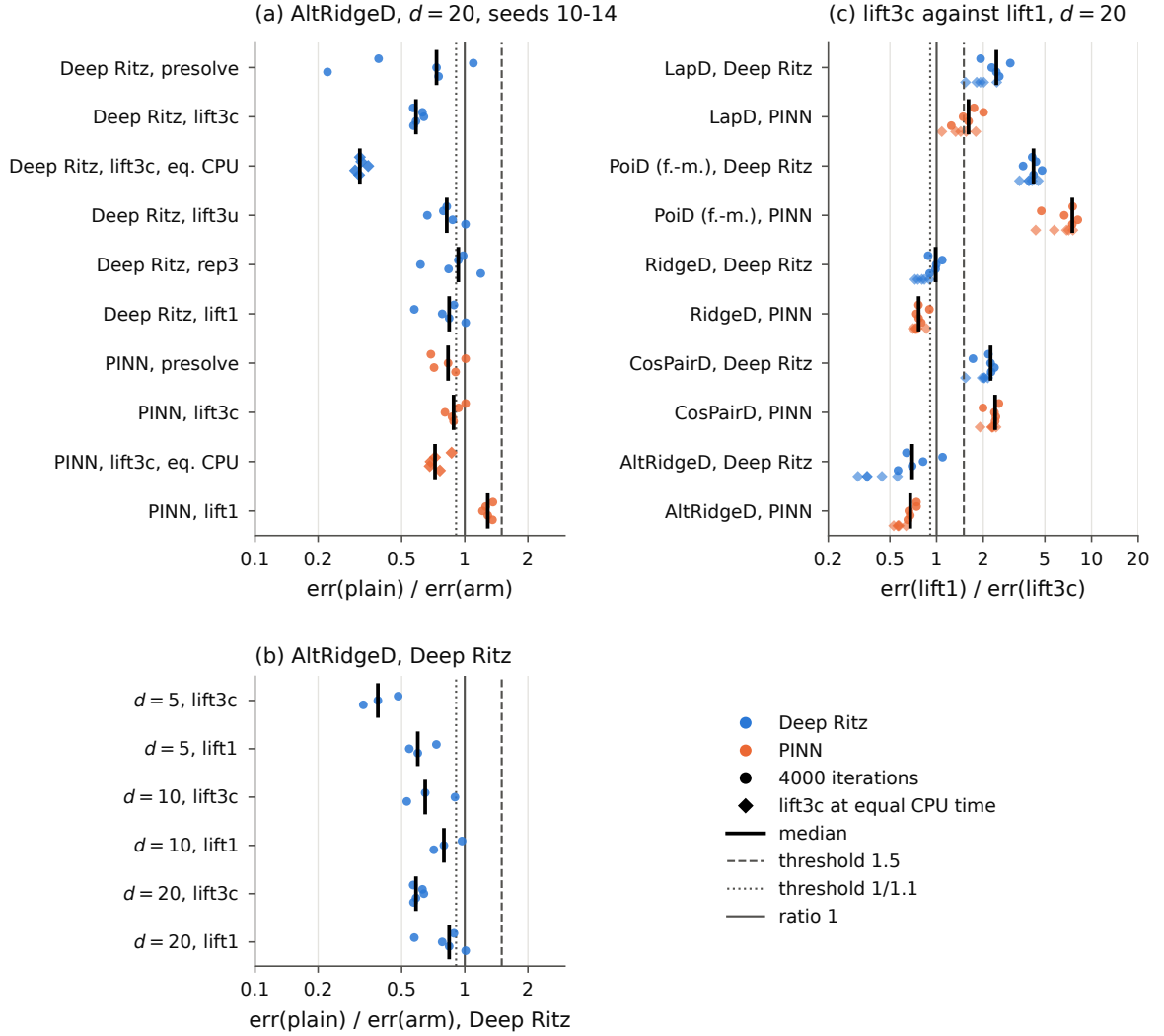


Figure S12: (a) Seed-paired ratios on AltRidgeD at $d = 20$; (b) AltRidgeD, Deep Ritz, lift3c and lift1 against plain at $d = 5, 10$ and 20 (three seeds at $d = 5$ and 10); (c) the lift against the centred-input network, $\varepsilon_2(\text{lift1})/\varepsilon_2(\text{lift3c})$, on the five problems at $d = 20$ (diamonds: lift3c at equal CPU time). Lines as in Figure S11.

Current-machine PINN replication. A separate cohort uses the same implementation, seeds 40–42 and $d = 20$ on an Apple M4 Pro CPU (Python 3.14.6, PyTorch 2.14.1). Each of two concurrent processes uses one PyTorch thread. The 24 networks comprise plain, presolve and lift3c on LapD at 4000 iterations; lift3c on LapD at a CPU-calibrated budget; and plain, presolve, lift3c and lift1 on SinLinD at 4000 iterations. The architecture, loss weights, training-point generators and held-out 50 000-point test set are those of the re-implementation. All weights and configurations are released, and reloading each checkpoint reproduces its recorded test error to less than 10^{-10} absolute difference. The records are kept separately from the earlier cohorts; these are retrospective descriptive comparisons, not new pre-registered tests.

On LapD, presolve and lift3c improve every seed at 4000 iterations (median error ratios 3.56 and 2.86). The median training times of the current-machine plain and lift3c runs set 2500 iterations for the calibrated comparison. Its realised median CPU ratio is 1.105, so it uses 10.5 per cent more CPU time than plain rather than an exactly matched budget. Its median error ratio is 1.31: all three seeds improve, but the ratio is below the 1.5 threshold used to label a clear gain. On SinLinD, presolve is less accurate in all three seeds (median ratio 0.87), whereas lift3c and lift1 improve all three (2.75 and 2.57). Thus the benefit of the polynomial features

remains problem-dependent, and centring alone accounts for much of the gain on **SinLinD**.

Table S14: Current-machine PINN replication at $d = 20$, seeds 40–42. ρ is the seed-paired error of plain divided by the arm’s error; wins count $\rho > 1$. CPU ratio is the ratio of median measured training times to plain at 4000 iterations, on the same machine. The calibrated budget is selected from the first-stage timing medians, not from historical timing.

Problem	Arm	Iterations	Median ρ	Wins	CPU ratio
LapD	presolve	4000	3.56	3/3	1.08
LapD	lift3c	4000	2.86	3/3	1.61
LapD	lift3c (calibrated)	2500	1.31	3/3	1.10
SinLinD	presolve	4000	0.87	0/3	1.09
SinLinD	lift3c	4000	2.75	3/3	1.58
SinLinD	lift1	4000	2.57	3/3	1.06

Counts over seven problems. Counting the two problems of the second round of the re-check (Deep Ritz only, three seeds), presolve is clearly more accurate than plain on two of seven problems (LapD, ExpCosD), close to plain on three (PoiD, CosPairD, SinLinD) and less accurate on two (RidgeD, AltRidgeD); lift3c is more accurate than plain on six and less accurate on one (AltRidgeD), but on two of the six (RidgeD and SinLinD) centring the input alone gives as much or more. On LapD, RidgeD, CosPairD and AltRidgeD plain ends below the affine floor (Deep Ritz 0.048, 0.037, 0.206, 0.023; PINN 0.048, 0.025, 0.246, 0.013; floors on the test set 0.102, 0.715, 1.000, 0.900). Of the problems added after the pre-registered test, only ExpCosD (Deep Ritz, three seeds, a different implementation, descriptive) is on the plateau ($\varepsilon_2 = 0.025$ for plain, affine floor 0.022), and presolve is 4.51 times more accurate there. A descriptive check at $d = 10$ with Deep Ritz and three seeds shows the same signs on LapD to CosPairD (presolve: 1.62 on LapD, 0.87 on PoiD, 0.74 on RidgeD, 1.00 on CosPairD; lift3c: 1.38, 4.87, 2.01, 4.68). Every run was made serially on a shared 12-core laptop whose one-minute load average ranged from about 7 to 166 during the runs, which affects the CPU-time measurements, and the cells at $d = 5$ and 10 have three seeds and are descriptive.

S6 Implementation pitfalls: details of the demonstrations

S6.1 Differentiating with respect to a slice of the input

Listing S1 shows the two patterns for u_{tt} ; the spatial second derivatives are treated alike. On the untrained network and 64 points, the faulty operator returns **None** in all three second-derivative calls and a residual that is exactly zero and does not even require a gradient, and the call without `allow_unused` raises the error that the differentiated tensor does not appear in the graph.

The training run. The network is 3–50–50–1 with tanh activations; the 10 000 training points are independent and uniform in $[0, 1] \times [-1, 1]^2$; the loss is the sum of the mean squared faulty residual, the mean squared difference from $e^{-t}S$ at the points of the batch, with $S = \sin(\pi x)\sin(\pi y)$, and the mean squared value at 400 points on the spatial boundary at $t = 0$. Batches of 64 are drawn by a permutation per pass; Adam with learning rate 10^{-3} , seed 0, 150 passes, that is 23 550 iterations, on one CPU thread. The least-squares amplitude of the mode, $a(t) = \langle u_\theta(t, \cdot), S \rangle / \langle S, S \rangle$, is 0.981, 0.364, 0.100 and 0.044 at $t = 0, 1, 2, 3$, against $e^{-t} = 1, 0.368, 0.135, 0.050$ and against the factor $\cos(\sqrt{2}\pi t) = 1, -0.266, -0.858, +0.723$ of the solution of Wave2 (Figure S13(a); the times 2 and 3 lie outside the sampled window). It never changes sign. The relative L^2 distance from the solution of Wave2 is 2.0 on the slice $t = 0.5$ and 2.4 on the

```

# faulty: txy[:, 0] is a new tensor on which u_t does not depend
h = torch.autograd.grad(u_t, txy[:, 0], grad_outputs=torch.ones_like(u_t),
                        create_graph=True, allow_unused=True)[0] # None
u_tt = torch.zeros_like(u_t) if h is None else h[:, 0]          # zeros
# correct: differentiate w.r.t. the whole input, then index
h = torch.autograd.grad(u_t.sum(), txy, create_graph=True)[0]  # (N, 3)
assert h is not None
u_tt = h[:, 0]

```

Listing S1: The second time derivative computed with respect to a slice of the input (faulty) and with respect to the whole input (correct). The comments at the right give what the faulty lines return. Both patterns are executed by `code/s7_pitfalls.py`.

slice $t = 1$. With 1500 passes the amplitudes are 0.997 and 0.368 at $t = 0$ and 1 (0.150 and 0.077 at $t = 2$ and 3), and the distance from $e^{-t}S$ on the held-out grid is 0.006. The PDE term is zero whatever the number of passes, because the missing derivative does not depend on the weights.

S6.2 The problem with data on one face

The five networks of Section 7.2 were trained as one stack on the GPU backend, with the loss $\text{mean}(\Delta u_\theta)^2 + \text{mean}(u_\theta - \text{datum})^2$ over the 9261 grid points and the 441 points of the face $x = \pi$, AdamW [11] with its default weight decay 0.01, single precision; their predictions are compared with the solution of Lap3 on its 40^3 cell-centred evaluation grid. The pointwise standard deviation of the five predictions over the seeds, averaged over that grid, is 4.49 times the root-mean-square value of the solution of Lap3.

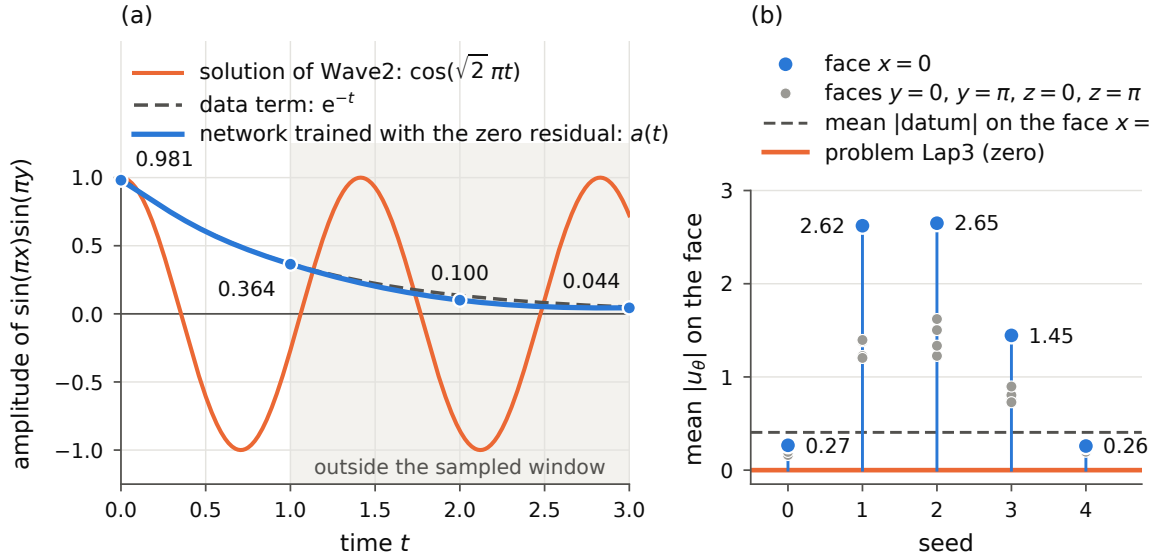


Figure S13: The two demonstrations of Section 7. (a) Residual made identically zero by differentiation with respect to a slice of the input: least-squares amplitude $a(t) = \langle u_\theta(t, \cdot), S \rangle / \langle S, S \rangle$ of the mode $S = \sin(\pi x)\sin(\pi y)$, on a 101×101 grid per time, for the network trained with the zero residual and the data term $e^{-t}S$ (seed 0, 150 passes, that is 23 550 Adam iterations), with the factor $\cos(\sqrt{2}\pi t)$ of the solution of Wave2; dots mark $t = 0, 1, 2, 3$. Training samples $t \in [0, 1]$; the shaded region lies outside that window. (b) Laplace problem with data on one face only: mean of $|u_\theta|$ over 20×20 points of the face $x = 0$ (blue, value printed) and of each of the four other faces without a condition (grey), for five seeds. The solution of Lap3, which has zero data on these faces, vanishes on all five; the dashed line is the mean of the datum $|\sin y \cos z|$ over the constrained face, $4/\pi^2 \approx 0.41$, for scale.

S7 Proofs

This section proves Propositions S1, S2 and 1 of Section 2, Proposition 2 of Section 3, Theorem 5, Proposition 6 and Corollary 7 of Section 6.7 together with the exact penalty bias (Section S7.5), the results on affine minimisers, floors and the reparametrisation control used in Section 6.8 (Section S7.6), and the two propositions of Section 7 (Proposition 8 on the decaying mode and Proposition 9 on the one-face problem). The derivatives, integrals and inequalities of Sections S7.1–S7.4 and S7.7 were also checked symbolically and numerically by the script `code/sa_proofs_checks.py`, those of Sections S7.5 and S7.6 by the scripts named in the source comments next to them; the proofs do not depend on these checks.

Two tools are used throughout. The first is Green's formula on a box. Let $B = \prod_i (a_i, b_i)$ be bounded and let $C^k(\overline{B})$ denote the functions whose partial derivatives up to order k extend continuously to $\overline{B}$. For $w \in C^2(\overline{B})$ and $\varphi \in C^1(\overline{B})$, integrating $\partial_i(\varphi \partial_i w)$ in x_i by the fundamental theorem of calculus, then over the other variables, and summing over i gives

$$\int_B \nabla w \cdot \nabla \varphi \, d\mathbf{x} + \int_B \varphi \Delta w \, d\mathbf{x} = \int_{\partial B} \varphi \partial_n w \, ds. \quad (\text{S1})$$

The second is completeness: the functions $\sin nz$, $n \geq 1$, form a complete orthogonal system of $L^2(0, \pi)$ [17, Sec. 5.4], and so do the products $\sin jy \sin nz$ in $L^2((0, \pi)^2)$. Indeed, if F is orthogonal to all of them, then for each n the function $y \mapsto \int_0^\pi F(y, z) \sin nz \, dz$ is orthogonal to every $\sin jy$ and vanishes for almost every y ; hence $F(y, \cdot) = 0$ for almost every y . By the same argument and a change of scale, the products $\prod_{i=1}^d \sin(\pi k_i x_i)$ with positive integers k_i are complete in $L^2((0, 1)^d)$.

S7.1 Exact solutions

Proof of Proposition S1. (a) Each u^* is a finite combination of products of polynomials and exponential, trigonometric and hyperbolic functions, hence infinitely differentiable on the closed domain. The equations and data are verified by differentiation. **Heat:** $\partial_t u^* = -\pi^2 u^* = \partial_{xx} u^*$, $u^*(0, x) = \sin \pi x$, and $\sin 0 = \sin \pi = 0$. **Wave1:** for $j \in \{1, 4\}$ both ∂_{tt} and $4 \partial_{xx}$ multiply the mode $\sin(j\pi x) \cos(2j\pi t)$ by $-4j^2\pi^2$; its time derivative vanishes at $t = 0$, and $\sin(j\pi x) = 0$ at $x = 0$ and $x = 1$. **Wave2:** with $S = \sin \pi x \sin \pi y$ one has $\Delta S = -2\pi^2 S$ and $\partial_{tt} \cos(\sqrt{2}\pi t) = -2\pi^2 \cos(\sqrt{2}\pi t)$; moreover $u^*(0, \cdot) = S$, $\partial_t u^*(0, \cdot) = 0$, and $S = 0$ where $x = \pm 1$ or $y = \pm 1$. **Lap2:** $x/(2\pi)$ is harmonic and $\Delta(\cos 2y \sinh 2x) = (4 - 4) \cos 2y \sinh 2x = 0$; $u^*(0, y) = 0$, $u^*(\pi, y) = \frac{1}{2} + \frac{1}{2} \cos 2y$, and $\partial_y u^*$ is a multiple of $\sin 2y$, which vanishes at $y = 0$ and $y = \pi$. **Lap3s:** $\Delta(\sin y \sin z \sinh(\sqrt{2}x)) = (2 - 1 - 1) \sin y \sin z \sinh(\sqrt{2}x) = 0$; the function vanishes where $x = 0$, $y \in \{0, \pi\}$ or $z \in \{0, \pi\}$ and equals $\sin y \sin z$ at $x = \pi$. **LapD:** $\Delta(x_{2k-1} x_{2k}) = 0$. **PoiD:** $\partial_{ii} \cos(\pi x_i) = -\pi^2 \cos(\pi x_i)$, so $-\Delta u^* = \pi^2 u^*$. For LapD and PoiD the boundary datum is the restriction of u^* .

(b) Throughout, n runs over the even integers $n \geq 2$ unless stated otherwise. Put $\rho_n(x) = \sinh(k_n x) / \sinh(k_n \pi)$ and $T_n = b_n \sin y \sin(nz) \rho_n(x)$. For $0 \leq x \leq \pi$ and $k \geq \sqrt{5}$,

$$\frac{\sinh kx}{\sinh k\pi} = e^{-k(\pi-x)} \frac{1 - e^{-2kx}}{1 - e^{-2k\pi}} \leq e^{-k(\pi-x)}, \quad \frac{\cosh kx}{\sinh k\pi} \leq \frac{2e^{-k(\pi-x)}}{1 - e^{-2k\pi}} \leq 3e^{-k(\pi-x)}. \quad (\text{S2})$$

Since $|b_n| \leq b_2 = 8/(3\pi) < 1$ and $n < k_n \leq 2n$, every derivative of T_n of order p in x , order q in z and any order in y is bounded on $K_\delta = [0, \pi - \delta] \times [0, \pi]^2$ by $3(2n)^{p+q} e^{-n\delta}$, and $\sum_n n^{p+q} e^{-n\delta} < \infty$. By the Weierstrass test the series and all its term-by-term derivatives converge uniformly on K_δ , so the sum is infinitely differentiable on K_δ and its derivatives are the term-by-term ones. As δ is arbitrary, $u^* \in C^\infty([0, \pi] \times [0, \pi]^2)$. Each term is harmonic, because $\Delta T_n = (k_n^2 - 1 - n^2) T_n = 0$, and vanishes for $x = 0$, for $y \in \{0, \pi\}$ and for $z \in \{0, \pi\}$; hence so does u^* .

For every integer $n \geq 1$, $\frac{2}{\pi} \int_0^\pi \cos z \sin nz \, dz = \frac{1}{\pi} \int_0^\pi (\sin(n+1)z + \sin(n-1)z) \, dz$, which equals $\frac{2}{\pi} (\frac{1}{n+1} + \frac{1}{n-1}) = b_n$ for even n and 0 for odd n . So the b_n are the sine coefficients of

$\cos z$ on $(0, \pi)$, and Parseval's identity gives $\frac{\pi}{2} \sum_n b_n^2 = \int_0^\pi \cos^2 z \, dz = \frac{\pi}{2}$, that is $\sum_n b_n^2 = 1$. For fixed $x < \pi$ the series $\sum_n b_n \rho_n(x) \sin nz$ converges uniformly in z , so the sine coefficients of $z \mapsto \sum_n b_n \rho_n(x) \sin nz - \cos z$ are $b_n(\rho_n(x) - 1)$. Parseval's identity in z and $\int_0^\pi \sin^2 y \, dy = \frac{\pi}{2}$ then give

$$\int_0^\pi \int_0^\pi (u^*(x, y, z) - \sin y \cos z)^2 \, dy \, dz = \frac{\pi^2}{4} \sum_n b_n^2 (1 - \rho_n(x))^2.$$

Each term tends to 0 as $x \rightarrow \pi^-$ and is at most b_n^2 , because $0 \leq \rho_n \leq 1$; as $\sum_n b_n^2 < \infty$, the sum tends to 0 by dominated convergence.

As $z \rightarrow 0$ and $z \rightarrow \pi$ the datum $\sin y \cos z$ on the face $\{x = \pi\}$ tends to $\sin y$ and $-\sin y$, whereas the datum on the faces $\{z = 0\}$ and $\{z = \pi\}$ is 0. For $0 < y < \pi$ these values differ, so the boundary data are discontinuous along the two edges.

Finally, let $D_\delta = (0, \pi - \delta) \times (0, \pi)^2$. On K_δ the gradient of u^* is the term-by-term sum, and each of the three families $\sin y \sin nz$, $\cos y \sin nz$, $\sin y \cos nz$ is orthogonal in $L^2((0, \pi)^2)$ with squared norms $\pi^2/4$. Using $k_n^2 = 1 + n^2$ and $\cosh^2 s + \sinh^2 s = \cosh 2s$, for every $x \leq \pi - \delta$

$$\begin{aligned} \int_0^\pi \int_0^\pi |\nabla u^*|^2 \, dy \, dz &= \frac{\pi^2}{4} \sum_n b_n^2 k_n^2 \frac{\cosh(2k_n x)}{\sinh^2(k_n \pi)}, \\ \int_{D_\delta} |\nabla u^*|^2 \, d\mathbf{x} &= \frac{\pi^2}{4} \sum_n b_n^2 k_n^2 \frac{\sinh(2k_n(\pi - \delta))}{2 \sinh^2(k_n \pi)}, \end{aligned}$$

the second by integrating the non-negative terms over $0 < x < \pi - \delta$. As $\delta \downarrow 0$ the n -th term of the second sum increases to $b_n^2 k_n \coth(k_n \pi) \geq b_n^2 k_n \geq 16/(\pi^2 n)$; the last inequality holds because $b_n^2 = 16n^2/(\pi^2(n^2 - 1)^2) \geq 16/(\pi^2 n^2)$ and $k_n \geq n$. By monotone convergence, for the integrals over D_δ and for the sums, $\int_{(0, \pi)^3} |\nabla u^*|^2 \, d\mathbf{x} \geq \sum_n 4/n = \infty$. $\square$

S7.2 Uniqueness

Proof of Proposition S2. (a) These are the classical energy arguments [5, Secs 2.2.5, 2.3.4, 2.4.3]. Let w be the difference of two solutions of the stated regularity; it solves the same problem with zero right-hand side and zero data.

Heat. Let $E(t) = \int_0^1 w(t, x)^2 \, dx$. Then $E'(t) = 2 \int_0^1 w w_t \, dx = 2 \int_0^1 w w_{xx} \, dx = -2 \int_0^1 w_x^2 \, dx \leq 0$; the boundary term of the integration by parts vanishes because $w = 0$ at $x = 0$ and $x = 1$. Since $E \geq 0$ and $E(0) = 0$, $E \equiv 0$ and $w \equiv 0$.

Wave1, Wave2. Let $c = 2$, $\Omega = (0, 1)$ for Wave1 and $c = 1$, $\Omega = (-1, 1)^2$ for Wave2, and $E(t) = \frac{1}{2} \int_\Omega (w_t^2 + c^2 |\nabla w|^2) \, d\mathbf{x}$. Differentiating under the integral and using (S1) with $\varphi = w_t$ at fixed t ,

$$E'(t) = \int_\Omega (w_t w_{tt} + c^2 \nabla w \cdot \nabla w_t) \, d\mathbf{x} = \int_\Omega w_t (w_{tt} - c^2 \Delta w) \, d\mathbf{x} + c^2 \int_{\partial\Omega} w_t \partial_n w \, ds = 0,$$

because $w = 0$ on $\partial\Omega$ for all t implies $w_t = 0$ there. As $w(0, \cdot) = 0$ and $w_t(0, \cdot) = 0$, $E(0) = 0$; hence $w_t \equiv 0$ and $\nabla w \equiv 0$, so w is constant, and the constant is $w(0, \cdot) = 0$.

Lap2, Lap3s, LapD, PoiD. Here $\Delta w = 0$, and (S1) with $\varphi = w$ gives $\int_\Omega |\nabla w|^2 \, d\mathbf{x} = \int_{\partial\Omega} w \partial_n w \, ds = 0$, since on every face either $w = 0$ or, on the sides $y = 0$ and $y = \pi$ of Lap2, $\partial_n w = 0$. So w is constant, and it vanishes on a face.

By Proposition S1(a), u^* is a solution of the stated regularity, hence the only one.

(b) Let w be the difference of two solutions with the three properties; then w has the first two properties and $w(x, \cdot, \cdot) \rightarrow 0$ in $L^2((0, \pi)^2)$ as $x \rightarrow \pi^-$. For integers $j, n \geq 1$ and $x \in [0, \pi)$ put

$$c(x) = \int_0^\pi \int_0^\pi w(x, y, z) \sin jy \sin nz \, dy \, dz.$$

As w is twice continuously differentiable on each compact set $[0, \pi - \delta] \times [0, \pi]^2$, $c \in C^2([0, \pi))$ and, for $0 < x < \pi$, $c'' = \iint w_{xx} \sin jy \sin nz = -\iint (w_{yy} + w_{zz}) \sin jy \sin nz$. Two integrations

by parts in y give $\int_0^\pi w_{yy} \sin jy \, dy = -j^2 \int_0^\pi w \sin jy \, dy$: the boundary terms contain $\sin jy$ or w , both zero at $y = 0$ and $y = \pi$. The same holds in z , so $c'' = (j^2 + n^2)c$. With $\lambda_{jn} = \sqrt{j^2 + n^2}$, therefore, $c(x) = A \sinh \lambda_{jn} x + A' \cosh \lambda_{jn} x$, and $A' = c(0) = 0$ because $w(0, \cdot, \cdot) = 0$. By the Cauchy–Schwarz inequality $|c(x)| \leq \frac{\pi}{2} \|w(x, \cdot, \cdot)\|_{L^2((0, \pi)^2)} \rightarrow 0$ as $x \rightarrow \pi^-$, whereas $A \sinh \lambda_{jn} x \rightarrow A \sinh \lambda_{jn} \pi$; so $A = 0$. Thus, for each $x < \pi$, every coefficient of $w(x, \cdot, \cdot)$ in the complete orthogonal system $\{\sin jy \sin nz\}$ vanishes, and $w(x, \cdot, \cdot) = 0$ by continuity. The series (4) has the three properties by Proposition S1(b). $\square$

S7.3 Trivial-predictor floors

Proof of Proposition 1. Write $\langle u \rangle = \int_\Omega u \, d\mathbf{x}$. Lebesgue measure on Ω is a probability measure under which the coordinates are independent and uniform on $(0, 1)$. The functions 1 and $x_i - \frac{1}{2}$, $i = 1, \dots, d$, are therefore orthogonal in $L^2(\Omega)$, with $\|x_i - \frac{1}{2}\|^2 = \frac{1}{12}$, and they span $\mathcal{P}_1$. By orthogonal projection, for $u \in L^2(\Omega)$,

$$\min_{c \in \mathcal{P}_0} \|u - c\|^2 = \langle u^2 \rangle - \langle u \rangle^2 =: V(u), \quad \min_{a \in \mathcal{P}_1} \|u - a\|^2 = V(u) - 12 \sum_{i=1}^d \langle u(x_i - \frac{1}{2}) \rangle^2. \quad (\text{S3})$$

(a) Let $p_k = x_{2k-1} x_{2k}$. Then $\langle p_k \rangle = \frac{1}{4}$, $\langle p_k^2 \rangle = \frac{1}{9}$ and $V(p_k) = \frac{1}{9} - \frac{1}{16} = \frac{7}{144}$. The p_k are independent, so $\langle u^* \rangle = m/4$, $V(u^*) = 7m/144$ and $\|u^*\|^2 = 7m/144 + m^2/16 = (9m^2 + 7m)/144$. Further, $\langle p_k(x_i - \frac{1}{2}) \rangle = \frac{1}{6} - \frac{1}{8} = \frac{1}{24}$ for $i \in \{2k-1, 2k\}$ and 0 for every other i , so the affine residual in (S3) is $7m/144 - 12 \cdot 2m/576 = m/144$. (Equivalently, $p_k = (x_{2k-1} - \frac{1}{2})(x_{2k} - \frac{1}{2}) + \frac{1}{2}x_{2k-1} + \frac{1}{2}x_{2k} - \frac{1}{4}$, where the first term is orthogonal to $\mathcal{P}_1$ and has squared norm $\frac{1}{144}$.) Dividing $7m/144$ and $m/144$ by $\|u^*\|^2$ gives the two squared ratios.

(b) $\langle \cos \pi x_i \rangle = 0$ and $\langle \cos^2 \pi x_i \rangle = \frac{1}{2}$, so by independence $\langle u^* \rangle = 0$ and $\|u^*\|^2 = V(u^*) = \frac{1}{2}$; the best constant is 0 and the first ratio is 1. Integrating by parts, $\langle \cos(\pi x_i)(x_i - \frac{1}{2}) \rangle = \int_0^1 x \cos \pi x \, dx = -2/\pi^2$, hence $\langle u^*(x_i - \frac{1}{2}) \rangle = -2/(\pi^2 \sqrt{d})$ and the affine residual is $\frac{1}{2} - 12d \cdot 4/(\pi^4 d) = \frac{1}{2} - 48/\pi^4$. The squared ratio is $1 - 96/\pi^4$, whatever d , and its square root is 0.1203. $\square$

S7.4 Penalised Ritz energy

In this subsection $\Omega = (0, 1)^d$, norms without subscript are those of $L^2(\Omega)$, $\|\cdot\|_\partial$ is the norm of $L^2(\partial\Omega)$, and $\|v\|_{H^1}^2 = \|v\|^2 + \|\nabla v\|^2$.

Lemma S5 (Trace inequalities on the unit cube). *Let F be one of the two faces of Ω on which x_i is constant. For every $v \in C^1(\overline{\Omega})$,*

$$(i) \quad \|v\|^2 \leq 2 \|v\|_{L^2(F)}^2 + 2 \|\partial_i v\|^2, \quad (ii) \quad \|v\|_{L^2(F)}^2 \leq 2 \|v\|^2 + 2 \|\partial_i v\|^2. \quad (\text{S4})$$

Moreover $C^\infty(\overline{\Omega})$ is dense in $H^1(\Omega)$. Consequently the restriction $v \mapsto v|_{\partial\Omega}$, defined on $C^1(\overline{\Omega})$, extends to a bounded linear operator $H^1(\Omega) \rightarrow L^2(\partial\Omega)$ (the trace, used in all boundary integrals below) with $\|v\|_\partial^2 \leq 4d \|v\|^2 + 4 \|\nabla v\|^2$; inequalities (i) and (ii) hold for all $v \in H^1(\Omega)$; and Green's formula (S1) holds for all $w \in C^2(\overline{\Omega})$ and $\varphi \in H^1(\Omega)$.

Proof. By symmetry take $F = \{x_d = 0\}$ and write $\mathbf{x} = (\mathbf{x}', s)$. For $v \in C^1(\overline{\Omega})$, $v(\mathbf{x}', s) - v(\mathbf{x}', 0) = \int_0^s \partial_d v(\mathbf{x}', \tau) \, d\tau$. With $(a+b)^2 \leq 2a^2 + 2b^2$ and the Cauchy–Schwarz inequality, each of $v(\mathbf{x}', s)^2$ and $v(\mathbf{x}', 0)^2$ is at most twice the other plus $2 \int_0^1 \partial_d v(\mathbf{x}', \tau)^2 \, d\tau$. Integrating over $(\mathbf{x}', s) \in \Omega$ gives (i) and (ii).

Density. Let $\mathbf{c}$ be the centre of Ω and, for $\mu > 1$ and a function g on Ω , let $g_\mu(\mathbf{x}) = g(\mathbf{c} + (\mathbf{x} - \mathbf{c})/\mu)$, which is defined on the larger cube $\Omega_\mu = \mathbf{c} + \mu(\Omega - \mathbf{c}) \supset \overline{\Omega}$. A change of variables gives $\|g_\mu\| \leq \mu^{d/2} \|g\|$, and $g_\mu \rightarrow g$ uniformly on Ω as $\mu \downarrow 1$ if $g \in C(\overline{\Omega})$; since $C(\overline{\Omega})$ is dense in $L^2(\Omega)$, $g_\mu \rightarrow g$ in $L^2(\Omega)$ for every $g \in L^2(\Omega)$. For $v \in H^1(\Omega)$ one has $v_\mu \in H^1(\Omega_\mu)$ with $\nabla v_\mu = \mu^{-1}(\nabla v)_\mu$, so $v_\mu \rightarrow v$ in $H^1(\Omega)$. Mollifying v_μ with a radius smaller than the distance from Ω to $\partial\Omega_\mu$ gives functions that are smooth on $\overline{\Omega}$ and converge to v_μ in $H^1(\Omega)$ [5, Sec. 5.3.1].

Summing (ii) over the $2d$ faces gives the trace bound on $C^1(\overline{\Omega})$, so the restriction extends by density and continuity. Both sides of (i), (ii) and (S1) are then continuous with respect to the $H^1(\Omega)$ norm of v , respectively φ , and the three relations extend likewise. $\square$

Proof of Proposition 2. (a) For $v, \varphi \in H^1(\Omega)$ let

$$a(v, \varphi) = \int_{\Omega} \nabla v \cdot \nabla \varphi \, d\mathbf{x} + 2\beta \int_{\partial\Omega} v \varphi \, ds, \quad \ell(\varphi) = \int_{\Omega} f \varphi \, d\mathbf{x} + 2\beta \int_{\partial\Omega} g \varphi \, ds,$$

so that $\mathcal{E}_{\beta}(v) = \frac{1}{2}a(v, v) - \ell(v) + \beta \|g\|_{\partial}^2$. By the trace bound of Lemma S5, a and ℓ are bounded on $H^1(\Omega)$, and by (i)

$$\|v\|_{H^1}^2 \leq 2 \|v\|_{\partial}^2 + 3 \|\nabla v\|^2 \leq \frac{3}{\min(1, 2\beta)} a(v, v).$$

Thus a is a symmetric, bounded and coercive bilinear form, and by the Lax–Milgram theorem [5, Sec. 6.2.1] (here the Riesz representation theorem for the inner product a) there is exactly one $u_{\beta} \in H^1(\Omega)$ with $a(u_{\beta}, \varphi) = \ell(\varphi)$ for all $\varphi \in H^1(\Omega)$; this is the displayed identity. For every $\varphi \in H^1(\Omega)$,

$$\mathcal{E}_{\beta}(u_{\beta} + \varphi) = \mathcal{E}_{\beta}(u_{\beta}) + (a(u_{\beta}, \varphi) - \ell(\varphi)) + \frac{1}{2}a(\varphi, \varphi) = \mathcal{E}_{\beta}(u_{\beta}) + \frac{1}{2}a(\varphi, \varphi),$$

which exceeds $\mathcal{E}_{\beta}(u_{\beta})$ unless $\varphi = 0$: u_{β} is the unique minimiser. The identity is the weak form of the Robin problem in the usual sense: if $u \in C^2(\overline{\Omega})$ satisfies $-\Delta u = f$ in Ω and $\partial_n u + 2\beta(u - g) = 0$ on $\partial\Omega$, then (S1) with $w = u$ shows that u satisfies the identity, and so $u = u_{\beta}$.

(b) By (S1) with $w = u^*$, $\int_{\Omega} \nabla u^* \cdot \nabla \varphi \, d\mathbf{x} = \int_{\Omega} f \varphi \, d\mathbf{x} + \int_{\partial\Omega} (\partial_n u^*) \varphi \, ds$ for all $\varphi \in H^1(\Omega)$. Subtracting this from the identity of (a) and using $g = u^*$ on $\partial\Omega$,

$$\int_{\Omega} \nabla w \cdot \nabla \varphi \, d\mathbf{x} + 2\beta \int_{\partial\Omega} w \varphi \, ds = - \int_{\partial\Omega} (\partial_n u^*) \varphi \, ds \quad \text{for all } \varphi \in H^1(\Omega). \quad (\text{S5})$$

The choice $\varphi = w$ is the energy identity. Write $A = \|w\|_{\partial}$ and $N = \|\partial_n u^*\|_{\partial}$. The right-hand side of the energy identity is at most NA , so $2\beta A^2 \leq NA$, that is $A \leq N/(2\beta)$, and $\|\nabla w\|^2 \leq NA - 2\beta A^2 \leq \max_{s \geq 0} (Ns - 2\beta s^2) = N^2/(8\beta)$.

(c) We show that

$$\|w\| \leq 2C_d \|w\|_{\partial}, \quad C_d = \sqrt{1 + \frac{1}{d\pi^2}}, \quad (\text{S6})$$

which with (b) gives $\|u_{\beta} - u^*\| \leq C_d N/\beta$; the constant decreases with d from $C_1 = 1.049$, and $C_2 = 1.025$. For $\mathbf{k} = (k_1, \dots, k_d)$ with positive integers k_i let $e_{\mathbf{k}}(\mathbf{x}) = 2^{d/2} \prod_i \sin(\pi k_i x_i)$. These functions form an orthonormal basis of $L^2(\Omega)$ (completeness was recalled above), vanish on $\partial\Omega$ and satisfy $-\Delta e_{\mathbf{k}} = \pi^2 |\mathbf{k}|^2 e_{\mathbf{k}}$ with $|\mathbf{k}|^2 = \sum_i k_i^2 \geq d$. Let $\hat{w}_{\mathbf{k}} = \int_{\Omega} w e_{\mathbf{k}} \, d\mathbf{x}$ and, for $M \in \mathbb{N}$,

$$w_M = \sum_{\max_i k_i \leq M} \hat{w}_{\mathbf{k}} e_{\mathbf{k}}, \quad \psi_M = \sum_{\max_i k_i \leq M} \frac{\hat{w}_{\mathbf{k}}}{\pi^2 |\mathbf{k}|^2} e_{\mathbf{k}},$$

so that $\psi_M \in C^\infty(\overline{\Omega})$, $\psi_M = 0$ on $\partial\Omega$ and $-\Delta \psi_M = w_M$. For fixed i and j , each of the families $\partial_i e_{\mathbf{k}}/(\pi k_i)$ and $\partial_i \partial_j e_{\mathbf{k}}/(\pi^2 k_i k_j)$, indexed by $\mathbf{k}$, is orthonormal in $L^2(\Omega)$, being of the same product form with some sines replaced by cosines. Hence, all sums being over $\max_i k_i \leq M$,

$$\|\nabla \psi_M\|^2 = \sum_{\mathbf{k}} \frac{\hat{w}_{\mathbf{k}}^2}{\pi^2 |\mathbf{k}|^2} \leq \frac{\|w_M\|^2}{d\pi^2}, \quad \sum_{i,j=1}^d \|\partial_i \partial_j \psi_M\|^2 = \sum_{\mathbf{k}} \hat{w}_{\mathbf{k}}^2 \frac{(\sum_i k_i^2)^2}{|\mathbf{k}|^4} = \|w_M\|^2.$$

On the two faces where x_i is constant, $\partial_n \psi_M = \pm \partial_i \psi_M$. Inequality (ii) of Lemma S5, applied to $v = \partial_i \psi_M$ on these two faces and summed over i , gives

$$\|\partial_n \psi_M\|_{\partial}^2 \leq 4 \|\nabla \psi_M\|^2 + 4 \sum_{i=1}^d \|\partial_i^2 \psi_M\|^2 \leq 4C_d^2 \|w_M\|^2.$$

Now take $\varphi = \psi_M$ in (S5): both boundary integrals vanish, so $\int_{\Omega} \nabla w \cdot \nabla \psi_M \, d\mathbf{x} = 0$. Green's formula (S1), with ψ_M in the role of w and $w \in H^1(\Omega)$ in the role of φ , then reads $-\int_{\Omega} w w_M \, d\mathbf{x} = \int_{\partial\Omega} w \partial_n \psi_M \, ds$. Since $\int_{\Omega} w w_M \, d\mathbf{x} = \|w_M\|^2$,

$$\|w_M\|^2 \leq \|w\|_{\partial} \|\partial_n \psi_M\|_{\partial} \leq 2C_d \|w\|_{\partial} \|w_M\|.$$

Hence $\|w_M\| \leq 2C_d \|w\|_{\partial}$, and letting $M \rightarrow \infty$, so that $w_M \rightarrow w$ in $L^2(\Omega)$, gives (S6).

(d) If $\partial_n u^* = 0$ on $\partial\Omega$, then $N = 0$ and (b) gives $\nabla w = 0$ and $\|w\|_{\partial} = 0$ for every $\beta > 0$, hence $w = 0$ by Lemma S5(i). Conversely, if $w = 0$ for one β , then (S5) gives $\int_{\partial\Omega} (\partial_n u^*) \varphi \, ds = 0$ for all $\varphi \in C^1(\overline{\Omega})$. Choose $\varphi(\mathbf{x}) = \chi(x_i) \theta(\mathbf{x}')$, where $\mathbf{x}'$ denotes the other coordinates, θ is smooth with compact support in the open face $F = \{x_i = 0\}$, and χ is smooth, equal to 1 near 0 and to 0 near 1. This φ vanishes on every face other than F , so $\int_F (\partial_n u^*) \theta \, ds = 0$ for all such θ ; as $\partial_n u^*$ is continuous on the closed face, it vanishes there, and likewise on every other face. So $\partial_n u^* = 0$, and then $u_{\beta} = u^*$ for every β by the first part.

Finally, LapD and PoiD satisfy the assumption of (b) by Proposition S1(a). For LapD, on the faces $x_{2k-1} \in \{0, 1\}$ one has $\partial_n u^* = \mp x_{2k}$ and on the faces $x_{2k} \in \{0, 1\}$, $\partial_n u^* = \mp x_{2k-1}$, each with squared L^2 norm $\int_0^1 s^2 \, ds = \frac{1}{3}$; when d is odd, $\partial_n u^* = 0$ on the two faces $x_d \in \{0, 1\}$. Summing the $4m$ contributions gives $\|\partial_n u^*\|_{\partial}^2 = 4m/3 > 0$, so $u_{\beta} \neq u^*$ for every β by (d). For PoiD, $\partial_i u^* = -\pi d^{-1/2} \sin(\pi x_i)$ vanishes at $x_i \in \{0, 1\}$, so $\partial_n u^* = 0$ and $u_{\beta} = u^*$. $\square$

Numerical observation S6 (The bounds against a computed Robin solution). For LapD at $d = 2$, where $N = \|\partial_n u^*\|_{\partial} = \sqrt{4/3}$ and $C_2 = \sqrt{1 + 1/(2\pi^2)} = 1.025$, part (c) of Proposition 2 gives $\|u_{\beta} - u^*\| \leq C_2 N / \beta = 1.184 / \beta$. In a second-order finite-difference solution of the Robin problem on a grid of 200 cells per side, $\beta \|u_{\beta} - u^*\|$ is 0.167, 0.183 and 0.185 for $\beta = 10, 100$ and 1000; divided by $\|u^*\| = \frac{1}{3}$, these are the values between $0.50/\beta$ and $0.55/\beta$ of Observation S3. The computed bias thus decays like $1/\beta$, the order of the bound, with a constant 6.4 to 7.1 times smaller than that of part (c); the sharper bound $\sqrt{U_2} N / (2\beta) = 0.312 / \beta$ of Corollary 7(b) is 1.69 to 1.87 times the computed values (Section 6.7). The trained Deep Ritz networks of Observation S3 are compared with these values there. The first bound of (b) is nearly attained: at $\beta = 1000$ the computed $\|w\|_{\partial}$ is 5.75×10^{-4} against $N/(2\beta) = 5.77 \times 10^{-4}$.

Numerical observation S7 (The bias in dimension d). For LapD in dimension $d = 2, 3, 5, 10, 20$ the relative bias $\|u_{\beta} - u^*\| / \|u^*\|$ was computed from the weak form (S5) by separation of variables: $w = u_{\beta} - u^*$ is a sum of products of the eigenfunctions of $-\varphi''$ on $(0, 1)$ with the Robin condition $\pm \varphi' + 2\beta \varphi = 0$ and of one hyperbolic factor, and $\|w\|^2$ becomes a sum over the eigenvalues of one or two coordinates and an expectation over the others, summed exactly for $d \leq 4$ and estimated by Monte Carlo for $d \geq 5$ (relative standard error at most 1.7×10^{-4} ; halving the number of modes kept changes no value). These are the values of Table S8: at $\beta = 100$ the bias is 5.48×10^{-3} , 4.70×10^{-3} , 2.99×10^{-3} , 1.46×10^{-3} and 7.37×10^{-4} for $d = 2, 3, 5, 10, 20$. At $d = 2$ they agree with the finite-difference solution of Observation S3 to 1×10^{-4} relative, and a Galerkin solution with tensor Legendre polynomials agrees with them to 1 per cent for $d \leq 5$ and $\beta \leq 100$ (at $d = 10$ and 20 it has not converged at the degrees that can be afforded). The exact representation of Proposition S11, computed with separate code, agrees with these values to within 4.3×10^{-4} relative in all 20 cells that the two computations share, including $d = 10$ and 20.

S7.5 The harmonic-extension constant of the unit cube

Throughout, $\Omega = (0, 1)^d$, $\|\cdot\|$ is the norm of $L^2(\Omega)$, $\|\cdot\|_{\partial}$ that of $L^2(\partial\Omega)$ with surface measure, and $|\partial\Omega| = 2d$. For $\mathbf{k} = (k_1, \dots, k_d)$ with positive integers k_i let $e_{\mathbf{k}} = 2^{d/2} \prod_i \sin(\pi k_i x_i)$ and $|\mathbf{k}|^2 = \sum_i k_i^2 \geq d$; these functions form an orthonormal basis of $L^2(\Omega)$ (Section S7), vanish on $\partial\Omega$ and satisfy $-\Delta e_{\mathbf{k}} = \pi^2 |\mathbf{k}|^2 e_{\mathbf{k}}$. A finite linear combination of them is called a sine polynomial.

$F_i^0 = \{x_i = 0\}$ and $F_i^1 = \{x_i = 1\}$ are the two faces normal to x_i . Green's formula (S1), with the roles of the two functions as needed, holds for $w \in H^1(\Omega)$ and $\psi \in C^2(\overline{\Omega})$ by Lemma S5:

$$\int_{\Omega} w \Delta \psi \, d\mathbf{x} = - \int_{\Omega} \nabla w \cdot \nabla \psi \, d\mathbf{x} + \int_{\partial\Omega} w \partial_n \psi \, ds. \quad (\text{S7})$$

The operator T . For a sine polynomial $\zeta = \sum a_{\mathbf{k}} e_{\mathbf{k}}$ let $\psi_{\zeta} = \sum a_{\mathbf{k}} e_{\mathbf{k}} / (\pi^2 |\mathbf{k}|^2)$, so that $-\Delta \psi_{\zeta} = \zeta$ and $\psi_{\zeta} = 0$ on $\partial\Omega$, and let $T\zeta = \partial_n \psi_{\zeta}|_{\partial\Omega}$. By Theorem 5 (whose proof below uses only sine polynomials) T is bounded, and it extends uniquely to a bounded operator $T : L^2(\Omega) \rightarrow L^2(\partial\Omega)$. For sine polynomials $\|\nabla \psi_{\zeta}\|^2 = \sum a_{\mathbf{k}}^2 / (\pi^2 |\mathbf{k}|^2) \leq \|\zeta\|^2 / (d\pi^2)$ and $\sum_{i,j} \|\partial_i \partial_j \psi_{\zeta}\|^2 = \|\zeta\|^2$ (the families $\partial_i e_{\mathbf{k}} / (\pi k_i)$ and $\partial_i \partial_j e_{\mathbf{k}} / (\pi^2 k_i k_j)$ are orthonormal), so for $\zeta \in L^2(\Omega)$ the sine series ψ_{ζ} converges in $H^2(\Omega)$ and $T\zeta$ is the trace of $\partial_n \psi_{\zeta}$ (Lemma S5 applied to $\partial_i \psi$). By (S7), $\int_{\Omega} w \zeta \, d\mathbf{x} = - \int_{\partial\Omega} w T\zeta \, ds$ for every harmonic $w \in C^2(\overline{\Omega})$ and every sine polynomial ζ ; thus $-T^*$ is the (very weak) harmonic extension from $L^2(\partial\Omega)$ to $L^2(\Omega)$, with norm $\|T\|$.

Lemma S8. $\kappa_d = \|T\|$, where κ_d is defined by (11).

Proof. Formula (S7) extends to $\psi \in H^2(\Omega)$: $C^2(\overline{\Omega})$ is dense in $H^2(\Omega)$ on the cube (dilation about the centre and mollification, as in the proof of Lemma S5), and both sides are continuous in the H^2 norm of ψ , the boundary term because $\|\partial_i \psi\|_{\partial}$ is bounded by the H^1 norm of $\partial_i \psi$. Let $u \in H^2(\Omega) \cap H_0^1(\Omega)$ and $\zeta = -\Delta u$. For each $\mathbf{k}$, (S7) with $(w, \psi) = (u, e_{\mathbf{k}})$ and with $(w, \psi) = (e_{\mathbf{k}}, u)$, together with $u = e_{\mathbf{k}} = 0$ on $\partial\Omega$, gives $\int u \Delta e_{\mathbf{k}} = - \int \nabla u \cdot \nabla e_{\mathbf{k}} = \int e_{\mathbf{k}} \Delta u$, that is $\pi^2 |\mathbf{k}|^2 \int u e_{\mathbf{k}} = \int \zeta e_{\mathbf{k}}$. So u and ψ_{ζ} have the same sine coefficients, $u = \psi_{\zeta}$ and $\partial_n u = T\zeta$. Conversely, for every $\zeta \in L^2(\Omega)$, $\psi_{\zeta} \in H^2(\Omega)$, and $\psi_{\zeta} \in H_0^1(\Omega)$ as an H^1 limit of sine polynomials. Hence $\{(\Delta u, \partial_n u) : u \in H^2 \cap H_0^1\} = \{(-\zeta, T\zeta) : \zeta \in L^2(\Omega)\}$, and the supremum defining κ_d is $\|T\|$. $\square$

Proof of Theorem 5. By Lemma S8 it suffices to bound $\|T\|$. Let $\zeta = \sum a_{\mathbf{k}} e_{\mathbf{k}}$ be a sine polynomial and $\psi = \psi_{\zeta}$.

Step 1 (face formula). Write $\mathbf{k}_{-i}$ for $\mathbf{k}$ without its i -th entry and let

$$F_{\mathbf{k}_{-i}} = 2^{(d-1)/2} \prod_{j \neq i} \sin(\pi k_j x_j);$$

these functions form an orthonormal family in $L^2(F_i^0)$ and in $L^2(F_i^1)$. Since

$$\partial_i e_{\mathbf{k}} = 2^{d/2} \pi k_i \cos(\pi k_i x_i) \prod_{j \neq i} \sin(\pi k_j x_j)$$

and the outward normal is $-e_i$ on F_i^0 and e_i on F_i^1 ,

$$\partial_n \psi|_{F_i^0} = - \sum_{\mathbf{k}} a_{\mathbf{k}} b_{\mathbf{k}}^{(i)} F_{\mathbf{k}_{-i}}, \quad \partial_n \psi|_{F_i^1} = \sum_{\mathbf{k}} (-1)^{k_i} a_{\mathbf{k}} b_{\mathbf{k}}^{(i)} F_{\mathbf{k}_{-i}}, \quad b_{\mathbf{k}}^{(i)} = \frac{\sqrt{2} k_i}{\pi |\mathbf{k}|^2}.$$

Step 2 (parity). Group the terms by $\mathbf{k}_{-i}$. For real $b_{\mathbf{k}}$, $(\sum_{\mathbf{k}} b_{\mathbf{k}})^2 + (\sum_{\mathbf{k}} (-1)^{k_i} b_{\mathbf{k}})^2 = 2E^2 + 2O^2$, with E and O the sums over even and odd k_i . By orthonormality,

$$\|T\zeta\|_{\partial}^2 = \frac{4}{\pi^2} \sum_{i=1}^d \sum_{\mathbf{k}_{-i}} (E_i(\mathbf{k}_{-i})^2 + O_i(\mathbf{k}_{-i})^2), \quad E_i, O_i = \sum_{k_i \text{ even, odd}} a_{\mathbf{k}} \frac{k_i}{|\mathbf{k}|^2}. \quad (\text{S8})$$

Step 3 (Cauchy-Schwarz). Fix i and $\mathbf{k}_{-i}$ and put $q = |\mathbf{k}_{-i}|^2 \geq d-1$, so $|\mathbf{k}|^2 = k_i^2 + q$. Writing $a_{\mathbf{k}} k_i / |\mathbf{k}|^2 = (a_{\mathbf{k}} k_i / |\mathbf{k}|) \cdot (1/|\mathbf{k}|)$,

$$O_i^2 \leq \left(\sum_{k_i \text{ odd}} a_{\mathbf{k}}^2 w_{\mathbf{k}}^{(i)} \right) S_{\text{odd}}(q), \quad w_{\mathbf{k}}^{(i)} = \frac{k_i^2}{|\mathbf{k}|^2}, \quad S_{\text{odd}}(q) = \sum_{j \text{ odd}} \frac{1}{j^2 + q},$$

and $E_i^2 \leq (\sum_{k_i \text{ even}} a_{\mathbf{k}}^2 w_{\mathbf{k}}^{(i)}) S_{\text{even}}(q)$ with $S_{\text{even}}(q) = \sum_{l \geq 1} 1/((2l)^2 + q) \leq S_{\text{odd}}(q)$ termwise. *Step 4 (partial fractions).* For $a > 0$, $\sum_{j \text{ odd}} 1/(j^2 + a^2) = \pi \tanh(\pi a/2)/(4a)$, from $\tanh z = \sum_{j \text{ odd}} 8z/(4z^2 + \pi^2 j^2)$ with $z = \pi a/2$; for $a = 0$ the sum is $\pi^2/8$. The function $a \mapsto \tanh(\pi a/2)/a$ decreases on $(0, \infty)$, since its derivative has the sign of $\pi a/2 - \sinh(\pi a/2) \cosh(\pi a/2) < 0$. Hence $S_{\text{odd}}(q) \leq \pi \tanh(\pi \sqrt{d-1}/2)/(4\sqrt{d-1}) = \pi^2 U_d/4$ for every $q \geq d-1$ (for $d = 1$, $q = 0$ and $\pi^2/8 = \pi^2 U_1/4$). *Step 5 (summation).* Inserting Steps 3 and 4 into (S8) and using $\sum_i w_{\mathbf{k}}^{(i)} = 1$, $\|T\zeta\|_{\partial}^2 \leq \frac{4}{\pi^2} \cdot \frac{\pi^2 U_d}{4} \sum_i \sum_{\mathbf{k}} a_{\mathbf{k}}^2 w_{\mathbf{k}}^{(i)} = U_d \|\zeta\|^2$. So $\|T\|^2 \leq U_d$.

Step 6 (lower bound L_d). Take $a_{\mathbf{k}} = j/(j^2 + d - 1)$ for $\mathbf{k} = (j, 1, \dots, 1)$ with j odd, $j \leq J$, and $a_{\mathbf{k}} = 0$ otherwise. Then $\|\zeta\|^2 = S_J := \sum_{j \leq J \text{ odd}} j^2/(j^2 + d - 1)^2$. In (S8) the face pair $i = 1$ alone, with $\mathbf{k}_{-1} = (1, \dots, 1)$, gives $O_1 = S_J$, so $\|T\zeta\|_{\partial}^2 \geq (4/\pi^2) S_J^2 = (4/\pi^2) S_J \|\zeta\|^2$. Letting $J \rightarrow \infty$ gives $\|T\|^2 \geq L_d$. *Step 7 (values and limits).* $L_1 = (4/\pi^2) \sum_{j \text{ odd}} j^{-2} = \frac{1}{2} = U_1$. With $x = \pi \sqrt{d-1}/2 > 0$, $U_d = \frac{1}{2} \tanh(x)/x < \frac{1}{2}$ for $d \geq 2$, and $\sqrt{d} U_d \rightarrow 1/\pi$ as $d \rightarrow \infty$. For $a = \sqrt{d-1}$ the function $f(x) = x^2/(x^2 + a^2)^2$ increases on $[0, a]$ to $1/(4a^2)$ and then decreases, so its total variation on $[0, \infty)$ is $1/(2a^2)$; the midpoint rule with step 2 gives $|\sum_{j \text{ odd}} f(j) - \frac{1}{2} \int_0^\infty f| \leq 1/(2a^2)$, and $\int_0^\infty f = \pi/(4a)$. Hence $\sqrt{d} L_d \rightarrow 1/(2\pi)$. *Step 8 (lower bound $1/(2d)$).* Let ζ_M be the L^2 projection of the constant 1 onto the span of the $e_{\mathbf{k}}$ with $\max_i k_i \leq M$, so $\zeta_M \rightarrow 1$ in $L^2(\Omega)$. The identity $\int_{\Omega} w \zeta = -\int_{\partial\Omega} w T \zeta$ with the harmonic function $w = 1$ gives $\int_{\Omega} \zeta_M = -\int_{\partial\Omega} T \zeta_M \leq |\partial\Omega|^{1/2} \|T \zeta_M\|_{\partial} \leq \sqrt{2d} \|T\| \|\zeta_M\|$; letting $M \rightarrow \infty$ gives $1 \leq \sqrt{2d} \|T\|$. $\square$

Proof of Proposition 6. By the proof of Lemma S8, the pairs $(\Delta u, \partial_n u)$ with $u \in H^2(\Omega) \cap H_0^1(\Omega)$ are exactly the pairs $(-\zeta, T\zeta)$, $\zeta \in L^2(\Omega)$. So $1/\delta_1 = \sup \|T\zeta\|_{\partial}^2 / \|\zeta\|^2 = \|T\|^2 = \kappa_d^2$ (functions with $\partial_n u = 0$ do not change the supremum), and the bounds follow from Theorem 5. $\square$

Remark S9. By Fichera's principle of duality, δ_1 is also the optimal constant in $\delta_1 \|h\|^2 \leq \|h\|_{\partial}^2$ over harmonic h ; for domains with a uniform outer ball condition, which includes convex domains, this is due to Bucur et al. [3], see Bucur and Gazzola [2, Sec. 3]. In the present notation it is the statement that $-T^*$ is the harmonic extension. Payne's bound $\delta_1 \geq 2/\ell$ for convex domains of minimal width ℓ [13], in the form stated in Bucur and Gazzola [2, Lemma 5.3], with the regularity assumptions removed by Philippin and Safoui [15], gives $\kappa_d^2 \leq \frac{1}{2}$ for the unit cube. An elementary bound of the same kind uses the torsion function ϑ of Ω , $-\Delta \vartheta = 1$ in Ω , $\vartheta = 0$ on $\partial\Omega$. For harmonic $h \in C^2(\overline{\Omega})$, Green's formula (S7) applied twice (to (h^2, ϑ) and to (ϑ, h^2) , with $\Delta(h^2) = 2|\nabla h|^2$ and $\vartheta = 0$ on $\partial\Omega$) gives $\int_{\Omega} h^2 = -2 \int_{\Omega} \vartheta |\nabla h|^2 - \int_{\partial\Omega} h^2 \partial_n \vartheta \leq m_d \|h\|_{\partial}^2$ with $m_d = \max_{\partial\Omega} |\partial_n \vartheta|$, since $\vartheta \geq 0$ (on the cube $\vartheta = \psi_1 \in H^2(\Omega)$ by the paragraph on T above, and (S7) holds for $\psi \in H^2(\Omega)$ by the proof of Lemma S8). This gives $\kappa_d^2 \leq m_d$. Indeed, for harmonic $h \in C^2(\overline{\Omega})$ the identity $\int_{\Omega} h \zeta = -\int_{\partial\Omega} h T \zeta$ of the paragraph on T extends by continuity from sine polynomials to all $\zeta \in L^2(\Omega)$, so $T^*(h|_{\partial\Omega}) = -h$ and $\|T^*(h|_{\partial\Omega})\|^2 \leq m_d \|h\|_{\partial}^2$. These traces are dense in $L^2(\partial\Omega)$: for $g \in C(\partial\Omega)$ let $h \in C(\overline{\Omega})$ be the harmonic function with boundary values g , which exists because the convex domain Ω satisfies an exterior sphere condition at every boundary point [7, Sec. 2.8]; with $\mathbf{c}$ the centre of Ω and $\mu > 1$, $h_{\mu}(\mathbf{x}) = h(\mathbf{c} + (\mathbf{x} - \mathbf{c})/\mu)$ is harmonic on the open cube $\mathbf{c} + \mu(\Omega - \mathbf{c}) \supset \overline{\Omega}$, hence in $C^2(\overline{\Omega})$, and $h_{\mu} \rightarrow h$ uniformly on $\overline{\Omega}$ as $\mu \downarrow 1$, since h is uniformly continuous; so g is a uniform limit of such traces, and $C(\partial\Omega)$ is dense in $L^2(\partial\Omega)$. As T^* is bounded, $\|T\|^2 = \|T^*\|^2 \leq m_d$, and $\kappa_d^2 \leq m_d$ by Lemma S8. On the cube $\vartheta(\mathbf{x}) = \int_0^\infty \prod_i u(t, x_i) dt$, where u solves the heat equation on $(0, 1)$ with $u(0, \cdot) = 1$ and zero boundary values; as $0 \leq u(t, y) \leq u(t, \frac{1}{2})$, the maximum is at the centres of the faces, $m_d = \int_0^\infty \partial_x u(t, 0) u(t, \frac{1}{2})^{d-1} dt$, with $m_1 = \frac{1}{2}$. By quadrature $m_d = 0.3377, 0.1953, 0.1703$ and 0.1366 for $d = 2, 10, 20$ and 100 (Table S9): m_d decreases with d , but more slowly than U_d . Lower bounds of δ_1 through the mean curvature [6, 14] or through curvature bounds and the inner radius [16] need a C^2 or smooth boundary and do not apply to the cube.

Numerical observation S10 (Rayleigh–Ritz lower bounds). The largest eigenvalue of T^*T restricted to the sine modes with $\max_i k_i \leq K$, computed from (S8) by the Lanczos method,

increases with K : for $d = 2$ it is 0.28366, 0.28569, 0.28670, 0.28721, 0.28746 for $K = 50, 100, 200, 400, 800$ ($U_2 = 0.29194$). T^*T preserves the parity pattern of $\mathbf{k}$, and for $d = 2, 3$ the top eigenvalue lies in the block in which all k_i are odd. Restricting further to all-odd $\mathbf{k}$ with at most $r \in \{1, 2\}$ entries different from 1 gives lower bounds for larger d ; the best values found are $0.985 U_d$ ($d = 2$), $0.964 U_d$ (3), $0.947 U_d$ (4), $0.932 U_d$ (5), $0.874 U_d$ (10), $0.796 U_d$ (20), $0.691 U_d$ (50) and $0.637 U_d$ (100). They are lower bounds of κ_d^2 up to rounding, not estimates of it. For $d = 2$ a Rayleigh–Ritz computation on the harmonic side, with the quotient $\|h\|^2/\|h\|_\partial^2$ over linear combinations h of the harmonic extensions of the face data $\sin(\pi k y)$, $k \leq 20$, on each of the four sides (zero on the other sides), made in the re-check (Section 3.9), gives the larger quotient $0.2877121 = 0.9855 U_2$, a lower bound up to rounding and quadrature error (quadrature with 200, 400 and 800 nodes agreeing to 10^{-13}).

Proof of Corollary 7. (a) Let $\hat{w}_\mathbf{k} = \int_\Omega w e_\mathbf{k}$, $\zeta_M = \sum_{\max k_i \leq M} \hat{w}_\mathbf{k} e_\mathbf{k}$ and $\psi_M = \psi_{\zeta_M} \in H_0^1(\Omega) \cap C^\infty(\bar{\Omega})$. By (S7) and the assumption, $\|\zeta_M\|^2 = \int_\Omega w \zeta_M = -\int_\Omega w \Delta \psi_M = \int_\Omega \nabla w \cdot \nabla \psi_M - \int_{\partial\Omega} w T \zeta_M = -\int_{\partial\Omega} w T \zeta_M$, so $\|\zeta_M\|^2 \leq \|w\|_\partial \|T\| \|\zeta_M\|$. Let $M \rightarrow \infty$ and use Lemma S8 and Theorem 5.

(b) By Proposition 2(b), $w = u_\beta - u^*$ satisfies (S5) and $\|w\|_\partial \leq \|\partial_n u^*\|_\partial/(2\beta)$; with $\varphi \in H_0^1(\Omega)$ in (S5) the boundary terms vanish, so w satisfies the assumption of (a).

(c) Put $e = v - u^* \in C^2(\bar{\Omega})$, $r = \Delta e = \Delta v + f$ and $r_\mathbf{k} = \int r e_\mathbf{k}$. The series $e_0 = -\sum r_\mathbf{k} e_\mathbf{k}/(\pi^2 |\mathbf{k}|^2)$ converges in $H_0^1(\Omega)$, because the $e_\mathbf{k}/(\pi |\mathbf{k}|)$ are orthonormal for $\int \nabla \cdot \nabla$, and $\|e_0\|^2 = \sum r_\mathbf{k}^2/(\pi^4 |\mathbf{k}|^4) \leq \|r\|^2/(d\pi^2)^2$. For each $\mathbf{k}$, $\int \nabla e_0 \cdot \nabla e_\mathbf{k} = -r_\mathbf{k}$. The sine polynomials are dense in $H_0^1(\Omega)$: a $\varphi \in H_0^1$ orthogonal to all of them in $\int \nabla \cdot \nabla$ satisfies $0 = \int \nabla \varphi \cdot \nabla e_\mathbf{k} = \pi^2 |\mathbf{k}|^2 \int \varphi e_\mathbf{k}$ by (S7), so $\varphi = 0$. Hence $\int \nabla e_0 \cdot \nabla \varphi = -\int r \varphi$ for all $\varphi \in H_0^1(\Omega)$, and by (S7) also $\int \nabla e \cdot \nabla \varphi = -\int r \varphi$. So $e - e_0$ satisfies the assumption of (a), and its trace is that of e , namely $v - g$. Therefore $\|e\| \leq \|e_0\| + \kappa_d \|v - g\|_\partial$. The second form follows from $|\Omega| = 1$ and $|\partial\Omega| = 2d$. $\square$

The exact penalty bias. Let $\alpha = 2\beta$, $q = \partial_n u^*$ on $\partial\Omega$, $N = \|q\|_\partial$ and $w = u_\beta - u^*$.

Proposition S11 (Exact bias on the cube). *Let $(\mu_n, \phi_n)_{n \geq 1}$ be the $L^2(0, 1)$ -normalised eigenpairs of $-\phi'' = \mu\phi$ with $\phi'(0) = \alpha\phi(0)$, $\phi'(1) = -\alpha\phi(1)$: the even ones $\phi \propto \cos(\omega(x - \frac{1}{2}))$ with $\omega \tan(\omega/2) = \alpha$, and the odd ones $\phi \propto \sin(\omega(x - \frac{1}{2}))$ with $\omega \cot(\omega/2) = -\alpha$; $\mu = \omega^2$. Put $\Phi_n = \prod_i \phi_{n_i}(x_i)$ and $\Lambda_n = \sum_i \mu_{n_i}$. If $u^* \in C^2(\bar{\Omega})$, then*

$$\|u_\beta - u^*\|^2 = \sum_n \frac{(\int_{\partial\Omega} q \Phi_n ds)^2}{\Lambda_n^2}.$$

For LapD , with $m = \lfloor d/2 \rfloor$, $D_n = \phi_n(1) - \phi_n(0)$, $X_n = \int_0^1 x \phi_n$ and $M_n = \int_0^1 \phi_n$,

$$\|u_\beta - u^*\|^2 = m \int_0^\infty t Q(t) G(t)^{d-2} dt, \quad G(t) = \sum_n M_n^2 e^{-t\mu_n},$$

$$Q(t) = 2 \left[\left(\sum_n D_n^2 e^{-t\mu_n} \right) \left(\sum_n X_n^2 e^{-t\mu_n} \right) + \left(\sum_n D_n X_n e^{-t\mu_n} \right)^2 \right].$$

Proof. (i) The form $a_1(\phi, \chi) = \int_0^1 \phi' \chi' + \alpha(\phi(0)\chi(0) + \phi(1)\chi(1))$ is symmetric, bounded and coercive on $H^1(0, 1)$, which embeds compactly in $L^2(0, 1)$; the solution operator of the one-dimensional Robin problem is therefore compact, self-adjoint and positive on $L^2(0, 1)$, and its eigenfunctions form an orthonormal basis [5, Sec. 6.5 and App. D]. Every eigenfunction is smooth with $\mu > 0$; splitting it into even and odd parts about $\frac{1}{2}$ gives the two equations, and $\theta \tan \theta = \alpha/2$ has exactly one root in each interval $(k\pi, k\pi + \pi/2)$, $\theta \cot \theta = -\alpha/2$ exactly one in each $(k\pi + \pi/2, (k+1)\pi)$ ($\omega = 2\theta$). Products of orthonormal bases are an orthonormal basis of $L^2(\Omega)$. (ii) $\Phi_n \in C^\infty(\bar{\Omega})$, $-\Delta \Phi_n = \Lambda_n \Phi_n$, and $\partial_n \Phi_n + \alpha \Phi_n = 0$ on every face (on F_i^0 this

is the condition of ϕ_{n_i} , the other factors being constant in the normal direction). By (S7), for $\varphi \in H^1(\Omega)$, $\int \nabla \varphi \cdot \nabla \Phi_{\mathbf{n}} + \alpha \int_{\partial\Omega} \varphi \Phi_{\mathbf{n}} = \Lambda_{\mathbf{n}} \int \varphi \Phi_{\mathbf{n}}$. (iii) The error identity (S5) with $\varphi = \Phi_{\mathbf{n}}$ gives $\Lambda_{\mathbf{n}} \int w \Phi_{\mathbf{n}} = - \int_{\partial\Omega} q \Phi_{\mathbf{n}}$, and Parseval's identity gives the first formula. (iv) For LapD, $q = \mp x_{2p}$ on $\{x_{2p-1} = 0\}$, $\{x_{2p-1} = 1\}$, $q = \mp x_{2p-1}$ on $\{x_{2p} = 0\}$, $\{x_{2p} = 1\}$, and $q = 0$ on the faces of x_d when d is odd. The four faces of the pair p contribute $(D_{n_{2p-1}} X_{n_{2p}} + X_{n_{2p-1}} D_{n_{2p}}) \prod_{j \notin \{2p-1, 2p\}} M_{n_j}$. $D_n = 0$ for even and $M_n = 0$ for odd eigenfunctions, so the term of pair p needs all coordinates outside p even and at least one coordinate of p odd (if both are odd, both products survive and give the cross term of Q); terms of different pairs never share an $\mathbf{n}$, the square of the sum is the sum of the squares, and by symmetry each pair gives the same value. (v) $\Lambda^{-2} = \int_0^\infty t e^{-t\Lambda} dt$; all terms are non-negative, so by Tonelli's theorem the sum over $\mathbf{n}$ may be taken inside the integral, where for $t > 0$ it converges absolutely and factorises over the coordinates; expanding the square of $D_{n_1} X_{n_2} + X_{n_1} D_{n_2}$ gives Q . $\square$

The numbers of Table S11 were computed by quadrature in t over $[10^{-11}, 80]$ with 2×10^5 roots per parity type. For large β the integrand, in $\log t$, decays only like $t^{1/2}$ as $t \rightarrow 0$; a check over $[10^{-18}, 400]$ changes the entries by at most 1.0×10^{-8} relative for $\beta \leq 1000$ and 4.5×10^{-5} for $\beta = 10^4, 10^5$.

By the Riesz theorem and Theorem 5 there is a unique $v_1 \in L^2(\Omega)$ with $\int_{\Omega} v_1 \zeta = \frac{1}{2} \int_{\partial\Omega} q T \zeta$ for all $\zeta \in L^2(\Omega)$; it is the very weak harmonic function with boundary values $-q/2$, and $\|v_1\| \leq \kappa_d N/2$.

Theorem S12 (First-order term of the bias). *If $u^* \in C^2(\overline{\Omega})$, then for every $\beta > 0$*

$$\left\| u_\beta - u^* - \frac{v_1}{\beta} \right\| \leq K_d N \beta^{-3/2}, \quad K_d = \frac{\sqrt{d} + 2/(\pi\sqrt{d})}{4\sqrt{2}};$$

in particular $\|u_\beta - u^\| - \|v_1\|/\beta \leq K_d N \beta^{-3/2}$.*

Proof. Let $\psi = \psi_\zeta$ for a sine polynomial ζ and $z = w - v_1/\beta$. (i) By (S7), $\int w \zeta = \int \nabla w \cdot \nabla \psi - \int_{\partial\Omega} w \partial_n \psi$. (ii) (S5) with $\varphi = \psi \in H_0^1$ gives $\int \nabla w \cdot \nabla \psi = 0$. (iii) By definition $\int v_1 \zeta = \frac{1}{2} \int_{\partial\Omega} q \partial_n \psi$; hence $\int z \zeta = - \int_{\partial\Omega} (w + q/(2\beta)) \partial_n \psi$. (iv) Let $\tilde{\zeta} = \sum_i (2x_i - 1) \partial_i \psi \in C^\infty(\Omega)$. On $F_i^0 \cup F_i^1$ the tangential derivatives of ψ vanish and $2x_i - 1 = \pm 1$ is the sign of the outward normal, so $\tilde{\zeta} = \partial_n \psi$ on $\partial\Omega$. (S5) with $\varphi = \tilde{\zeta}$ gives $\int \nabla w \cdot \nabla \tilde{\zeta} + 2\beta \int_{\partial\Omega} w \partial_n \psi = - \int_{\partial\Omega} q \partial_n \psi$, that is $\int_{\partial\Omega} (w + q/(2\beta)) \partial_n \psi = -(2\beta)^{-1} \int \nabla w \cdot \nabla \tilde{\zeta}$. (v) $\partial_j \tilde{\zeta} = 2\partial_j \psi + \sum_i (2x_i - 1) \partial_i \partial_j \psi$ and $|2\mathbf{x} - \mathbf{1}| \leq \sqrt{d}$ give $\|\nabla \tilde{\zeta}\| \leq 2\|\nabla \psi\| + \sqrt{d} (\sum_{i,j} \|\partial_i \partial_j \psi\|^2)^{1/2} \leq (2/(\pi\sqrt{d}) + \sqrt{d}) \|\zeta\|$ by the two identities of the paragraph on T . (vi) With $\|\nabla w\| \leq N/\sqrt{8\beta}$ (Proposition 2(b)), $|\int z \zeta| \leq (2\beta)^{-1} N (8\beta)^{-1/2} (\sqrt{d} + 2/(\pi\sqrt{d})) \|\zeta\| = K_d N \beta^{-3/2} \|\zeta\|$. Sine polynomials are dense in $L^2(\Omega)$, which gives the claim; the second statement is the triangle inequality. $\square$

S7.6 Affine minimisers, floors and the reparametrisation control

Here $\Omega = (0, 1)^d$ with the uniform measures, $\phi(\mathbf{x}) = (1, x_1, \dots, x_d)^\top$, and an affine function is $p_c = c^\top \phi$, $c \in \mathbb{R}^{d+1}$. The problems are $-\Delta u^* = f$ in Ω , $u^* = g$ on $\partial\Omega$, with LapD, PoiD as in (5) and (6) and, with $m = \lfloor d/2 \rfloor$ and $t_i = 2x_i - 1$, RidgeD: $u^* = \cos(2s)$, $s = d^{-1/2} \sum_i t_i$, $f = 16u^*$; CosPairD: $u^* = \sum_{k \leq m} \cos(\pi x_{2k-1}) \cos(\pi x_{2k})$, $f = 2\pi^2 u^*$; AltRidgeD: $u^* = \cos(\pi s' + 1)$, $s' = d^{-1/2} \sum_i e_i t_i$, $e_i = +1$ for odd and -1 for even i , $f = 4\pi^2 u^*$. The sources follow from $\partial_{ii} \cos(2s) = -(16/d) \cos(2s)$ and $\partial_{ii} \cos(\pi s' + 1) = -(4\pi^2/d) \cos(\pi s' + 1)$. The population losses whose Monte-Carlo estimates are minimised are the penalised Ritz energy $\mathcal{E}_\beta$ of (8) and the population PINN loss $\mathcal{L}_\lambda^{\text{pop}}(v) = \int_{\Omega} (\Delta v + f)^2 + \frac{\lambda}{2d} \int_{\partial\Omega} (v - g)^2$ of Remark 3.

Lemma S13 (Boundary Gram matrix). *$M_\partial = \int_{\partial\Omega} \phi \phi^\top ds$ has the entries $(M_\partial)_{00} = 2d$, $(M_\partial)_{0i} = d$, $(M_\partial)_{ii} = 1 + \frac{2(d-1)}{3}$ and $(M_\partial)_{ij} = \frac{d}{2}$ for $i \neq j$, $i, j \geq 1$. Its eigenvalues are $(d+2)/6$, with multiplicity $d-1$, and the two eigenvalues of $B = \begin{pmatrix} 2d & d\sqrt{d} \\ d\sqrt{d} & a+(d-1)d/2 \end{pmatrix}$, $a = 1 + \frac{2(d-1)}{3}$, with $\det B = d(d+2)/3$. In particular M_∂ is positive definite.*

Proof. $\partial\Omega$ is the union of the $2d$ faces $\{x_k = s\}$, $s \in \{0, 1\}$, each of unit area with the uniform measure in the other coordinates. On $\{x_k = s\}$: $\int 1 = 1$; $\int x_i = \frac{1}{2}$ for $i \neq k$ and s for $i = k$; $\int x_i^2 = \frac{1}{3}$ for $i \neq k$ and s^2 for $i = k$; $\int x_i x_j = \int x_i \int x_j$ for $i \neq j$. Summing over the faces gives the entries; for $i \neq j$ the faces of x_i and of x_j give $\frac{1}{2}$ each and the other $2(d-2)$ faces $\frac{1}{4}$ each. With $b = d/2$ the lower block is $(a-b)I + b\mathbf{1}\mathbf{1}^\top$, so every $(0, v)$ with $v \perp \mathbf{1}$ is an eigenvector with eigenvalue $a-b = (d+2)/6$; on the span of e_0 and $(0, \mathbf{1}/\sqrt{d})$, M_∂ acts as B , and $\det B = 2d(a + (d-1)\frac{d}{2}) - d^3 = d(d+2)/3 > 0$, $\text{tr } B > 0$. $\square$

Proposition S14 (Affine minimisers). *Let $E = \text{diag}(0, 1, \dots, 1)$, $r_f = \int_\Omega f \phi$ and $r_g = \int_{\partial\Omega} g \phi$. For $\beta > 0$ and $\lambda > 0$ the minimisers of $\mathcal{E}_\beta$ and of $\mathcal{L}_\lambda^{\text{pop}}$ over affine functions are unique; their coefficient vectors solve $(E + 2\beta M_\partial)c = r_f + 2\beta r_g$ and $M_\partial c = r_g$, respectively. The PINN minimiser is the $L^2(\partial\Omega)$ projection of g onto the affine functions and depends neither on λ nor on f .*

Proof. For $v = p_c$, $\frac{1}{2} \int |\nabla p_c|^2 = \frac{1}{2} c^\top E c$, $\int f p_c = c^\top r_f$ and $\int_{\partial\Omega} (p_c - g)^2 = c^\top M_\partial c - 2c^\top r_g + \int_{\partial\Omega} g^2$, so $\mathcal{E}_\beta(p_c)$ is a quadratic in c with Hessian $E + 2\beta M_\partial$, positive definite by Lemma S13; its unique minimiser is the stationary point. For the PINN, $\Delta p_c = 0$, so $\mathcal{L}_\lambda^{\text{pop}}(p_c) = \int f^2 + \frac{\lambda}{2d}(c^\top M_\partial c - 2c^\top r_g + \int_{\partial\Omega} g^2)$, minimised uniquely at $M_\partial c = r_g$. $\square$

Proposition S15 (Structure of the affine minimisers). *(a) On CosPairD, $r_f = r_g = 0$, so both affine minimisers are 0. (b) On RidgeD both affine minimisers are constant. (c) On LapD the PINN minimiser is $p = \sum_{k \leq m} \frac{1}{2}(x_{2k-1} + x_{2k}) - \frac{m}{4}$, which is also the best $L^2(\Omega)$ approximation of u^* by affine functions.*

Proof. (a) f and g are sums of multiples of $\cos(\pi x_a) \cos(\pi x_b)$, $a \neq b$. Since $\int_0^1 \cos(\pi x) dx = 0$, every interior integral against 1 or x_i contains such a factor, and so does every face integral on a face of a coordinate other than a and b . On $\{x_a = s\}$ the term is $\cos(\pi s) \cos(\pi x_b)$; against 1 or x_i , $i \neq b$, it contains $\int \cos(\pi x_b) = 0$, and against x_b it equals $\cos(\pi s) (-2/\pi^2)$, which cancels between $s = 0$ and $s = 1$. The faces of x_b are symmetric. So $r_f = r_g = 0$ and $c = 0$. (b) The reflection $\sigma(\mathbf{x}) = \mathbf{1} - \mathbf{x}$ preserves Ω , $\partial\Omega$ and both measures, and $s \circ \sigma = -s$, so u^* , f and g are invariant and so are $\mathcal{E}_\beta$ and $\mathcal{L}_\lambda^{\text{pop}}$. If p_c is the unique minimiser, so is $p_c \circ \sigma = (c_0 + \sum_i c_i) - \sum_i c_i x_i$; uniqueness gives $c_i = 0$ for $i \geq 1$. (c) $x_a x_b = (x_a - \frac{1}{2})(x_b - \frac{1}{2}) + \frac{1}{2}(x_a + x_b) - \frac{1}{4}$, so $u^* - p = \sum_k q_k$ with $q_k = (x_{2k-1} - \frac{1}{2})(x_{2k} - \frac{1}{2})$. Each $q = (x_a - \frac{1}{2})(x_b - \frac{1}{2})$ is orthogonal to 1 and to every x_i in $L^2(\Omega)$ (a factor $\int_0^1 (x - \frac{1}{2}) dx = 0$ remains) and in $L^2(\partial\Omega)$: on faces of other coordinates the same factor remains; on $\{x_a = s\}$, $q = (s - \frac{1}{2})(x_b - \frac{1}{2})$ gives 0 against 1 and x_i , $i \neq b$, and $(s - \frac{1}{2})/12$ against x_b , which cancels between $s = 0, 1$. So p satisfies the normal equations of both projections. $\square$

Proposition S16 (Floors of affine and lifted features). *Let $\mathcal{A}$ be the affine functions, $\mathcal{F}$ the span of 1 and the $P_k(t_i)$, $k = 1, 2, 3$, $i = 1, \dots, d$ (P_k the Legendre polynomials; this is the span of the lift3c and of the lift3u features), and $\mathcal{S} = \{c_0 + \sum_i w_i(x_i)\}$ the additively separable functions. With $\text{fl}_\mathcal{V}(u) = \min_{v \in \mathcal{V}} \|u - v\|/\|u\|$ and $\mathbb{E}$ the mean over Ω ,*

$$\begin{aligned} \text{fl}_\mathcal{A}(u)^2 &= 1 - \frac{(\mathbb{E}u)^2 + 12 \sum_i (\mathbb{E}[(x_i - \frac{1}{2})u])^2}{\mathbb{E}u^2}, \\ \text{fl}_\mathcal{F}(u)^2 &= 1 - \frac{(\mathbb{E}u)^2 + \sum_i \sum_{k=1}^3 (2k+1) (\mathbb{E}[P_k(t_i)u])^2}{\mathbb{E}u^2}, \end{aligned}$$

and $\mathcal{A} \subset \mathcal{F} \subset \mathcal{S}$. Moreover: (a) *LapD*: $\text{fl}_\mathcal{A} = \text{fl}_\mathcal{F} = \text{fl}_\mathcal{S} = ((m/144)/(m/9 + m(m-1)/16))^{1/2}$; (b) *CosPairD*: $\text{fl}_\mathcal{A} = \text{fl}_\mathcal{F} = \text{fl}_\mathcal{S} = 1$; (c) *RidgeD*: $\text{fl}_\mathcal{A}$ equals the floor of the constants, $(1 - (\mathbb{E}u^*)^2/\mathbb{E}u^{*2})^{1/2}$, with $\mathbb{E}u^* = \text{sinc}(2/\sqrt{d})^d$, $\mathbb{E}u^{*2} = \frac{1}{2}(1 + \text{sinc}(4/\sqrt{d})^d)$ and $\text{sinc } a = \sin a/a$; (d) *PoiD*: $\mathbb{E}u^{*2} = \frac{1}{2}$; (e) *AltRidgeD*: with $a = \pi/\sqrt{d}$ and $\chi(a) = (\sin a - a \cos a)/a^2$, $\mathbb{E}u^* = \cos(1) \text{sinc}(a)^d$, $\mathbb{E}[(x_i - \frac{1}{2})u^*] = -\frac{1}{2}e_i \sin(1)\chi(a) \text{sinc}(a)^{d-1}$ and $\mathbb{E}u^{*2} = \frac{1}{2}(1 + \cos(2) \text{sinc}(2a)^d)$, hence $\text{fl}_\mathcal{A}^2 = 1 - (\cos^2(1) \text{sinc}(a)^{2d} + 3d \sin^2(1)\chi(a)^2 \text{sinc}(a)^{2d-2})/\mathbb{E}u^{*2}$.

Proof. Under the uniform measure, 1 and the $x_i - \frac{1}{2}$ are pairwise orthogonal with $\|x_i - \frac{1}{2}\|^2 = \frac{1}{12}$, and 1 and the $P_k(t_i)$ are pairwise orthogonal (for different i by independence and $\mathbb{E}P_k(t_i) = 0$, for the same i by the orthogonality of the Legendre polynomials) with $\|P_k(t_i)\|^2 = \frac{1}{2k+1}$. The best approximation from the span of an orthogonal system is the sum of the projections, and $\|u - \Pi u\|^2 = \|u\|^2 - \|\Pi u\|^2$; this gives both formulas. $\mathcal{A} \subset \mathcal{F}$ because $x_i = (P_1(t_i) + 1)/2$. The projection of u onto $\mathcal{S}$ is $\mathbb{E}u + \sum_i (\mathbb{E}[u | x_i] - \mathbb{E}u)$: the remainder has zero conditional mean given each single coordinate and is orthogonal to every $w_i(x_i)$. (a) $\mathbb{E}[u^* | x_i]$ is affine in x_i , so the three projections coincide; the remainder is $\sum_k q_k$ (Proposition S15(c)), with pairwise orthogonal q_k of squared norm $\frac{1}{144}$, and $\mathbb{E}u^{*2} = m/9 + m(m-1)/16$. (b) $\mathbb{E}[\cos(\pi x_a) \cos(\pi x_b) | x_i] = 0$ for every i , so the $\mathcal{S}$ -projection is 0. (c) u^* is invariant under $t \mapsto -t$ and $x_i - \frac{1}{2}$ is odd, so $\mathbb{E}[(x_i - \frac{1}{2})u^*] = 0$. With $a = 2/\sqrt{d}$ and $Z = \prod_j e^{iat_j}$, $u^* = \text{Re } Z$ and $\mathbb{E}e^{iat} = \text{sinc}(a)$ for t uniform on $[-1, 1]$; $u^{*2} = \frac{1}{2}(1 + \cos 4s)$. (d) $\mathbb{E}\cos^2(\pi x) = \frac{1}{2}$ and $\mathbb{E}[\cos(\pi x_i) \cos(\pi x_j)] = 0$ for $i \neq j$. (e) $u^* = \text{Re}(e^i Z)$ with $Z = \prod_j e^{iae_j t_j}$; sinc is even, so $\mathbb{E}u^* = \cos(1) \text{sinc}(a)^d$. Since $\mathbb{E}[te^{ibt}] = i \text{sign}(b) \chi(|b|)$, $\mathbb{E}[t_i u^*] = \text{Re}(e^i i e_i \chi(a)) \text{sinc}(a)^{d-1} = -e_i \sin(1) \chi(a) \text{sinc}(a)^{d-1}$, and $x_i - \frac{1}{2} = t_i/2$. $u^{*2} = \frac{1}{2}(1 + \cos(2\pi s' + 2))$ and the same argument with $2a$ give $\mathbb{E}u^{*2}$; insert into the formula for fl_A . $\square$

Lemma S17 (Laplacian of a lifted network). *Let $\mathcal{N}$ be a network of affine layers and tanh activations, $\psi : \mathbb{R} \rightarrow \mathbb{R}^3$ a C^2 map applied to each coordinate, $\Psi(\mathbf{x}) = (\psi_a(x_j))_{a,j} \in \mathbb{R}^{3d}$ and $u = \mathcal{N} \circ \Psi + p_c$. Propagate (h, J, L) by $h_0 = \Psi(\mathbf{x})$, $(J_0)_{i,(a,j)} = \delta_{ij} \psi'_a(x_j)$, $(L_0)_{(a,j)} = \psi''_a(x_j)$; through an affine layer $h \mapsto Wh + b$: $J \mapsto JW^\top$, $L \mapsto WL$; through a tanh layer, with $\tau = \tanh(h)$: $L \mapsto (1 - \tau^2) \odot L - 2\tau(1 - \tau^2) \odot \sum_i J_{i,\cdot}^2$, $J \mapsto J \text{diag}(1 - \tau^2)$, $h \mapsto \tau$. Then at the output $\Delta u = L$ and $\nabla u = J + (c_1, \dots, c_d)$.*

Proof. Induction over the layers with the invariant $J_{i,\cdot} = \partial_i h$ and $L = \sum_i \partial_{ii} h$ componentwise; it holds at the input because the component (a, j) depends on x_j only. Affine layers act linearly on derivatives; for tanh, $\partial_{ii} \tanh(h) = \tanh''(h)(\partial_i h)^2 + (1 - \tau^2) \partial_{ii} h$ with $\tanh'' = -2\tau(1 - \tau^2)$. The affine p_c adds $(c_1, \dots, c_d)$ to the gradient and nothing to the Laplacian. The plain network is the case $\psi = \text{id}$ (the forward Laplacian of Section 3.5). $\square$

Proposition S18 (The control rep3). *Let $\mathbf{t} = 2\mathbf{x} - \mathbf{1}$ and let the first layer of a network be $z \mapsto Wz + b$. Write the first-layer weight of the rep3 network $\mathcal{N}_3(\mathbf{t}, \mathbf{t}, \mathbf{t})$ as $W = [W^{(1)} W^{(2)} W^{(3)}]$ and put $W_{\text{eff}} = W^{(1)} + W^{(2)} + W^{(3)}$. (a) $\mathcal{N}_3(\mathbf{t}, \mathbf{t}, \mathbf{t}) = \mathcal{N}_1(\mathbf{t})$, where $\mathcal{N}_1$ has the layers of $\mathcal{N}_3$ except the first weight, which is W_{eff} . (b) If $\mathcal{N}_3$ is trained by Adam with step size σ (no weight decay, any momentum parameters and any ϵ) on any loss that depends on the network only through the function it computes, the iterates of $\mathcal{N}_1$ built by (a) are exactly those of Adam applied to $\mathcal{N}_1$ with step size 3σ for W_{eff} and σ for all other parameters, from the initial values given by (a). (c) Under the PyTorch default initialisation, uniform on $\pm 1/\sqrt{\text{fan-in}}$ for weights and biases, the entries of W_{eff} have variance $1/(3d)$, as those of the first weight of $\text{lift1} = \mathcal{N}(\mathbf{t})$, while the first bias has variance $1/(9d)$ in rep3 and $1/(3d)$ in lift1.*

Proof. (a) $W(\mathbf{t}, \mathbf{t}, \mathbf{t})^\top = W_{\text{eff}} \mathbf{t}$. (b) By (a) the loss depends on the $W^{(k)}$ only through W_{eff} , so $\nabla_{W^{(k)}} J = \nabla_{W_{\text{eff}}} J =: G$ for $k = 1, 2, 3$ at every step, where J is the training loss. The moment estimates of Adam for each $W^{(k)}$ are those of G , each $W^{(k)}$ moves by the same $-\sigma \hat{m}_G / (\sqrt{\hat{v}_G} + \epsilon)$, and W_{eff} moves by three times that, which is the Adam step with step size 3σ for the gradient sequence G . The other parameters see the same gradients in both networks; induction over the steps. (c) The uniform distribution on $\pm c$ has variance $c^2/3$. The fan-in of rep3 is $3d$, so each $W_{ij}^{(k)}$ has variance $1/(9d)$ and the sum of three independent copies $1/(3d)$; lift1 has fan-in d ; the biases have variances $1/(9d)$ and $1/(3d)$. $\square$

S7.7 The decaying mode and the one-face problem

Proof of Proposition 8. We use $\Delta S = -2\pi^2 S = -\omega^2 S$. (a) $v_{tt} = v$ and $\Delta v = -2\pi^2 v$, so $v_{tt} - \Delta v = (1 + 2\pi^2) e^{-t} S$, which is non-zero wherever $S \neq 0$. (b) The first assertion is Propositions S1(a) and S2(a). Since $v - u^* = (e^{-t} - \cos \omega t) S$ and $u^* = \cos(\omega t) S$, the factor $\int_{\Omega} S^2 d\mathbf{x} = 1$ cancels in the ratio, which leaves the stated quotient of time integrals. Elementary integration gives

$$\int_0^1 \cos^2 \omega t dt = \frac{1}{2} + \frac{\sin 2\omega}{4\omega}, \quad \int_0^1 e^{-t} \cos \omega t dt = \frac{1 + e^{-1}(\omega \sin \omega - \cos \omega)}{1 + \omega^2},$$

and $\int_0^1 e^{-t} dt = \frac{1}{2}(1 - e^{-2})$. With $\omega = \sqrt{2}\pi$ the denominator is 0.52888 and the numerator $\int_0^1 (e^{-t} - \cos \omega t)^2 dt$ is 1.00726, so the ratio is $\sqrt{1.00726/0.52888} = 1.380$. $\square$

Proof of Proposition 9. All functions below are restrictions to $\overline{D}$ of functions that are smooth on $\mathbb{R}^3$. (a) $\Delta u_0 = (2 - 1 - 1) u_0 = 0$, and $\cosh 0 = 1$ gives the datum on $\{x = \pi\}$. (b) If $h_{yy} + h_{zz} = 0$, then $\Delta((x - \pi)h) = (x - \pi)(h_{yy} + h_{zz}) = 0$ and $(x - \pi)h$ vanishes on $\{x = \pi\}$, so $u + (x - \pi)h$ is a solution. The difference of two solutions solves the problem with zero datum, so the solutions form the affine space $u_0 + V$, where V is the linear space of the functions in $C^\infty(\overline{D})$ that are harmonic in D and vanish on $\{x = \pi\}$. V contains $(x - \pi) \operatorname{Re}(y + iz)^j$ for $j = 0, 1, 2, \dots$; these are linearly independent, being non-zero polynomials of distinct degrees, so V has infinite dimension. (c) This is (b) with $h \equiv A$: at $(0, \pi/2, \pi/2)$, $u_0 = 0$ because $\cos(\pi/2) = 0$, and $A(x - \pi) = -A\pi$. (d) $\Delta u_N = (2 - 1 - 1) u_N = 0$ and $u_N(\pi, y, z) = \sin y \cos z$; u_N contains the factors $\sinh(\sqrt{2}x)$ and $\sin y$, which vanish at $x = 0$ and at $y \in \{0, \pi\}$, and $\partial_z u_N$ contains the factor $\sin z$, which vanishes at $z \in \{0, \pi\}$. Finally $u_N \neq u_0$, since $u_0(0, y, z) = \cosh(\sqrt{2}\pi) \sin y \cos z \neq 0 = u_N(0, y, z)$. $\square$

S8 Implementation details, validation and full tables

This section lists the settings of every experiment, validates the driver that trains many networks at once, gives the complete tables behind Sections 4 and 5, and records what is reproducible and what was re-checked. Its tables follow the text.

S8.1 Settings of the experiments

Table S15 covers Sections 4–7. The script that writes it imports network sizes, point sets and iteration counts from the released code, reads seeds, stack sizes and devices from the stored run records, and stops if a script no longer contains a setting that the table prints.

Single networks are trained with the L-BFGS [10] of PyTorch [12] (strong-Wolfe line search, memory 50). Stacks use our own version: one two-loop history per network; Armijo backtracking with constant 10^{-4} , the step of a network being halved until that network accepts it (at most 20 trials); and a curvature pair (s, y) stored only if $y \cdot s > 10^{-10} \|y\| \|s\|$. In exact arithmetic nothing couples the networks of a stack; in floating point it does, through rounding (Section S8.4).

S8.2 Validation of the stacked driver

The 175 networks of the benchmark were trained in six stacks. The loss of a stack is the sum of the losses of its S networks and Adam [8] treats every parameter separately, so in exact arithmetic one Adam run on a stack is S independent runs. Three checks cover rounding and L-BFGS.

Loss. For all five problems the loss assembled in one forward pass equals its term-by-term evaluation, and the loss of a stack of three networks equals the single-network loss (largest relative difference 0).

Adam. On Heat and Lap2 ($N_r = 1024$, five strategies, seeds 0 and 1: 20 configurations, CPU) the training losses of the stacked and the single-network run differ relatively by 0 at iteration 1,

by at most 6.6×10^{-8} at iteration 50 and by at most 1.5×10^{-7} at iteration 100. Rounding then separates the trajectories: after 600 iterations the ratio of the two final errors ε_2 lies between 0.72 and 1.88.

L-BFGS. In the same configurations, 300 iterations of the stacked L-BFGS were compared with 300 iterations of that of PyTorch, each from its own Adam state at iteration 300. The ratio of the final errors ε_2 , stacked over single, has geometric mean 1.10 (range 0.94 to 1.25) on **Heat** and 1.04 (0.87 to 1.37) on **Lap2**: the stacked version is 4 to 10 per cent less accurate in geometric mean, a difference smaller than the spread that rounding alone produced under Adam.

These checks have two limits: they used $N_r = 1024$ and $300 + 300$ iterations, and they were run before the loss was re-assembled into one forward pass (the first check shows that the re-assembly leaves the loss unchanged). At the settings of the benchmark, whose runs used the re-assembled loss, the verification code **V-bench** of Section S8.5 (one network, CPU, different derivative code, the L-BFGS of PyTorch) shares 27 configurations with the stacked driver; the ratio of its Adam $\rightarrow$ L-BFGS errors to those of the stacked driver has geometric mean 0.98 (range 0.60 to 1.26).

S8.3 Full tables

Table S16 adds the geometric means and the L^∞ errors to Table 4. Table S17 gives every paired comparison of a strategy with **random** and the per-seed test of the optimiser arms (Section 3.7). Table S18 is the convergence study of the finite-difference baselines: the observed order is 1.99 to 2.01 for **Heat**, **Wave1**, **Wave2** and **Lap2**, and 1.97, 2.70 and 1.89 for **Lap3** (Section 5).

Table S19 keeps three time measurements apart. Measurement A timed 400 Adam and 200 L-BFGS iterations of the single-network reference driver on four CPU threads and scaled them to $1500 + 1500$, while the load average of the 12-core machine rose from 95 to 167. Measurement B is the time of the 36 runs of **V-bench** at the benchmark’s N_r , at a load average of about 20 to 45; 12 of them (on **Heat** and **Lap2**) stopped on the L-BFGS tolerance before iteration 1500, the earliest after 1178 iterations. Measurement C is the training time of the Adam $\rightarrow$ L-BFGS arm of the GPU stacks that produced the errors of the benchmark, divided by the number of networks in the stack (25 or 50): 6.9 to 13.9 s per network. It is reached only when that many networks are trained at once, and the load during these runs was not recorded. A is 5.1 to 8.3 times the median of B, and no timing was repeated for this article. The ratio of the PINN time to that of the cheapest finite-difference solve of at least its accuracy is 1.7×10^4 to 7.7×10^5 with A, 1.9×10^3 to 1.3×10^5 with B and 4.8×10^2 to 2.1×10^4 with C; Section 5 quotes all three.

S8.4 Reproducibility

Seeds. In the low-dimensional experiments the seed s initialises the generator of PyTorch before the network is built and, as the pair $(s, 7919)$, the NumPy generator of the point sets; the strategies and arms of a seed thus share an initial network. In d dimensions s fixes the network and $1000 + s$ the training points, for both methods alike; the weights λ and β were chosen with seed 100.

Bit-level reproducibility. The single-thread CPU runs of Section 6 are reproducible bit for bit: the verification repeated seven of them, one of 16 000 iterations, and recovered the stored ε_2 to all ten printed digits. The stacked runs, in single precision on the GPU backend, are also reproducible bit for bit on the machine used. Three stacks were repeated with unchanged code: the benchmark stacks of **Heat** (50 networks) and **Wave2** (25 networks), and stack B of the control experiment of Section 4.3 (20 networks). Every stored final value (300, 150 and 120 of them), every logged curve and the number of loss evaluations of L-BFGS were recovered exactly. The result of one network does, however, depend on the stack in which it is trained. Stack A of the control experiment (40 networks) and the benchmark stack of **Heat** (50 networks) contain the same 20 pairs of strategy and seed (**random** and **grid-c**, seeds 0 to 9), trained by the same

code. Their Adam arms agree bit for bit, but their L-BFGS arms differ from the first logged iteration after the switch (iteration 1550), and the final errors differ by factors 0.72 to 1.09. In a direct test (Heat, random, seed 0, 300 Adam and 300 L-BFGS iterations) the network gave identical results in three stacks of two networks, with different partners and orders, and in a repetition, but not in a stack of five: $\varepsilon_2 = 7.82 \times 10^{-2}$ against 8.46×10^{-2} after Adam and 1.08×10^{-2} against 8.59×10^{-3} after Adam $\rightarrow$ L-BFGS. On the CPU all five stacks gave identical results. The mechanism, presumably the rounding of operations over the whole stack on the GPU backend, was not investigated, and other hardware was not tested. A seed therefore fixes a run only together with the size of the stack in which it is trained; the label of one strategy effect depends on it (Section 4.3). A first block of the heat benchmark (seeds 0 to 4) differs from the stored one in the code path (the loss assembled from separate forward passes) and in the stack size (25 networks instead of 50), and was superseded by the 10-seed block. For equal seeds the loss at iteration 1 is identical in all 25 networks and the final error in none: the ratio of the two values of ε_2 has geometric mean 0.98 (range 0.60 to 1.95) after Adam $\rightarrow$ L-BFGS and 0.76 (0.03 to 9.9) after Adam. The superseded block enters no result.

Software and compute. All runs used Python 3.14.6, PyTorch 2.14.1, NumPy 2.5.3 and SciPy 1.18.1 [18] under macOS 26.7 on one shared 12-core Apple-silicon machine (versions read from the environment when the tables were generated). The six stacks of the benchmark took 241 to 790 s each (2623 s together); the run notes put all low-dimensional computations at about 95 minutes, which cannot be re-measured. The 88 runs of the dimension study sum to 5051 s of wall-clock time under load. The longer runs, the control experiments and the demonstrations of Section 7 are not included.

S8.5 Verification

The benchmark and the dimension study were re-checked with the aim of refuting them, with the verification codes V-bench and V-dim, whose scope and limits are stated in Section 3.9; the longer runs, the control experiments and the demonstrations listed there are not covered. For the benchmark, the verification re-aggregated the 350 rows from the raw run files and reproduced the statistics of Tables S16 and S17; trained 40 configurations with V-bench (derivatives propagated forward through the layers), among them a grid-n control; and re-derived the finite-difference errors, except those of Wave1, from closed-form solutions of the discrete schemes and from its own sparse solver. For the dimension study, it re-aggregated the 88 run records; re-evaluated the 88 stored networks on its own 50 000 test points (largest change of ε_2 : 1.4 per cent; no ordering changed); repeated seven runs bit for bit; trained 73 networks with V-dim and other seeds; re-measured the cost per iteration at lower load; and solved the Robin problem of Observation S3 by finite differences. No key result was reversed. Where the verification gave a different value, the article reports both or states only what both support: L-BFGS won 39 of the 40 pairs of V-bench; the strategy effects on Lap2, which are smaller in V-bench; the finite-difference speed-up of three to five orders of magnitude for a single network; and the statements on the mechanism at $d = 20$ and on the penalty weights, which are restricted to what both codes show (Section 6). The total compute could not be verified.

The analyses of Sections 6.7 and 6.8 were pre-registered and then re-checked in two rounds with separately written code within the same project (Section 3.9); no gap was found in the proofs. The first round re-evaluated all 88 networks of the bound (efficiency within -0.78 to $+0.61$ per cent of the stored values, the same decisions). It also retrained 24 networks with seeds 11 to 13, on PoiD and on a harmonic problem outside the study at $d = 5$ and 10: the bound held in all, with η between 1.97 and 5.19 (no standard errors were recorded for these networks). Its second round retrained 21 further networks with another separately written trainer (its forward Laplacian agrees with the trace of the automatic-differentiation Hessian to 2.2×10^{-15} relative) and seeds 31 to 33: LapD at $d = 20$ with the PINN, with Deep Ritz and with Deep Ritz at the CPU time of the PINN (12 900 to 14 600 iterations), PoiD at $d = 2$, and the problem

$u^* = \sin(\pi d^{-1/2} \sum_i x_i)$, outside the study, at $d = 5$. The bound was above the error in all 21 networks, by at least 92.5 standard errors, with η from 1.92 to 1.93 (**LapD**, PINN, $d = 20$), 2.06 to 2.08 (Deep Ritz) and 2.14 to 2.17 (Deep Ritz at equal CPU time), 8.1 to 11.2 and 9.2 to 18.4 (**PoiD** at $d = 2$, PINN and Deep Ritz) and 1.63 to 2.68 (the new problem). The second round also computed a Rayleigh quotient for the unit square on the harmonic side (Section S5.7) and $dc(d)$ up to $d = 10^6$ without mode truncation (Conjecture S4). For the remedies, the first round retrained Deep Ritz (three seeds) on **LapD**, **RidgeD** and **AltRidgeD**, with the arms **presolve**, **lift3c**, **lift3u**, **rep3** and **lift1**, and the PINN on **AltRidgeD**; its median ratios against plain lie on the same side of 1 as those of the study’s runs in every cell that both contain, except **rep3** for Deep Ritz on **AltRidgeD** at $d = 20$ (1.21, two of three seeds better, against 0.93). The second round (Deep Ritz, three seeds, descriptive) ran **SinLinD** and **ExpCosD**. The separate current-machine cohort adds 24 PINN networks, with the full three-seed comparisons on **LapD** and **SinLinD**, released weights and checkpoint-level test-error checks (Section S5.9).

Table S15: Settings of every experiment; this part: the low-dimensional problems. “ 2×128 ” means two hidden layers of 128 units; all networks have tanh activations and the default initialisation of PyTorch. Constraint points are the initial and boundary points of Table 2; evaluation points are held out (Table 1). “Stack of S ”: S networks trained at once by the stacked driver (Section S8.2). GPU: the Metal backend of PyTorch on an Apple-silicon machine. The longer runs, the control experiments and the demonstrations are listed as configured in their scripts.

Experiment	Network (parameters); N_r and strategy; constraint points	Optimiser; iterations	Seeds	Driver, device	Evaluation points
<i>Low-dimensional problems</i> (package pinnbench ; Sections 4 and 5). Loss (7) with $\lambda = 1$; Adam with constant learning rate 10^{-3} ; L-BFGS with memory 50; single precision. “As benchmark”: as in the benchmark row of the same problem.					
Benchmark, Heat	2×128 (17 025); 256, five strategies; 300	Adam: 3000; Adam $\rightarrow$ L-BFGS: 1500 + 1500	0–9	stack of 50, GPU	201×201
Benchmark, Lap2	3×128 (33 537); 256; 400	as for Heat	0–9	2 stacks of 25, GPU	201×201
Benchmark, Wave1	3×64 (8577); 1024; 300	as for Heat	0–4	stack of 25, GPU	201×201
Benchmark, Wave2	2×50 (2801); 512; 1365	as for Heat	0–4	stack of 25, GPU	$21 \times 81 \times 81$
Benchmark, Lap3	2×100 (10 601); 512; 864	as for Heat	0–4	stack of 25, GPU	40^3
Budget sweep, Heat	as benchmark; 64 and 1024; grid-c, random, Sobol	as benchmark	0–4	stack of 15, GPU	as benchmark
Verification code V-bench (Section S8.5)	as benchmark; for Heat also grid-n and $N_r = 1024$	as benchmark, PyTorch L-BFGS	Heat: 0–5; Lap2: 0–2; else 0	single network, CPU, 4 threads	as benchmark
Grid control (Section 4.3), Heat	as benchmark; 256: random, grid-c, grid-n, grid-c shifted in t ; 272: grid-c with one further column	as benchmark	0–9	stacks of 40 and 20, GPU	as benchmark
Time slices, Heat	as benchmark; 256; random, Sobol	as benchmark, PyTorch L-BFGS	0–9	single network, CPU	as benchmark ^a
Longer runs, Wave2	as benchmark; 512; Sobol	Adam: 10 000; Adam 1500, then PyTorch L-BFGS up to 10 000 in total	0–2	single network, CPU	as benchmark
Longer runs, Wave1	as benchmark; 1024; Sobol	Adam: 41 000; Adam 1500, then PyTorch L-BFGS up to 41 000 in total	0–2	single network, CPU	as benchmark
Compatible data, Lap3s and Lap3	as Lap3; 512; Sobol	as benchmark	0–4	stack of 5, CPU	as Lap3
Driver validation (Section S8.2), Heat, Lap2	as benchmark; 1024; five strategies	Adam: 600; Adam $\rightarrow$ L-BFGS: 300 + 300	0, 1	stack of 10 and single network, CPU	as benchmark
Timing A (Table S19), five problems	as benchmark; Sobol	400 Adam and 200 PyTorch L-BFGS iterations timed, scaled to 1500 + 1500	0	single network, CPU, 4 threads	—
Finite differences, five problems	n cells per direction (Table S18)	direct solve or time stepping, double precision; time: best of 3	—	NumPy, SciPy, CPU	as benchmark

^a The trained single-precision networks are evaluated in double precision.

Table S15: (continued) The d -dimensional problems and the demonstrations of Section 7.

Experiment	Problems, dimensions, weights; or network, points	Optimiser; iterations	Seeds	Runs; driver, device	Evaluation points
<i>Problems LapD and PoiD in d dimensions</i> (module <code>hd_core</code> ; Section 6). Network 3×64 with $64d + 8449$ parameters; $N_r = N_b = 1024$, redrawn at every iteration; Adam, learning rate 10^{-3} , multiplied by 0.1 after 50 and after 75 per cent of the run; one network per run, single precision, CPU, one thread.					
Choice of weights	$d = 5$; $\lambda, \beta \in \{1, 10, 100, 1000\}$	4000	100	16 runs ^b	5000 validation points
Main study	$d = 2, 3, 5, 10, 20$; $\lambda = 1000$; $\beta = 100$ (LapD), 1 (PoiD)	4000	0–2	60 runs	50 000 test points, seed $10\,000 + d$
Longer schedule	LapD, $d = 10$, both methods	16 000	0–2	6 runs	as main study
Equal CPU time	Deep Ritz, $d = 10$	10 000 (LapD), 9000 (PoiD)	0–2	6 runs	as main study
Longer schedule, $d = 20$	LapD and PoiD, $d = 20$, both methods	16 000	0–2	12 runs	as main study
Cost per iteration	$d = 2, 3, 5, 10, 20, 50, 100$; Deep Ritz, PINN with forward and with nested Laplacian	5 warm-up iterations, then 7 rounds of 10 timed iterations for each variant	0	CPU time of the process	—
<i>Demonstrations of Section 7; single precision.</i>					
Zero residual (Section 7.1)	3–50–50–1; 10 000 uniform points, batches of 64; faulty residual, data term $e^{-t}S$, 400 boundary points at $t = 0$	Adam, learning rate 10^{-3} ; 150 passes, 23 550 iterations (also 1500 passes)	0	single network, CPU, 1 thread	$21 \times 81 \times 81$ of Wave2
Data on one face (Section 7.2)	2×100 (10 601); closed 21^3 grid (9261 points); datum on the 441 points with $x = \pi$ only	AdamW, learning rate 10^{-3} , weight decay 0.01 (the default); 2000	0–4	stack of 5, GPU	40^3 , against u^* of Lap3

^b The first 13 of these 16 runs used four threads, and the five PINN runs among them the nested Laplacian; the loss is the same (Section S2.4).

Table S16: Held-out errors of the benchmark: mean, sample standard deviation (s.d.) and geometric mean (geo.) over seeds of the relative L^2 error ε_2 , and mean and s.d. of the L^∞ error ε_∞ , for the network after the last iteration. Both arms have 3000 iterations. The notation $2.98\text{e-}2$ means 2.98×10^{-2} .

Strategy	Adam					Adam $\rightarrow$ L-BFGS				
	ε_2			ε_∞		ε_2			ε_∞	
	mean	s.d.	geo.	mean	s.d.	mean	s.d.	geo.	mean	s.d.
Heat: $N_r = 256$, 10 seeds										
grid-c	2.98e-2	2.98e-2	2.13e-2	1.48e-2	3.47e-3	6.16e-3	4.01e-3	5.01e-3	6.33e-3	4.02e-3
random	4.33e-2	6.43e-2	2.47e-2	1.79e-2	9.67e-3	1.25e-3	3.86e-4	1.19e-3	1.69e-3	7.51e-4
resample	3.70e-2	1.90e-2	3.29e-2	1.97e-2	7.12e-3	1.45e-3	1.00e-3	1.25e-3	1.88e-3	1.05e-3
Sobol	2.41e-2	2.00e-2	1.77e-2	1.11e-2	3.41e-3	1.22e-3	4.05e-4	1.15e-3	1.34e-3	6.20e-4
RAD	1.48e-2	5.23e-3	1.40e-2	1.59e-2	8.72e-3	1.16e-3	5.19e-4	1.07e-3	1.06e-3	3.97e-4
Lap2: $N_r = 256$, 10 seeds										
grid-c	6.00e-3	4.56e-3	4.91e-3	1.05e-2	4.39e-3	2.41e-4	8.27e-5	2.28e-4	7.37e-4	2.66e-4
random	8.66e-3	9.88e-3	6.22e-3	1.11e-2	6.19e-3	4.87e-4	1.65e-4	4.63e-4	1.02e-3	3.98e-4
resample	5.09e-2	3.38e-2	3.96e-2	3.23e-2	1.87e-2	4.40e-4	1.58e-4	4.19e-4	9.87e-4	3.87e-4
Sobol	4.08e-3	1.62e-3	3.80e-3	9.05e-3	3.07e-3	2.55e-4	7.19e-5	2.48e-4	7.34e-4	2.56e-4
RAD	1.12e-2	1.47e-2	6.20e-3	1.27e-2	8.74e-3	4.35e-4	1.62e-4	4.08e-4	8.11e-4	2.85e-4
Wave1: $N_r = 1024$, 5 seeds										
grid-c	4.83e-1	2.13e-2	4.83e-1	6.70e-1	3.70e-2	4.35e-1	1.51e-2	4.35e-1	6.52e-1	5.43e-2
random	4.81e-1	2.22e-2	4.81e-1	6.62e-1	3.25e-2	4.42e-1	1.37e-2	4.42e-1	6.59e-1	2.43e-2
resample	4.86e-1	2.22e-2	4.86e-1	6.81e-1	3.77e-2	4.41e-1	1.90e-2	4.41e-1	6.58e-1	6.67e-2
Sobol	4.82e-1	2.22e-2	4.81e-1	6.60e-1	3.50e-2	4.43e-1	1.12e-2	4.42e-1	6.68e-1	4.63e-2
RAD	4.84e-1	2.22e-2	4.84e-1	6.69e-1	3.60e-2	4.49e-1	1.15e-2	4.49e-1	6.90e-1	3.78e-2
Wave2: $N_r = 512$, 5 seeds										
grid-c	2.11e-1	6.77e-2	2.03e-1	2.67e-1	6.98e-2	4.92e-2	1.26e-2	4.80e-2	7.99e-2	1.77e-2
random	2.13e-1	6.54e-2	2.05e-1	2.66e-1	6.27e-2	4.89e-2	1.92e-2	4.64e-2	7.75e-2	2.67e-2
resample	2.35e-1	6.52e-2	2.29e-1	2.87e-1	6.04e-2	5.01e-2	8.69e-3	4.94e-2	7.92e-2	2.49e-2
Sobol	2.09e-1	5.69e-2	2.03e-1	2.59e-1	5.01e-2	4.85e-2	1.72e-2	4.61e-2	8.07e-2	2.70e-2
RAD	2.03e-1	4.44e-2	2.00e-1	2.66e-1	4.66e-2	4.79e-2	1.32e-2	4.65e-2	8.06e-2	2.19e-2
Lap3: $N_r = 512$, 5 seeds										
grid-c	1.18e-1	2.24e-2	1.16e-1	2.14e-1	1.75e-2	4.88e-2	1.21e-2	4.76e-2	1.00e-1	2.92e-2
random	3.00e-1	3.43e-1	2.05e-1	2.58e-1	7.01e-2	4.63e-2	8.77e-3	4.56e-2	8.33e-2	2.39e-2
resample	1.55e-1	2.02e-2	1.54e-1	2.41e-1	1.21e-2	4.28e-2	3.74e-3	4.27e-2	1.01e-1	7.26e-2
Sobol	1.31e-1	1.54e-2	1.30e-1	2.22e-1	1.14e-2	3.56e-2	3.66e-3	3.55e-2	7.73e-2	1.50e-2
RAD	1.33e-1	1.76e-2	1.32e-1	2.27e-1	1.51e-2	4.03e-2	6.23e-3	3.99e-2	7.01e-2	9.76e-3

Table S17: Paired comparisons (10 seeds for **Heat** and **Lap2**, 5 for the other problems). (a) Each strategy against **random** at equal N_r , paired by seed: geometric mean of the per-seed ratio of ε_2 (below 1: better than **random**) and, in brackets, the number of seeds in which the strategy is better and the two-sided p -value of the exact Wilcoxon signed-rank test (notation as in Table S16). (b) The two optimiser arms. “Runs” counts all pairs of a strategy and a seed; for the test the errors of the five strategies are first averaged geometrically within each seed. No p -value is corrected for multiple comparisons; with 5 seeds the smallest attainable value is 0.0625.

(a)

Problem	grid-c	resample	Sobol	RAD	
Arm Adam $\rightarrow$ L-BFGS					
Heat	4.21 (0/10; 0.002)	1.05 (6/10; 1.000)	0.97 (6/10; 0.695)	0.90 (5/10; 0.695)	
Lap2	0.49 (10/10; 0.002)	0.91 (6/10; 0.322)	0.54 (9/10; 0.004)	0.88 (8/10; 0.160)	
Wave1	0.98 (4/5; 0.438)	1.00 (3/5; 0.812)	1.00 (3/5; 1.000)	1.02 (1/5; 0.188)	
Wave2	1.04 (3/5; 1.000)	1.07 (2/5; 0.812)	0.99 (2/5; 1.000)	1.00 (3/5; 1.000)	
Lap3	1.04 (2/5; 0.812)	0.93 (3/5; 0.438)	0.78 (5/5; 0.0625)	0.88 (3/5; 0.625)	
Arm Adam					
Heat	0.86 (5/10; 0.846)	1.33 (4/10; 0.275)	0.71 (6/10; 0.432)	0.57 (5/10; 0.432)	
Lap2	0.79 (6/10; 0.322)	6.36 (0/10; 0.002)	0.61 (7/10; 0.084)	1.00 (5/10; 1.000)	
Wave1	1.00 (4/5; 0.625)	1.01 (1/5; 0.125)	1.00 (2/5; 0.812)	1.01 (2/5; 0.312)	
Wave2	0.99 (3/5; 0.625)	1.11 (1/5; 0.188)	0.99 (3/5; 0.625)	0.97 (3/5; 0.812)	
Lap3	0.57 (4/5; 0.188)	0.75 (2/5; 0.812)	0.64 (3/5; 0.438)	0.64 (2/5; 0.812)	

(b)

Problem	Seeds	geometric mean of ε_2			Adam $\rightarrow$ L-BFGS better in		Wilcoxon p (per seed)
		Adam	Adam $\rightarrow$ L-BFGS	Ratio	runs	seeds	
Heat	10	2.12e−2	1.56e−3	0.074	50/50	10/10	0.002
Lap2	10	7.78e−3	3.39e−4	0.044	50/50	10/10	0.002
Wave1	5	4.83e−1	4.42e−1	0.915	25/25	5/5	0.0625
Wave2	5	2.08e−1	4.73e−2	0.228	25/25	5/5	0.0625
Lap3	5	1.45e−1	4.20e−2	0.291	25/25	5/5	0.0625

Table S18: Second-order finite-difference baselines on the held-out evaluation points of the networks. n : cells per space direction; “values”: number of stored values of the discrete solution: all space–time nodes for Heat and Wave1, all nodes for Lap2 and Lap3, and for Wave2 the nodal values at the 21 output times, $21(n+1)^2$ (the scheme itself has $n+1$ time levels and so $(n+1)^3$ space–time nodes); “order”: $\log_2$ of the ratio of ε_2 on two consecutive grids; “time”: best of three solves. For Lap3 the evaluation points are interpolated for $n=20, 40$ and are nodes for $n=80, 160$.

n	values	ε_2	ε_∞	order	time (s)	n	values	ε_2	ε_∞	order	time (s)
Heat: Crank–Nicolson, $\Delta t = \Delta x = 1/n$						Lap2: five-point Laplacian, $\Delta x = \pi/n$					
25	676	8.16e−3	4.35e−3		5.46e−4	25	676	1.20e−3	9.75e−4		2.22e−3
50	2601	2.02e−3	1.08e−3	2.01	1.09e−3	50	2601	3.02e−4	2.50e−4	1.99	6.31e−3
100	10 201	5.05e−4	2.71e−4	2.00	2.21e−3	100	10 201	7.51e−5	6.22e−5	2.01	2.42e−2
200	40 401	1.26e−4	6.71e−5	2.00	5.18e−3	200	40 401	1.89e−5	1.51e−5	1.99	1.21e−1
400	160 801	3.15e−5	1.68e−5	2.00	1.25e−2	400	160 801	4.72e−6	3.78e−6	2.00	6.55e−1
800	641 601	7.87e−6	4.19e−6	2.00	3.24e−2						
Wave1: leapfrog, Courant number 0.5, $\Delta x = 1/n$						Wave2: leapfrog, Courant number 0.5, $\Delta x = 2/n$					
25	2626	5.14e−2	9.45e−2		4.45e−4	20	9261	5.12e−3	8.83e−3		3.55e−4
50	10 251	1.28e−2	2.36e−2	2.01	1.04e−3	40	35 301	1.28e−3	2.20e−3	2.00	1.24e−3
100	40 501	3.19e−3	5.88e−3	2.00	1.87e−3	80	137 781	3.19e−4	5.50e−4	2.00	4.49e−3
200	161 001	7.96e−4	1.47e−3	2.00	3.87e−3	160	544 341	7.98e−5	1.38e−4	2.00	2.65e−2
400	642 001	1.99e−4	3.67e−4	2.00	8.69e−3	320	2 163 861	1.99e−5	3.44e−5	2.00	1.86e−1
800	2 564 001	4.98e−5	9.19e−5	2.00	2.29e−2						
Lap3: seven-point Laplacian, $\Delta x = \pi/n$											
20	9261	3.90e−2	1.71e−1		2.90e−4						
40	68 921	9.97e−3	4.34e−2	1.97	1.61e−3						
80	531 441	1.54e−3	4.58e−3	2.70	1.00e−2						
160	4 173 281	4.13e−4	1.25e−3	1.89	8.62e−2						

Table S19: Cost of one Adam $\rightarrow$ L-BFGS run (1500 + 1500 iterations) against finite differences. Measurement A: single-network reference driver, four CPU threads, load average 95 to 167; milliseconds per iteration, loss-and-gradient evaluations per L-BFGS iteration (ev./it), and the run time estimated from them. Measurement B: runs of the verification code V-bench at a load average of about 20 to 45 (to iteration 3000 or to the tolerance of L-BFGS); smallest and largest time, number of runs in square brackets. Measurement C: training time of the Adam $\rightarrow$ L-BFGS arm of the GPU stacks of the benchmark per network (range over the stacks of a problem). FD: solve time of the cheapest grid of Table S18 that is at least as accurate as the best strategy of the benchmark (smallest geometric mean among the five strategies, Table S16); that strategy, its error and the grid are given in Table 8. Time ratio: PINN time over finite-difference time ($9.7\text{e}4$ means 9.7×10^4). Absolute times indicate orders of magnitude only.

Problem	measurement A				B	C	FD	time ratio		
	ms/it Adam	ms/it L-BFGS	ev./it	run (s)				A	B	C
Heat	72	72	1.06	216	20–37 [21]	12.4	2.2e−3	9.7e4	9.1e3–1.7e4	5.6e3
Lap2	129	139	1.06	402	46–83 [12]	11.5–13.9	2.4e−2	1.7e4	1.9e3–3.4e3	4.8e2–5.7e2
Wave1	94	108	1.065	302	60 [1]	6.9	4.4e−4	6.8e5	1.3e5	1.5e4
Wave2	86	96	1.045	273	36 [1]	7.4	3.5e−4	7.7e5	1.0e5	2.1e4
Lap3	111	117	1.055	341	50 [1]	11.7	1.6e−3	2.1e5	3.1e4	7.3e3

Table S20: The two controls of Section 6.8, run for Deep Ritz at $d = 20$ with seeds 10–14 and 4000 iterations (pre-registered), with the plain network for reference: lift3u, the span of lift3c with the uncentred linear feature x_i in place of $P_1(t_i)$, and rep3, the input $(\mathbf{t}, \mathbf{t}, \mathbf{t})$ (Proposition S18). Quantities as in Table S12: ε_2 in units of 10^{-2} , mean (sample standard deviation), and the seed-paired ratio $\rho = \varepsilon_2(\text{plain})/\varepsilon_2(\text{arm})$, median [minimum, maximum] and the number of seeds with $\rho > 1$. f.-m.: feature-matched.

problem	method	arm	iterations	ε_2 , mean (s.d.)	ρ : median [min, max], wins
LapD	Deep Ritz	plain	4000	4.75 (0.37)	–
LapD	Deep Ritz	lift3u	4000	9.18 (0.20)	0.52 [0.47, 0.57] 0/5
LapD	Deep Ritz	rep3	4000	5.42 (0.16)	0.88 [0.81, 0.99] 0/5
PoiD (f.-m.)	Deep Ritz	plain	4000	12.6 (0.037)	–
PoiD (f.-m.)	Deep Ritz	lift3u	4000	1.87 (0.039)	6.71 [6.59, 6.91] 5/5
PoiD (f.-m.)	Deep Ritz	rep3	4000	1.58 (0.035)	8.10 [7.78, 8.15] 5/5
RidgeD	Deep Ritz	plain	4000	3.74 (0.33)	–
RidgeD	Deep Ritz	lift3u	4000	3.08 (0.34)	1.26 [0.97, 1.40] 4/5
RidgeD	Deep Ritz	rep3	4000	1.50 (0.14)	2.37 [2.06, 3.21] 5/5
CosPairD	Deep Ritz	plain	4000	20.6 (0.79)	–
CosPairD	Deep Ritz	lift3u	4000	12.4 (1.2)	1.60 [1.51, 2.03] 5/5
CosPairD	Deep Ritz	rep3	4000	7.77 (0.57)	2.65 [2.50, 2.94] 5/5

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